

Electoral Integrity, Electoral Access, and Biometric Voter Registration in Brazil

Gabriel S. Gaspar¹

¹*London Business School*

Abstract

Reforms designed to strengthen electoral integrity can also make electoral participation more administratively demanding. This paper studies that tradeoff through the staggered rollout of biometric voter registration across Brazilian municipalities between 2008 and 2018. Using administrative electoral records, I show that biometric registration reduced the size of the registered electorate at adoption and disproportionately lowered the share of less-educated voters on the rolls. I then combine those administrative results with LAPOP survey data to examine whether exposure to the reform is associated with changes in trust in institutions, democratic attitudes, and perceptions of electoral integrity. The survey evidence suggests higher trust in institutions among lower-education respondents in treated municipalities, while effects on broader democratic attitudes are weaker and evidence of backlash among Bolsonaro-aligned respondents is limited. The results highlight a central tension of electoral modernization: reforms intended to secure the voter registry can also reshape who remains represented within it.

1. Introduction

Trust in political institutions is both an input into democratic stability and an output of how democracy is administered. Elections are the most visible point of contact between citizens and the state, so procedures that govern registration, identification, and vote

verification can shape confidence in democratic institutions well beyond election day itself. These issues have become especially salient as electoral authorities around the world have modernized how citizens are registered and identified through biometric rolls and voter identification requirements, typically with the stated goal of strengthening electoral integrity and reducing fraud (Wolf et al., 2017). Yet these same reforms can also impose new administrative burdens, exclude citizens who fail to comply in time, and alter who remains politically represented (Electoral Commission, 2024). This paper studies that tension: when the state modernizes voter identification, does greater security translate into greater trust in democratic institutions, or does it instead narrow the electorate and invite political backlash?

I study this question through the staggered rollout of biometric voter registration (BVR) across Brazilian municipalities between 2008 and 2018. BVR required voters to register biometric information with the electoral authority and shifted polling-place verification away from the voter registration card and toward fingerprint authentication. In principle, this reform could improve registry accuracy, reduce impersonation and administrative error, and strengthen confidence in the electoral process. But it could also shrink the electorate if some citizens failed to complete the new registration procedure in time. I assemble a municipality-by-election panel from administrative electoral records and combine it with survey evidence from LAPOP to study this second wave of electoral modernization in Brazil, after the earlier transition to electronic voting at ballot casting (Fujiwara, 2015).

The administrative evidence shows that BVR reduced the size of the registered electorate at adoption, consistent with a combination of registry cleaning and incomplete compliance with the new registration requirements. The reform also changed the composition of the electorate, disproportionately reducing the share of lower-education voters on the rolls while increasing the relative share of more educated voters. At the same time, survey evidence suggests that exposure to this institutional modernization is associated with higher trust in institutions and in democracy, especially among lower-education respondents who remain in the electorate. These patterns suggest that biometric registration affected several margins at once: it made the voter registry more secure and arguably more accurate, but it also changed who was included in it. I also examine a distinct concern that has become more prominent in recent years—whether electoral technologies become focal points for distrust and partisan backlash. In the Brazilian context, I test whether Bolsonaro-aligned respondents in treated locations are systemat-

ically less trusting of vote-counting integrity or ballot secrecy, and I find little evidence that greater exposure to BVR amplified such distrust.

To organize these results, the paper introduces a compact formal model of electoral modernization before turning to the data. The model treats biometric registration as a security technology that raises the cost of remaining on the rolls, with those compliance costs borne more heavily by less-educated citizens. It yields four predictions. First, more demanding verification reduces registration disproportionately among low-education citizens. Second, because education and income are positively correlated, selective compliance shifts the post-registration median voter toward lower redistribution. Third, higher observed institutional trust among post-reform respondents can arise mechanically from selection into the surviving electorate rather than from persuasion, which is the key lens through which the LAPOP results should be read. Fourth, the planner’s optimal biometric intensity is interior and depends on how much electoral integrity is valued relative to inclusion.

The paper’s empirical leverage comes from the staggered municipal rollout of BVR, which creates variation in the timing of adoption across locations. I use event-study and difference-in-differences designs suited to staggered treatment timing to trace the dynamic effects of the reform on electorate size and composition, and I then pair those administrative results with survey evidence on institutional trust, democratic attitudes, and perceptions of electoral integrity. This setting allows the paper to make three contributions. First, it extends the literature on voting technology from ballot-casting technology to registration and voter-identification technology. Second, it contributes to the literature on administrative burden and political inclusion by showing that a security-oriented reform can improve the integrity of the registry while also excluding citizens who do not comply with new procedural requirements. Third, it speaks to political economy by showing that electoral modernization can shape not only the formal quality of election administration but also the composition of the electorate, citizens’ trust in democratic institutions, and the scope for backlash against electoral technology. More broadly, the results suggest that the relationship between electoral security and democratic legitimacy runs through both the credibility of the process and the inclusiveness of the electorate.

Literature overview.

Registration costs and the composition of participation. A first strand studies how electoral rules alter the cost of becoming, or remaining, an eligible voter. Classic evidence from the United States shows that registration requirements depress turnout, especially when deadlines are early, offices are inconvenient, or reregistration is required after moving Rosenstone and Wolfinger (1978); Ansolabehere and Konisky (2005). Subsequent work complicates the distributional claim: education gaps need not always widen mechanically when registration rules change Nagler (1991), while reforms that lower administrative frictions, such as preregistration and automatic reregistration, can raise participation among targeted groups Holbein and Hillygus (2014); Kim (2022). Comparative work similarly emphasizes that turnout reflects both rules and mobilization incentives Cox (2015); Frank and Martínez i Coma (2023); Gaebler et al. (2020). The voter-ID literature pushes this argument toward institutional design that is explicitly justified by electoral integrity: strict ID rules can screen out voters without compliant documentation, with especially large burdens for racial and ethnic minorities and durable effects even when rules are later relaxed Fraga and Miller (2021); Hajnal et al. (2017); Grimmer and Yoder (2021). Other evidence bounds the aggregate consequences more tightly, showing that in Florida and Michigan only a very small share of ballots were cast without ID and that even extreme disenfranchisement assumptions would rarely change election outcomes Hoekstra and Koppa (2019). This paper studies a related but distinct reform: biometric voter registration does not merely require identification at the polling place, but changes the administrative stock of registered voters itself, making it possible to measure both aggregate registry exit and compositional change.

Election technology and biometric state capacity. A second strand asks whether election technology expands effective access or instead substitutes one form of error for another. In Brazil, electronic voting reduced residual votes and increased the political weight of less-educated voters, with downstream effects on representation and infant health Fujiwara (2015). Yet the same broad technological modernization agenda can introduce new mistakes: electronic voting also increased party-label votes in ways consistent with ballot-interface confusion Zucco and Nicolau (2016). The closest existing study of Brazilian biometric registration finds that biometric recadastramento reduced the number of registered voters but did not significantly reduce actual turnout, interpreting the gap partly as the removal of invalid registrations Forquesato and Rodrigues (2023). Outside electoral administration, biometric monitoring has been studied as a state-capacity tool: biometric smartcards in India reduced leakage and improved pay-

ment delivery without clear evidence of beneficiary exclusion, while biometric monitoring in health care improved adherence and data accuracy Muralidharan et al. (2016); Bossuroy et al. (2024). The unresolved issue is whether the integrity gains from biometric administration come with politically meaningful changes in who remains registered. This paper addresses that gap by decomposing registry change into exit and education relabeling at the municipality-cohort level.

Unequal turnout, representation, and welfare. A third branch links participation to representation and welfare. If turnout is uneven, policy need not reflect the preferences of the full eligible population: in city politics, higher turnout among underrepresented groups can shift descriptive and substantive representation Hajnal and Trounstein (2005). Political-economy models give the mechanism theoretical bite: voting costs, candidate entry, and the location of the decisive voter shape equilibrium policy, redistribution, and the welfare consequences of participation rules Besley and Coate (1997); Meltzer and Richard (1981); Börgers (2004); Krasa and Polborn (2008). A related public-finance literature shows why administrative restrictions can improve targeting in principle while also screening out real beneficiaries in practice Nichols and Zeckhauser (1982); Deshpande and Li (2019). Recent evidence from Brazil reinforces the stakes: restrictions on fraudulent registration can alter political competition, candidate selection, public goods, and social outcomes Karim (2025). The contribution here is to bring these welfare and representation concerns to a measured electoral-administration shock: rather than assuming that biometric recadastramento only removes ghosts or only burdens real voters, the paper estimates the size and composition of the affected electorate and uses those estimates in a welfare framework.

Electoral integrity, administration, and institutional trust. A fourth strand studies how citizens interpret electoral administration itself. Electoral reforms may increase procedural integrity, but they can also change beliefs about the fairness or competence of institutions. Mobilization around electoral procedures in Kenya increased turnout but lowered trust in the electoral commission, particularly among politically disappointed voters and in violent constituencies Marx et al. (2021). International and domestic actors also value electoral process independently of outcomes, which is why process integrity can become politically salient even when candidate stakes are high Bubeck and Marinov (2017). Evidence from local election administration suggests that

partisan administrators do not necessarily shift vote shares in favor of their party, but the question of perceived neutrality remains central to democratic legitimacy Ferrer et al. (2023). More broadly, electoral accountability can discipline corruption when citizens can observe and sanction incumbents Ferraz and Finan (2011). Biometric voter registration sits at the intersection of these concerns: it is advertised as an anti-fraud reform, but it may also make registration loss more visible, create confusion, or change who is represented in the electorate. The LAPOP component of this paper therefore treats trust in institutions and democratic attitudes as outcomes, not merely as background conditions.

Gap and contribution. Taken together, existing work shows that administrative frictions affect turnout, that election technologies can enfranchise some voters while generating new errors, that unequal participation matters for representation and welfare, and that procedural reforms can reshape trust. What remains less well understood is how a large biometric registration reform changes the electorate before votes are cast, especially outside the U.S. voter-ID setting and in a context where biometric registration is rolled out gradually across municipalities. This paper fills that gap by combining Brazilian administrative data with survey evidence: it estimates municipality-level registry exit, education relabeling, downstream voting patterns, and institutional trust, while distinguishing strict and hybrid implementation regimes. The result is a bridge between the voter-ID literature’s concern with burdens, the state-capacity literature’s focus on administrative integrity, and political-economy models of representation and welfare.

Roadmap. Section 2 describes the institutional setting, the rollout of biometric registration, and the data. Section 3 presents the empirical design. Section 4 presents the main administrative evidence on the voter registry, covering both electorate size and composition. Section 5 decomposes the registry shift into exit and education-record re-labeling. Section 6 turns to political and downstream outcomes. Section 8 turns to survey evidence on political trust, democratic attitudes, electoral-integrity beliefs, and backlash. Section 9 develops the welfare framework, Section 10 calibrates optimal electoral monitoring, and Section 11 concludes.

2. Background and Data

Brazil is a useful setting for studying biometric voter registration because elections combine centralized administration, compulsory voting for most adults, and long-running investments in election technology. The *Tribunal Superior Eleitoral* (TSE) and the state-level *Tribunais Regionais Eleitorais* (TREs) maintain the voter registry, organize elections, and supervise polling-day procedures. Voting itself is electronic and has been nationwide since 2000, a reform that substantially reduced residual votes and expanded de facto enfranchisement among poorer and less educated citizens, even as it also generated some new forms of voter error (Fujiwara, 2015; Zucco and Nicolau, 2016). In that institutional environment, biometric registration is best understood not as an isolated administrative tweak, but as a later stage of electoral modernization that changes both how voters are authenticated and how the electoral state manages the registry.

2.1. Voting Administration and Biometric Registration

Before biometric identification, voters typically authenticated themselves at the polling place with a voter registration card and another identity document, after which poll workers checked the information and released the electronic ballot. Biometric registration changed that interface. Under the biometric revision process, the electoral courts collect a voter’s photograph, signature, and fingerprints in advance, and those data can then be used to verify identity at the polling place (Tribunal Superior Eleitoral, 2024c). The program started with three pilot municipalities in 2008, formalized in TSE Resolution No. 22,713 of February 28, 2008 (Tribunal Superior Eleitoral, 2008), and expanded over subsequent election cycles. By the 2024 municipal elections, the TSE reported that 129.2 million electors, or 82.87% of the electorate, had biometric records on file (Tribunal Superior Eleitoral, 2024a).

The reform matters substantively because its effects are ambiguous ex ante. On one hand, biometric registration may improve record quality, reduce impersonation risk, and increase confidence in electoral administration. On the other hand, it requires voters to interact with the bureaucracy before election day, which can create compliance costs, especially where access to registration offices or information is uneven. Those two possibilities motivate the paper’s focus on both administrative outcomes, such as

electorate size and composition, and attitudinal outcomes, such as trust in elections and support for democracy.

Figure 1: Electoral Identification Materials



(a) Voter identification card



(b) Biometric verification at the polling place

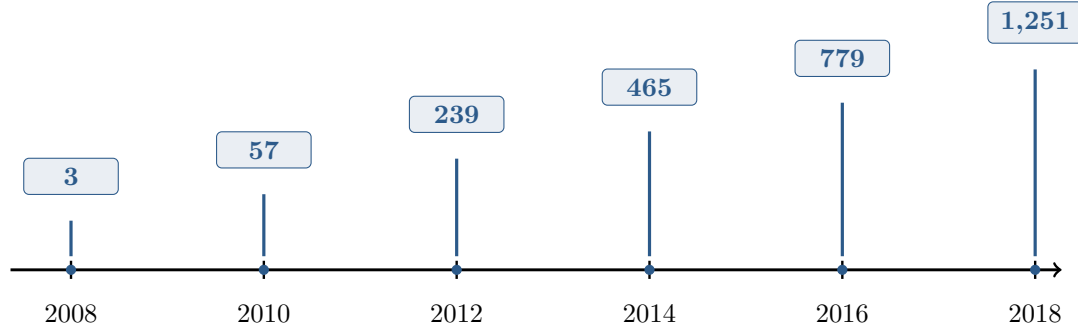
2.2. Rollout and Treatment Coding

The empirical design exploits the staggered municipal rollout of biometric registration. I define treatment timing as the first election year in which a municipality appears in the clean municipality-level biometric rollout file assembled from official TSE resolutions, annexes, election-use attachments, and election-year electorate files. The clean first-treatment file contains 2,794 treated municipalities by 2018, while 2,777 municipalities remain untreated in the municipal panel through that year. Newly treated municipalities enter in waves: 3 in 2008, 57 in 2010, 239 in 2012, 465 in 2014, 779 in 2016, and 1,251 in 2018.

Figure 2 summarizes those treatment waves directly. The visual makes clear that rollout was not only staggered, but increasingly concentrated in the later election cycles: adoption remains modest through 2012, then accelerates sharply in 2014, 2016, and especially 2018. By the end of the period, 2,794 municipalities in the clean panel, roughly half of all municipalities in the analysis sample, had entered the biometric regime. This timing structure is the core source of identifying variation in the paper. A companion spatial map appears in Appendix Figure 16, which shades municipalities by first observed BVR year, including hybrid implementation, and marks hybrid municipalities with diagonal hatching.

One complication is partial or hybrid implementation. The project therefore separates the municipality-level first-treatment file from hybrid-status information. In the clean

Figure 2: Municipal Rollout of Biometric Registration



Note: Each labeled count reports the number of municipalities whose first biometric election occurred in that election cycle. Counts are constructed from the project’s clean municipality-level first-treatment file built from official TSE rollout documents and electorate files. By 2018, 2,794 of the 5,571 municipalities in the clean panel had entered the biometric regime.

panel, 1,533 municipality-year observations in 2018 are flagged as hybrid, and those cases can be excluded or handled separately in robustness checks. This distinction matters because a municipality observed under a hybrid system is not equivalent to one that had fully transitioned to biometric verification earlier. The baseline treatment timing used in the municipal estimators is therefore conservative: it is anchored in the municipality-level first-treatment file rather than in a looser indicator of any biometric presence.

2.3. Administrative Data

The main administrative dataset is a municipality-election panel built from official TSE electorate files and the project’s cleaned biometric-rollout file. The panel covers ten election years from 2000 through 2018 and contains 55,661 municipality-election observations across 5,571 municipalities in all 27 units of the federation (Tribunal Superior Eleitoral, 2024b). At the municipality-election level, the core variables are the number of registered voters, its logarithm, the share of registered voters with lower schooling, the share with higher schooling, and parallel gender shares. Municipality identifiers are harmonized with the IBGE-TSE crosswalk stored in the repository, and the resulting panel is validated so that the final key is unique on ‘year_election + municipality_id’ (Instituto Brasileiro de Geografia e Estatística, 2024).

The electorate-composition measures come from TSE tabulations by municipality, edu-

cation, and gender. I classify “low education” as the set of categories below completed secondary schooling: illiterate, reads and writes, incomplete primary, and complete primary. “High education” includes incomplete secondary, complete secondary, incomplete higher education, and complete higher education. Importantly, the denominator for the reported shares is always the total municipality-level electorate, including cases with missing or unknown schooling. For that reason, the low- and high-education shares need not sum to one. This choice is deliberate: it prevents changes in missingness from being mechanically reallocated across the substantive education categories.

The administrative panel is designed to keep treatment coding and outcome construction transparent. Event time is defined relative to the first election year in which a municipality enters the biometric regime, while untreated municipalities remain in the sample as controls. Because the same cleaned panel also carries hybrid-status information, the administrative analysis can report a main specification and then assess whether results are sensitive to excluding municipality-years with documented hybrid implementation.

2.4. Survey Data and Attitudinal Outcomes

To study trust, democratic support, and beliefs about electoral integrity, the paper uses harmonized Brazil waves from the LAPOP AmericasBarometer (LAPOP Lab, 2023). The core survey file currently available in the repository contains 10,009 respondents across six waves, 2008, 2010, 2012, 2014, 2017, and 2019, covering 25 states and 238 municipalities. The harmonized file preserves municipality and state names, interview timing, core demographics, and a consistent set of attitudinal variables. Those include trust in elections, trust in municipal government, broader institutional support, and several measures of democratic support and satisfaction.

The survey data complement the administrative panel in two ways. First, they allow the paper to ask whether biometric registration changed how citizens understand the integrity of elections and the legitimacy of democratic institutions, not just who remains on the voter rolls. Second, they make it possible to compare the timing of administrative change to shifts in attitudes measured after adoption. The municipal linkage is necessarily less comprehensive than in the TSE panel, but it is still rich enough to support descriptive and difference-in-differences analyses that use the same treatment-timing logic while remaining explicit about survey coverage and precision limits.

Taken together, the administrative and survey data allow the paper to study biometric registration as both a registry reform and a political event. The administrative panel is well suited to measuring immediate changes in electorate size and composition around adoption, while the survey data allow the paper to examine whether those administrative changes coincide with greater confidence, skepticism, or backlash. Keeping those modules distinct at the data-construction stage is useful analytically: it makes clear which results come from population-wide administrative records and which come from sampled attitudes, and it helps keep causal claims proportional to the underlying evidence.

3. Empirical Design

Staggered rollout. The empirical design uses the staggered rollout of biometric voter registration across Brazilian municipalities. Municipalities entered BVR in different election years, while a large set of municipalities remained never treated through the end of the sample. This timing creates a panel design in which each treated cohort can be compared to municipalities that had not entered the biometric regime, with municipality histories anchoring the pre-treatment path and election-year comparisons absorbing national shocks common to all municipalities in a given election cycle.

The main registry and composition figures use the Callaway–Sant’Anna group-time difference-in-differences estimator (Callaway and Sant’Anna, 2020). For treatment cohort g and election year t , the primitive estimand is

$$ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g], \quad (1)$$

where G is the first treatment year and $Y_t(0)$ is the no-BVR potential outcome. I aggregate these group-time effects by event time $k = t - g$, using event time -2 as the reference period. The displayed window is eight years before and after adoption, so $k \in \{-8, -6, -4, -2, 0, 2, 4, 6, 8\}$. The reference period is included in the plots at zero to make the normalization explicit. Standard errors are clustered at the municipality level, and the preferred implementation uses never-treated municipalities as the control group.

Estimator choice. I use the Callaway–Sant’Anna estimator as the main estimator

because the treatment is staggered and treatment effects are likely to vary across adoption cohorts and over event time. A conventional dynamic TWFE event study is useful as a transparent diagnostic, but under staggered adoption it can combine treatment effects with non-convex weights when effects are heterogeneous. The group-time estimator instead estimates cohort-specific treatment effects against a clean comparison group and then aggregates them into the common event-time profile shown in the paper. Appendix E compares the main Callaway–Sant’Anna path to dynamic TWFE, Borusyak–Jaravel–Spiess, and de Chaisemartin–D’Haultfoeuille alternatives for the core registry outcomes. The estimated trajectories are qualitatively similar across estimators.

Strict and hybrid BVR. The preferred treatment contrast is exposure to full biometric registration, which corresponds to municipalities that entered the strict BVR regime. Hybrid implementation is a lower-intensity form of exposure: some voters could update biometric records before full enforcement, but the municipality had not fully transitioned into the strict regime. For this reason, the main estimates use the strict-BVR treatment timing and exclude municipalities ever flagged as hybrid, leaving 2,480 treated municipalities and 1,245 never-treated controls in the registry-size specification. Appendix D repeats the exercise with strict and hybrid municipalities included in the treatment definition. The hybrid-inclusive estimates preserve the sign and shape of the results but are attenuated on impact: the registry-size effect falls from about -0.117 log points in the strict sample to about -0.068 log points when hybrid municipalities are included, while the low-education and high-education effects fall from about 8.9 percentage points to about 5.1 percentage points.

Outcomes. The first set of outcomes measures the electorate itself: log registered voters, the share of registered voters with low education, and the share with high education. These are the outcomes in Section 4 and Section 4.3. The decomposition in Section 5 then separates the registry contraction into exit from the voter roll and education re-labeling. Downstream analyses apply the same timing logic to turnout, blank and null voting, presidential vote shares, municipal fiscal outcomes, health outcomes, and survey-based measures of trust and electoral legitimacy where the data support the design.

Diagnostics and robustness. Several checks discipline the interpretation of the event studies. The figures show pre-treatment coefficients relative to event time -2 , so visible pre-trends would be evidence against the design. Appendix E checks that the core

registry results are not driven by estimator-specific features of the Callaway–Sant’Anna aggregation. Appendix D shows that the same patterns appear when hybrid municipalities are included, with smaller magnitudes consistent with a lower-intensity treatment. Appendix F applies HonestDiD-style sensitivity tests to the pre-trend restrictions, and Appendix G reports state-by-year fixed effects, within-state support restrictions, permutation tests, negative-weight diagnostics, and cohort-balance checks. Across these exercises, the central registry and composition results remain stable.

4. BVR and the Voter Registry

The most immediate administrative consequence of biometric registration should appear in the voter registry itself. If the reform improved record quality and eliminated duplicate or outdated registrations, the electorate should contract when municipalities adopt it. If the recadastramento requirement also imposed meaningful compliance costs, that contraction may reflect not only registry cleaning but also temporary or persistent exclusion from the rolls. Those same forces can change the social composition of the electorate, not just its size. For that reason, it is useful to study registry size and registry composition together. In the model, this is the first margin: Proposition 1 predicts that once compliance costs are steeper for lower-education citizens, any positive biometric burden should shrink their registration rate more sharply.

This section studies that administrative margin directly. I first describe the event-study specification used for the registry outcomes. I then show that strict BVR produces an immediate contraction in the log number of registered voters. Next, I examine the education margin using both education-specific log counts and education shares, because those two objects distinguish the size of the affected registry groups from the composition of the surviving electorate. I then ask whether the effect is larger in municipalities with weaker pre-treatment connectivity, a descriptive test of the compliance-cost interpretation. The section closes by summarizing the main robustness exercises that support the registry results.

4.1. Econometric Specification

The registry analysis uses a municipality-election panel. Let G_m denote municipality m 's first year of strict BVR exposure, with municipalities that never enter the strict regime serving as the never-treated comparison group. The main sample excludes municipalities ever coded as hybrid, so the baseline contrast is full biometric registration against municipalities that had not adopted BVR. For each outcome Y_{mt} , I estimate Callaway–Sant’Anna group-time treatment effects,

$$ATT(g, t) = \mathbb{E}[Y_{mt}(g) - Y_{mt}(0) \mid G_m = g],$$

and aggregate them by event time $k = t - g$. The figures display event times from eight years before adoption to eight years after adoption, in two-year election-cycle steps. Event time -2 is the reference period and is shown at zero in every plot. Standard errors are clustered by municipality.

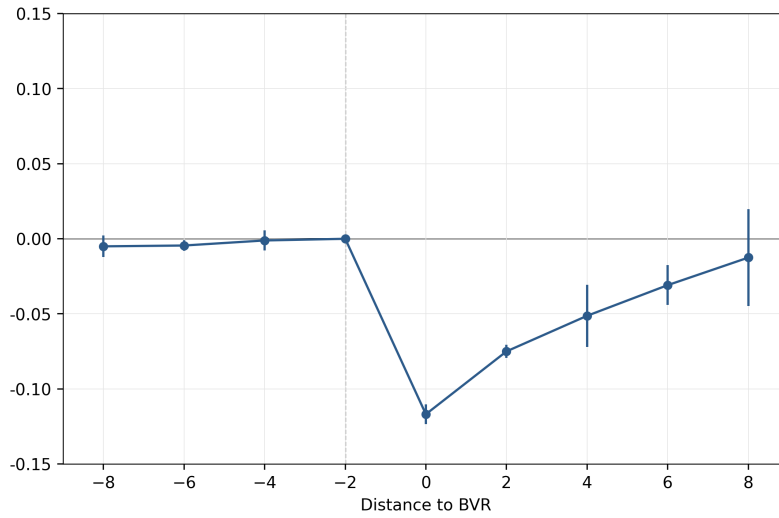
The outcomes are chosen to separate three registry objects. The first is the log number of registered voters, which measures the aggregate size of the electorate. The second set is the log number of low- and high-education registrants, which shows how the count of each education group changes around adoption. The third set is the share of the total registered electorate in each education group. These shares are defined over the full electorate, including registrants with missing or unknown schooling, so the low- and high-education shares need not sum mechanically to one.

4.2. Effect on Electorate Size

Figure 3 shows the Callaway–Sant’Anna dynamic effects of strict biometric registration on the log number of registered voters. The pattern is sharp. At event time zero, the registry falls by about 0.117 log points, or roughly 11 percent, relative to two years before adoption. The decline then narrows steadily: it is still substantial after one election cycle, but much smaller by six years after adoption, and close to zero by the longest horizon. This sequence is consistent with an immediate wave of removals at the time of recadastramento, followed by gradual re-entry into the registry over subsequent elections. Pre-treatment coefficients are generally small over the immediate pre-period, which is reassuring, although the longest pre- and post-treatment horizons should be

interpreted more cautiously because they are identified from the earliest cohorts only.

Figure 3: Biometric Registration and Electorate Size



Note: The figure plots Callaway-Sant’Anna dynamic event-study estimates for the log number of registered voters, using strict BVR municipalities as the treated sample and never-treated municipalities as the control group. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality. Appendix Figure 26 compares the same outcome across alternative staggered-adoption estimators.

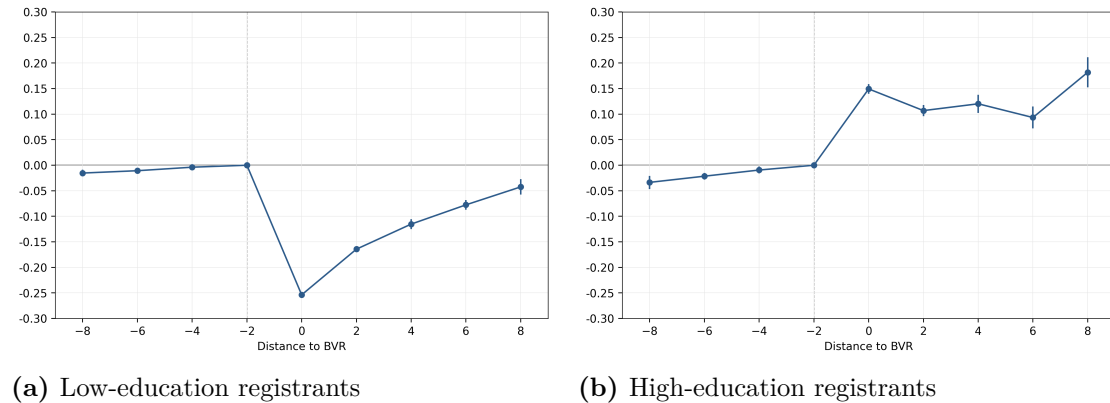
4.3. Effects by Education Level

Biometric registration did not simply change the number of voters on the rolls; it also changed who remained on them. That distinction matters for political economy because a reform that improves registry integrity but imposes unequal compliance costs can re-allocate political voice even if aggregate registration later recovers. The TSE electorate files make it possible to track registered voters by schooling at the municipality-election level. Following the definitions in Section 2.3, I classify low education as all categories below completed secondary schooling and high education as incomplete secondary schooling or more. The Brazilian electorate was already becoming more educated over time: the electorate-weighted share of low-education registrants fell from 72.8% in 2000 to 63.9% in 2008 and 46.2% in 2018, while the corresponding high-education share rose from 27.0% to 36.0% and 53.8%. Those aggregate trends make simple cross-sectional comparisons uninformative. The relevant question is whether municipalities change discontinuously around biometric adoption relative to comparable municipalities that have

not yet adopted the reform. That compositional margin is also what drives Proposition 2: if the surviving electorate becomes relatively more educated, the downstream equilibrium policy should tilt away from redistribution even when the aggregate registry later recovers.

Figure 4 reports the education-specific log-count event studies. The low-education registry contracts sharply at adoption, falling by about 0.254 log points at event time zero. By contrast, the high-education count rises by about 0.149 log points at the same horizon. The two series therefore show that the aggregate registry contraction is concentrated among low-education records and is accompanied by an increase in high-education records. That pattern is consistent with differential compliance costs, but it also points to the possibility of education-record updating during recadastramento, a mechanism that the decomposition section investigates more directly.

Figure 4: Biometric Registration and Education-Specific Registry Counts

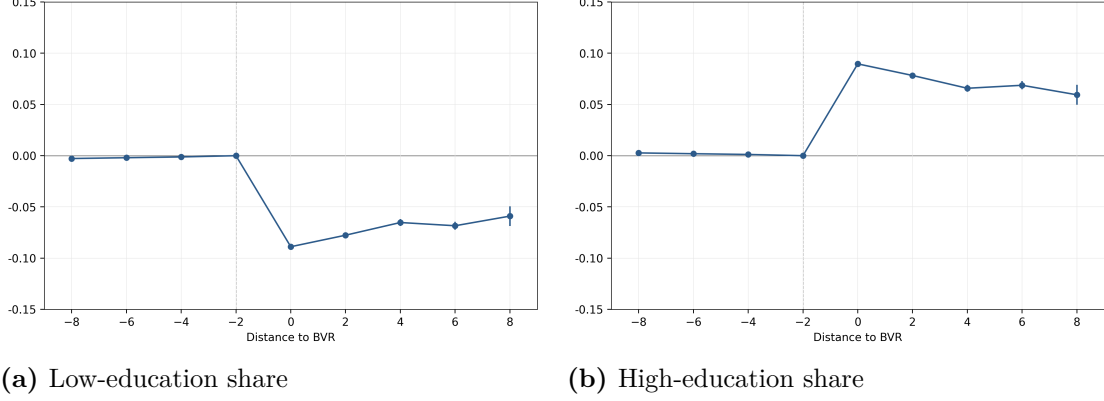


Note: Each panel plots Callaway–Sant’Anna dynamic event-study estimates for log education-specific registered-voter counts, using strict BVR municipalities as the treated sample and never-treated municipalities as the control group. Municipalities ever coded as hybrid are excluded. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality. Both panels use the same y-axis range and tick marks.

Figure 5 reports the corresponding Callaway–Sant’Anna event-study estimates for the low- and high-education shares. The effects are large on impact. At event time zero, biometric registration lowers the low-education share by 8.9 percentage points and raises the high-education share by 9.0 percentage points. The estimates then partially unwind as the registry recovers, but they do not disappear. Eight years after adoption, the low-education share remains 5.9 percentage points below its pre-treatment level, while the high-education share remains 5.9 points above it. Because these education shares are

defined over the total electorate, including voters with missing schooling, they need not mechanically sum to one. The near mirror-image responses therefore point to a genuine reweighting of the registered electorate toward more educated voters rather than a purely mechanical denominator effect.

Figure 5: Biometric Registration and Education Composition



Note: Each panel plots Callaway–Sant’Anna dynamic event-study estimates, using strict BVR municipalities as the treated sample and never-treated municipalities as the control group. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality. Panel (a) uses the share of registered voters with schooling below completed secondary education as the outcome. Panel (b) uses the share with incomplete secondary schooling or more. Appendix Figure 27 reports the corresponding comparison figures for alternative staggered-adoption estimators.

One plausible channel is differential compliance: the in-person recadastramento requirement may have imposed larger administrative burdens on lower-education voters, leading to more cancellations or slower re-entry in those groups. A second channel is information updating. Re-registration may have improved the quality of education records, so some of the persistent shift could reflect cleaner measurement rather than exclusion alone. Taken together with the earlier evidence on registry size, the most credible interpretation is that biometric registration both cleaned the rolls and changed their composition.

4.4. Effects by Connectivity

The main registry result raises a natural mechanism question: did the same biometric requirement impose different compliance costs across municipalities with different information infrastructure? Biometric recadastramento required citizens to learn about

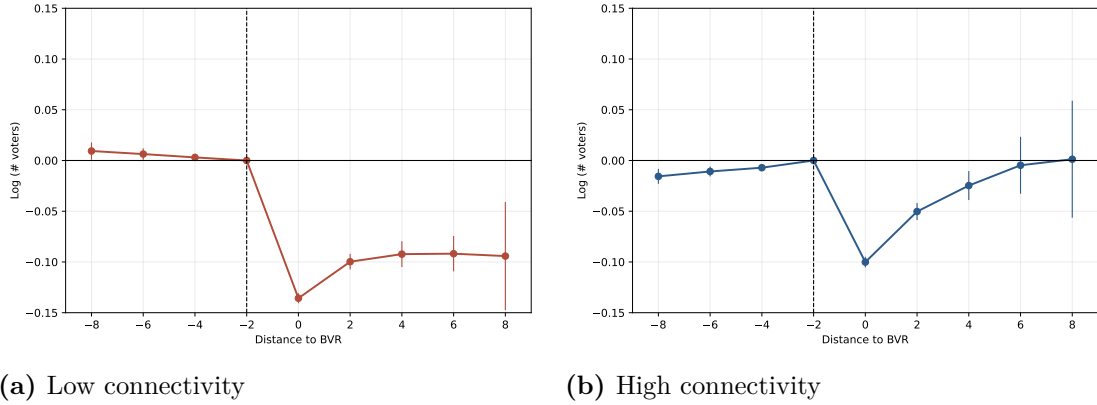
the revision process, appear before the electoral bureaucracy, and update administrative records before the next election. Those requirements are likely to be less costly in places where citizens and local organizations can more easily receive information, coordinate appointments, or monitor administrative deadlines. This is the same logic that motivates the literature on registration costs and administrative burdens: procedural rules need not be facially discriminatory to have unequal participation effects when the cost of compliance is unevenly distributed (Rosenstone and Wolfinger, 1978; Ansolabehere and Konisky, 2005; Herd and Moynihan, 2018).

To examine this possibility, I split municipalities by pre-treatment connectivity. I use fixed broadband density in 2008, before the main expansion of biometric registration, and compare municipalities in the bottom third of the distribution to municipalities in the top third. I leave the middle third out of the figure so that the comparison is between municipalities with clearly different pre-existing communication infrastructure. This split should be read as heterogeneity in exposure to an administrative environment, not as a causal estimate of the effect of broadband access itself. It is useful because the theory predicts that BVR should contract the registry most where compliance costs are high, but the exercise is descriptive because connectivity is bundled with many other municipal characteristics.

Figure 6 reports the dynamic registry effects separately for low- and high-connectivity municipalities. The specification mirrors the main Callaway–Sant’Anna event study for electorate size: strict BVR is the treatment, municipalities ever coded as hybrid are excluded, never-treated municipalities form the control group, event time -2 is the reference period, and standard errors are clustered by municipality. The axes are held fixed across the two panels so that the magnitudes can be compared visually.

The pattern is consistent with connectivity moderating the administrative burden of BVR. At adoption, the estimated registry contraction is about 0.136 log points in low-connectivity municipalities and about 0.100 log points in high-connectivity municipalities. The difference persists after adoption. Two election cycles after treatment, the low-connectivity estimate remains close to -0.092 , while the high-connectivity estimate has narrowed to about -0.025 . By the longest horizon, the high-connectivity series is essentially back to zero, whereas the low-connectivity series remains around -0.094 . This contrast suggests that the initial registry cleaning and subsequent re-entry process were more costly, slower, or less complete in places with weaker pre-existing connectivity.

Figure 6: Biometric Registration and Electorate Size by Pre-BVR Connectivity



Note: Each panel plots dynamic event-study estimates for the log number of registered voters, splitting the strict-BVR sample by 2008 fixed broadband density. Panel (a) keeps municipalities in the bottom third of the 2008 connectivity distribution, and panel (b) keeps municipalities in the top third. The middle third is omitted from the figure. Never-treated municipalities within the corresponding connectivity group are the control group. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality. Both panels use the same y-axis range and tick marks.

This heterogeneity matters for the paper’s interpretation because it links the aggregate registry contraction to a concrete mechanism. The main estimates show that BVR reduced the number of registered voters; the connectivity split suggests that the effect was not only a uniform administrative purge of invalid records. If the reform were only removing obsolete or duplicate registrations with no meaningful compliance burden for valid voters, there would be less reason to expect the contraction to be systematically larger in low-connectivity municipalities. The stronger effect in less connected places is instead consistent with a compliance-cost channel: information frictions, travel burdens, bureaucratic uncertainty, and weaker local administrative support may have made it harder for eligible citizens to complete recadastramento on time.

The result also clarifies how biometric registration differs from earlier electoral technologies in Brazil. Electronic voting reduced ballot error and increased the political weight of less-educated voters by making vote casting easier at the polling place (Fujiwara, 2015). Biometric registration is a different kind of modernization. It can improve the credibility and quality of the registry, but it operates before election day and therefore requires citizens to pass through an administrative process in order to remain eligible. In that sense, the relevant state-capacity question is not only whether the technology works, but whether the infrastructure around the technology allows citizens to comply with it.

This connects the result to broader evidence that biometric and digital state-capacity investments can improve service delivery while raising distributional concerns about who can access the new administrative interface (Muralidharan et al., 2016; Bossuroy et al., 2024).

At the same time, the connectivity split has important shortcomings. Low- and high-connectivity municipalities are not otherwise comparable places. Appendix Table 10 shows that high-connectivity municipalities are larger, richer, much more likely to be in the South and Southeast, and have substantially lower baseline low-education shares. Low-connectivity municipalities are concentrated in the North and Northeast and have lower 2008 GDP per capita. These differences matter because fixed broadband density is a proxy for a broader bundle of municipal development and state capacity, not an isolated source of quasi-random variation. The same factors that predict connectivity may also predict the quality of electoral administration, the availability of registry offices, the intensity of local outreach, and citizens' ability to absorb bureaucratic shocks (Besley and Persson, 2011; Acemoglu and Robinson, 2019).

For this reason, I interpret Figure 6 as mechanism evidence rather than as an independent causal design. The exercise shows that BVR's registry effects are larger where pre-existing communication infrastructure is weaker, which is exactly the direction predicted by a compliance-cost interpretation. It does not prove that broadband access itself protected voters from exclusion. A stronger version of the analysis would residualize connectivity with respect to baseline municipal characteristics, compare municipalities within narrower regional support, or exploit supply-side changes in telecommunications infrastructure. The present result is therefore best viewed as a disciplined descriptive check: it supports the claim that administrative modernization can have unequal consequences across local state-capacity environments, while also making clear that those environments differ along many dimensions.

Finally, the result helps discipline the welfare interpretation developed below. If BVR's costs were borne evenly across space, the main normative question would be the aggregate tradeoff between registry integrity and total registry contraction. The connectivity gradient points to a more distributional problem. Modernization may improve the average quality of electoral administration while imposing its largest participation costs on places least able to navigate the transition. That is why the welfare analysis treats compliance costs as heterogeneous and why the empirical decomposition focuses not only

on how many registrants disappear, but also on which citizens and which municipalities are most exposed to the administrative burden.

4.5. Robustness

The appendix reports several complementary checks. First, Appendix Section D repeats the Callaway–Sant’Anna event studies after including hybrid municipalities in the treatment definition. The resulting estimates have the same signs and dynamic shape but are smaller at impact, consistent with hybrid BVR being a lower-intensity version of the reform. Second, Appendix Figure 26 reports the estimator-comparison exercise for registry size, and Appendix Figure 27 does the same for the education shares. Across Callaway–Sant’Anna, dynamic TWFE, Borusyak–Jaravel–Spiess, and de Chaisemartin–D’Haultfoeuille estimators, the main qualitative conclusions are stable: the registry contracts at adoption and then recovers, while the electorate shifts away from lower-education registrants and toward higher-education registrants. Third, Appendix Section F applies Rambachan–Roth sensitivity analysis to the event-time 0 coefficients; under the relative-magnitudes restriction, the breakdown value exceeds $\bar{M} = 2$ for all three outcomes, so the impact estimates remain statistically separated from zero even when post-treatment confounding is allowed to be more than twice as large as the worst observed pre-trend deviation. Fourth, Appendix Section G reports state $\times$ election-year fixed effects, within-state-support restrictions, permutation tests, negative-weight diagnostics, and cohort-balance checks, which together strengthen the case for relying on within-municipality event-time comparisons rather than simple cross-sectional contrasts.

5. Decomposing Registry Change: Exit versus Re-labeling

Section 4 shows that BVR contracts the registered electorate at adoption and shifts its measured composition away from lower-education voters. That empirical pattern admits two conceptually different channels. The first is *exit*: voters fail to complete biometric recadastramento and disappear from the registry. The second is *re-labeling*: voters do re-register, but the recadastramento process updates an education record that had been stale since the voter’s original registration. The distinction matters because exit changes

electoral participation, while re-labeling changes only administrative measurement.

This section proceeds in the order implied by the strength of the evidence. I first use the count event studies to establish what is identified with minimal additional structure. Those aggregate count results already imply that re-labeling is strictly positive. I then introduce a cohort framework whose purpose is narrower: it allocates the aggregate change across municipalities, regimes, and age groups using the joint age-by-education panel available from 2008 onward. Because that framework cannot distinguish BVR-induced cancellation from source-municipality registry loss due to transfers and migration, I treat its exit measure as an *exit-channel loss* and as an upper-bound proxy for causal exclusion. Appendix C reports implementation details, diagnostics, and additional descriptive figures.

5.1. Aggregate count accounting and identified bounds

The cleanest decomposition evidence comes from the count event studies. Let $y_{j,mt} = N_{j,mt}/N_{m,2006}$ denote the count of registered voters in municipality m and year t , normalized by the municipality's 2006 baseline electorate, for $j \in \{N, L, H\}$ corresponding to total, low-education, and high-education registrants. Let γ_j^0 be the event-time-zero coefficient from the staggered-adoption specification in Section 4. Write the adoption-time accounting identities as

$$\gamma_L^0 = -E_L^0 - R^0, \quad \gamma_H^0 = -E_H^0 + R^0, \quad (2)$$

where $E_L^0, E_H^0 \geq 0$ are the adoption-time exit shares from each education group and R^0 is the share of the baseline electorate that is re-labeled from low to high education. Under non-negativity of exits, (2) yields sharp bounds:

$$R_{\min}^0 = \max\{0, \gamma_H^0\}, \quad R_{\max}^0 = -\gamma_L^0.$$

The lower bound is the key identified object. If the high-education count rises at adoption, pure exit is impossible.

Table 1 is the aggregate anchor for the rest of the section. It establishes three facts that do not depend on the cohort framework. First, BVR contracts the registry on impact.

Table 1: Count-based decomposition at event time zero

Object	Value	Interpretation
Total count effect, γ_N^0	-12.68%	Registry contraction at adoption
Low-education count effect, γ_L^0	-17.82%	Low-education registrants fall sharply
High-education count effect, γ_H^0	+5.22%	Pure exit is infeasible
Re-labeling lower bound, $R_{\min}^0$	5.22%	Identified by the rise in high-education counts
Re-labeling upper bound, $R_{\max}^0$	17.82%	Entire low-education decline attributed to re-labeling
Scenario B benchmark	$R^0 = 5.22\%$	All exit allocated to low education
Scenario C benchmark	$R^0 = 10.40\%$	Exit allocated proportionally by education

Note: Coefficients are the adoption-time estimates from the count event studies in Section 4. The slight difference between γ_N^0 and $\gamma_L^0 + \gamma_H^0$ reflects the small unknown-education residual, about 0.08 percentage points at adoption. Scenario B is the conservative benchmark used below; Scenario C is an alternative allocation reported for comparison.

Second, low-education counts fall sharply. Third, re-labeling must be positive because the high-education count rises. I therefore treat positive re-labeling as identified in the aggregate and use the cohort framework only to allocate that change across municipalities and age groups.

5.2. Cohort framework for municipality-level allocation

The cohort framework is an allocation device, not the source of the paper’s cleanest identification. Its value is that it uses the joint age-by-education panel from 2008 onward to say where the aggregate effects are concentrated. Let $N_{m,t,c,e}$ denote the number of registrants in municipality m , election year t , age cohort c , and education group $e \in \{L, H\}$. Let $a_{cc'}$ be the mechanical-aging weight that maps source cohort c' at time t into target cohort c at time $t+2$. By construction, $a_{cc'} \in [0, 1]$ and $\sum_c a_{cc'} = 1$ for each source cohort c' . For the five-year bins, the maintained approximation is $a_{cc} = 3/5$ and $a_{c^+,c} = 2/5$ over one electoral cycle; Appendix C.1 records the full transition matrix.

Let $\hat{\sigma}_{m,c'}$ denote the pre-treatment survival parameter for source cohort c' in municipality m , and let $\hat{I}_{m,t+2,c}$ denote estimated inflows into the youngest cohorts. The no-BVR

counterfactual count for target cohort c is

$$\hat{N}_{m,t+2,c}^0 = \sum_{c'} a_{cc'} \hat{\sigma}_{m,c'} N_{m,t,c'} + \hat{I}_{m,t+2,c}. \quad (3)$$

For cohorts above age 24, the maintained approximation is $\hat{I}_{m,t+2,c} = 0$: delayed first-time registration outside the youngest bins is assumed to be small relative to aging flows over one election cycle.

I define cohort-level exit-channel loss as the positive part of the gap between the no-BVR prediction and the observed post-treatment count,

$$\hat{E}_{m,t+2,c} = \max\{0, \hat{N}_{m,t+2,c}^0 - N_{m,t+2,c}\}, \quad \hat{E}_{m,t+2} = \sum_c \hat{E}_{m,t+2,c}. \quad (4)$$

The non-negativity floor rules out negative exit. Because source-municipality registry loss can reflect either BVR-induced cancellation or post-treatment transfer and migration, $\hat{E}_{m,t+2}$ should be interpreted as an *exit-channel loss* and as an upper-bound proxy for causal exclusion.

To recover re-labeling, I carry forward low-education mass within cohort and allow for pre-treatment schooling progression. Let $\hat{\delta}_{c'}$ denote the pre-treatment probability that a low-education registrant in source cohort c' naturally progresses to high education over one electoral cycle. The no-BVR counterfactual for low-education mass is

$$\hat{N}_{m,t+2,c,L}^0 = \sum_{c'} a_{cc'} \hat{\sigma}_{m,c'} (1 - \hat{\delta}_{c'}) N_{m,t,c',L} + \hat{I}_{m,t+2,c}^L. \quad (5)$$

My preferred conservative allocation is Scenario B, which assigns all within-cohort exit to low education. Re-labeling in cohort c is then

$$\hat{R}_{m,t+2,c}^B = \max\{0, \hat{N}_{m,t+2,c,L}^0 - N_{m,t+2,c,L} - \hat{E}_{m,t+2,c}\}, \quad \hat{R}_{m,t+2}^B = \sum_c \hat{R}_{m,t+2,c}^B. \quad (6)$$

Scenario C instead allocates exit proportionally to the cohort's counterfactual education shares. I report it as an alternative allocation, but treat Scenario B as the preferred benchmark because it is mechanically conservative on re-labeling. Appendix C.2 describes the survival-rate hierarchy, biometric-uptake cap, and implementation details.

5.3. Cohort-based results by regime and age

Table 2 reports the decomposition at the first treated election. To keep the main text focused on magnitudes that are comparable across regimes, the table reports rates as percentages of the relevant 2008 baseline electorate. Raw counts are moved to Appendix Table 6.

Table 2: Decomposition at the first treated election

	Strict-first	Hybrid-first	All treated
Exit-channel loss (upper-bound interpretation)	13.64%	3.28%	8.35%
Re-labeling, Scenario B (preferred)	7.93%	1.16%	4.47%
Re-labeling, Scenario C (alternative)	10.89%	1.51%	6.10%

Note: Entries are percentages of the 2008 baseline electorate for municipalities in the relevant first-treatment regime. Scenario B allocates all within-cohort exit to low education and is the preferred conservative benchmark. Scenario C allocates exit proportionally across education groups and is reported as an alternative allocation. Because source-municipality transfers and BVR-induced cancellation are observationally similar in the cohort framework, the exit-channel measure should be read as an upper-bound proxy for causal exclusion.

Three patterns matter for the paper. First, strict enforcement drives most of the administrative change. Strict-first municipalities account for roughly 49% of the treated baseline electorate but for about 80% of estimated exit-channel loss and 87% of conservative re-labeling. Summing exit-channel loss and re-labeling gives a useful summary measure of the share of the baseline electorate *administratively affected* by BVR through either removal or record updating: 21.57% in strict municipalities versus 4.44% in hybrid municipalities under Scenario B. I use that stacked total only as a summary of administrative change, not as a decomposition of net registry contraction.

Second, re-labeling is concentrated where the identifying assumptions are most credible. The age-profile in Appendix Figure 20 shows that the 25–44 range accounts for 67.71% of total Scenario B re-labeling, with the 35–44 band alone accounting for 40.39%. By contrast, the 65+ group accounts for only 1.90% of total re-labeling. This is the pattern one would hope to see if the framework is mostly picking up record updating in cohorts that are old enough for natural schooling progression to be limited but young enough that survival-rate noise is manageable.

Third, exit-channel loss is more diffuse across ages and therefore harder to interpret

cohort by cohort. Working-age cohorts 25–54 account for 47% of total exit-channel loss, while the 65+ group accounts for 25.29%. Because the oldest bins are also the noisiest part of the cohort framework, I treat the age pattern in exit as descriptive rather than definitive. Appendix Table 7 is designed to report working-age restrictions precisely so the aggregate decomposition is not read as hinging on the youngest and oldest cohorts.

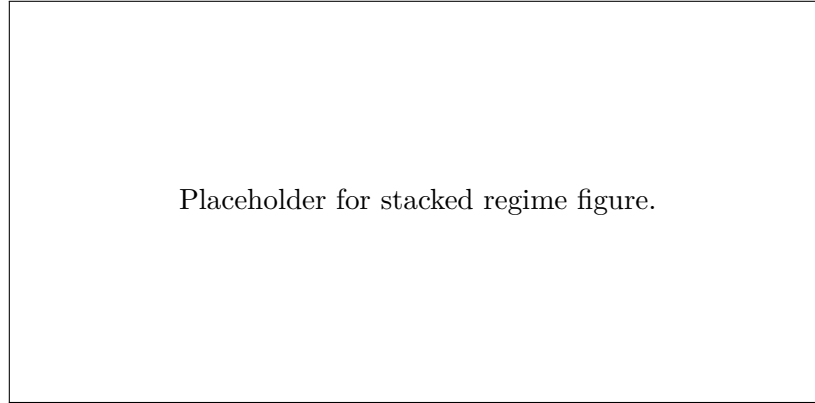


Figure 7: Exit-channel loss and conservative re-labeling by first-treatment regime

Note: The lower segment is exit-channel loss and the upper segment is Scenario B re-labeling, both expressed as shares of the relevant 2008 baseline electorate. The stacked total should be read as the share of the baseline electorate administratively affected through either removal or record updating. It is not a decomposition of net registry contraction.

Appendix Figures 17–19 report additional heterogeneity by baseline education share and treatment regime. I keep those figures out of the main text because they are supportive rather than central: the count bounds above remain the cleanest decomposition evidence.

5.4. Diagnostics, scope, and interpretation

The most important diagnostic is the comparison between the aggregate count bounds and the cohort allocation. The count accounting uses a 2006 denominator, whereas the cohort framework starts in 2008 because the joint age-by-education panel begins then. After rescaling the cohort totals to the 2006 treated baseline, the framework implies exit-channel loss of 8.65%, Scenario B re-labeling of 4.63%, and Scenario C re-labeling of 6.32%. The rescaled Scenario B estimate sits slightly below the identified lower bound of 5.22% from Table 1. I view that near miss as evidence that the cohort procedure is conservative on re-labeling, not as a substitute for the cleaner aggregate bound. For that

reason, the paper should anchor the aggregate decomposition on the count accounting and use the cohort framework for allocation and heterogeneity.

Two additional diagnostics belong in the appendix and should be treated as part of the section’s objective evidence. The first is a pre-treatment placebo backtest that runs the full cohort algorithm on pre-BVR transitions. If the framework is well calibrated, placebo re-labeling in older working-age cohorts should be near zero and placebo exit should be modest relative to post-treatment estimates. The second is a working-age restriction that re-computes the aggregate decomposition on age windows such as 25–64 and 35–64. Those restrictions matter because they directly test whether the headline decomposition is being driven by the mechanically weakest bins. Appendix Tables 8 and 7 provide the slots for those diagnostics.

The remaining limitations are interpretation rather than identification. Small municipalities require survival-rate fallback rules; Appendix Table 9 should therefore report the electorate-weighted share of cells using municipality-, state-, and national-level survival estimates, as well as how often the biometric-uptake cap and non-negativity floor bind. And because the within-municipality cross-cohort interaction coefficient is much larger than the corresponding cross-municipality coefficients, I treat that coefficient plot as descriptive appendix evidence rather than a calibration object.

Taken together, the section supports a disciplined reading of the reform. Positive re-labeling is identified in the aggregate from the count event studies. The cohort framework then shows that strict enforcement drives most of the administratively affected share and that re-labeling is concentrated in working-age cohorts where the identifying assumptions are strongest. The part of the section that deserves the most weight is therefore the aggregate count accounting and the conservative allocation it motivates, not the fine-grained municipality-by-cohort channel split.

6. Downstream Outcomes

The earlier administrative sections show that biometric voter registration changed the electorate on two margins that matter politically: it shrank the registry and shifted the surviving electorate toward more educated voters. A natural next question is whether

those compositional changes propagated into downstream political or policy outcomes. If BVR altered the local electorate in a substantively meaningful way, one might expect changes in turnout, in partisan vote shares, in the composition of local public spending, or even in downstream health outcomes. But none of those channels is automatic. Electoral composition can change without changing the identity of the median voter; fiscal mandates can mute policy responses even when representation shifts; and national political cycles can overwhelm local electoral-composition effects.

To study those downstream margins, I build a municipality-election panel keyed on (year_election, municipality_id) using the existing treatment-timing file in `data/clean/tse_bvr/municipa`. The panel combines three families of outcomes. First, from the TSE I construct turnout, blank-and-null vote rates, and presidential party vote shares. Second, from historical FINBRA files I construct log per-capita health, education, social-assistance, discretionary-spending, IPTU, and FPM outcomes, aligning odd fiscal years to the subsequent election year so that fiscal year 2001 maps to the 2002 election observation. Third, from DATA-SUS I construct infant mortality and the share of births with seven or more prenatal visits. A neonatal series was attempted but is not reported because the current public extractor returned the same series as total under-one mortality and therefore needs a cleaner age-band pull before it can be interpreted. For these more exploratory downstream panels, I continue to report the dynamic TWFE event-study summaries alongside Callaway–Sant’Anna and Borusyak–Jaravel–Spiess specifications. Event time runs from -8 to $+8$, with event time -2 omitted as the reference period. Standard errors are clustered by municipality and election year for TWFE and BJS; the current Callaway–Sant’Anna implementation supports municipality clustering only, so those estimates should be read as a robustness check rather than a perfectly matched standard-error benchmark.

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6.1. Political Outcomes

The cleanest downstream political result is mechanical rather than ideological. Turnout, defined here as valid votes divided by registered voters, rises sharply when municipalities adopt BVR. The unweighted TWFE estimate at event time 0 is about 0.079, and it remains positive and statistically precise at later horizons. The weighted version is slightly smaller but still large: about 0.072 on impact. This is exactly the kind of downstream quantity that should be interpreted carefully. Because BVR removes voters from the registry, it changes the denominator of the turnout ratio even if voting behavior conditional on registration does not move much. I therefore read the turnout result as evidence that the registry became tighter, not as evidence that political participation conditional on being eligible suddenly increased.

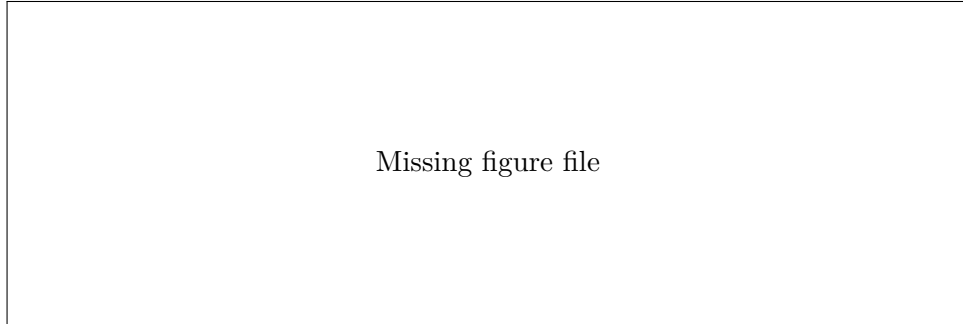
Blank and null voting does not show a comparable post-treatment break. The unweighted event study is close to zero throughout, with only a modest negative coefficient at event time 4. The weighted event study is similarly small on impact, and in fact shows somewhat noisy positive pre-trends. That pattern does not suggest a clean behavioral change in protest or confusion voting after adoption.

The more substantive political test is whether BVR shifted presidential party support in a way consistent with the education-tilted electorate result. The answer is, at best, mixed. For PT vote share, the unweighted TWFE event study shows negative coefficients at event time 0 and especially at event time 4, which is the sign one would expect if the post-BVR electorate became less favorable to the PT. But that pattern is not robust. The Callaway–Sant’Anna and BJS estimates are not centered on the same negative effect, the population-weighted TWFE specification is close to zero at the main post-treatment horizons, and a direct state-by-year fixed-effects check removes the negative interpretation entirely: once I absorb state-specific national-cycle shocks, the event-time-0 PT coefficient becomes positive rather than negative. PSDB vote share is similarly unstable across estimators and horizons. Taken together, the presidential-party results do not support a strong claim that BVR generated a durable partisan reallocation consistent with a simple “education shift $\rightarrow$ anti-PT shift” mechanism.

This is an important null. The electorate-composition effect is real, but it does not automatically translate into a clear municipality-level partisan downstream effect once

national election dynamics are allowed to matter. That weakens the strongest version of the median-voter story on the electoral side.

Figure 8: Estimator Comparison for Key Downstream Outcomes



Note: The figure overlays dynamic TWFE, Callaway–Sant’Anna, and Borusyak–Jaravel–Spiess estimates for three focal outcomes: PT presidential vote share, log health spending per capita, and infant mortality. Event time is measured relative to first biometric adoption and event time -2 is omitted in the TWFE baseline. The Callaway–Sant’Anna estimates are clustered at the municipality level because the current implementation does not support two-way clustering.

6.2. Fiscal Outcomes

If BVR changed who remained in the electorate, one possible downstream channel is fiscal policy. The strongest version of that argument would predict that a more educated and administratively attached electorate might tilt away from redistributive spending, consistent with a reverse Meltzer-Richard mechanism. The fiscal results do not support that stronger claim.

Across the logged per-capita spending outcomes, the dominant pattern is stability. Health spending is essentially flat in the unweighted event study: the point estimate at event time 0 is about -0.024 log points, and later coefficients remain small and imprecise. Education spending is similarly close to zero. Social-assistance spending and total discretionary spending also show no sharp post-treatment decline. The weighted event studies actually tilt positive for most spending categories, but those estimates are imprecise and do not isolate a clear break at adoption.

The revenue-side outcomes reinforce the caution. IPTU collection and FPM transfers both display notable pre-treatment movement, especially in the weighted specifications. FPM is especially useful as a placebo because the federal transfer formula should not

mechanically respond to BVR rollout. The absence of a clean post-treatment FPM response is reassuring, but the pre-trends mean these fiscal event studies should not be overinterpreted as precise causal estimates of post-adoption municipal budgeting. More modestly, the fiscal evidence shows that there is no strong, robust, and immediate shift in local policy spending after BVR adoption. That null is substantively meaningful given the size of the registry-composition result in Section 4.

One reason may be institutional. Brazilian municipalities face strong constitutional and statutory floors for education and health spending. Those constraints likely compress the amount of policy movement that a registration-induced electorate shift can generate, especially in the short run. In that environment, even a genuine compositional change in the electorate may have limited scope to produce large observed budget reallocations.

6.3. Real Outcomes

The most demanding downstream test is whether BVR ultimately changes what the local state delivers, rather than simply how elections are administered or how budgets are recorded. I use infant mortality and prenatal care as reduced-form measures of municipal service delivery.

Here too the main result is null. The unweighted infant-mortality event study is centered very near zero at impact and at the early post-treatment horizons, and the BJS and Callaway–Sant’Anna estimates tell the same basic story. The weighted event study becomes somewhat more negative at long horizons, but it also shows non-trivial pre-treatment movement, which weakens any causal reading of the late negative coefficients. Neonatal mortality behaves similarly, and the share of births with seven or more prenatal visits is essentially flat throughout.

This is the most policy-relevant null in the downstream module. Even if BVR changed who stayed on the rolls, there is no clean evidence in these data that it changed a high-stakes real outcome like infant mortality. That suggests the reform’s consequences are concentrated in electoral administration and representation rather than in an immediate remapping of municipal service delivery.

6.4. Interpretation

Taken together, the downstream module points to a more restrained conclusion than a simple median-voter story would imply. BVR clearly changed the registry and the composition of registered voters. It also mechanically raises turnout measured as valid votes divided by registered voters. But the more substantively consequential downstream outcomes do not move in a clean, robust way. Presidential party support is not stable across estimators or once state-specific national cycles are absorbed. Local spending does not show a systematic post-adoption shift once the outcomes are logged and the pre-trends are taken seriously. Real health outcomes are largely null.

The most plausible interpretation is therefore that BVR mattered first at the electoral-administration margin. It changed who remained in the formal electorate, but those shifts were either too modest, too offsetting, or too institutionally constrained to generate a strong downstream policy response at the municipality level. That does not make the compositional result unimportant. On the contrary, it clarifies what kind of reform BVR was. Its most durable consequence appears to be on electoral access and registry composition, not on a rapid transformation of local partisan competition or public-service delivery.

7. Policy Responses and Mechanisms

The registry results raise a policy question that is distinct from the administrative question studied above. Sections 4 and 4.3 show that strict biometric registration suppressed the registered electorate and shifted its recorded composition away from lower-education registrants and toward higher-education registrants. Section 5 further shows that part of the compositional shift may reflect education-record updating at the moment voters pass through recadastramento. The relevant downstream question is therefore not only whether fewer low-education voters remain on the rolls. It is also whether politicians and local administrators face a different, cleaner, and more educated electorate in their administrative records.

To examine this possibility, I estimate an expanded downstream screen using the BJS imputation estimator (Borusyak et al., 2024). The sample excludes municipalities ever

classified as hybrid BVR municipalities, uses strict BVR timing, clusters standard errors by municipality, and normalizes every event-study path to event time zero. I report only figures for outcomes that pass the pretrend screen: at least three pre-period estimates, no pre-period coefficient with $p < 0.05$, and a maximum absolute pre-period t-statistic no larger than 1.96. Outcomes that nearly pass the screen are discussed only when they discipline the interpretation; they are not plotted here.

The expanded screen remains exploratory. It estimates many outcomes, and most do not pass the pretrend screen. The figures in this section therefore do not establish a single definitive mechanism. They instead show which downstream patterns are most consistent with the paper’s main representation channel: BVR changes the active and administratively visible electorate, and local policy may then respond to the preferences, needs, or administrative legibility of that changed electorate.

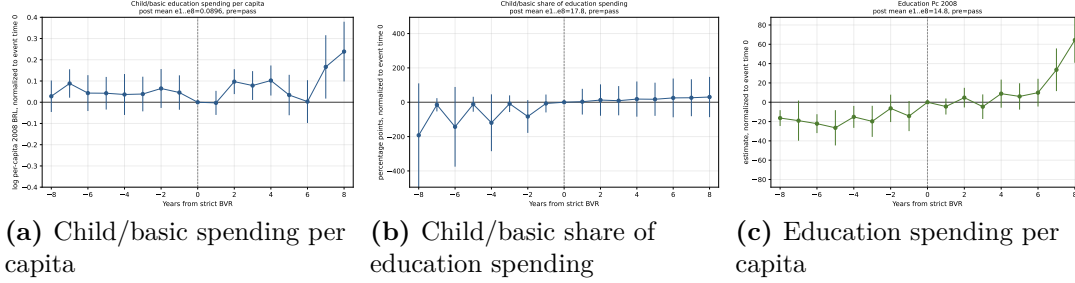
7.1. Education and Child-Oriented Policy

The clearest downstream pattern is in child-oriented education outcomes. Figure 9 reports the pass-pretrend spending outcomes that most directly speak to this mechanism. The child/basic education spending panel rises after BVR in both levels and, more noisily, as a share of education spending. The education-screen SICONFI outcome also rises in per-capita terms. These estimates should be read as screening evidence rather than as final fiscal aggregates, because related education-spending transformations in the dedicated SICONFI screen do not all pass pretrends. Still, the pass-pretrend evidence points toward greater attention to regular child/basic schooling after strict BVR.

Figure 10 shows that the education evidence is not purely an accounting pattern. Several school-flow outcomes move in directions consistent with better progression through regular schooling. Municipal final-years elementary approval rises after BVR, while abandonment and reprobation fall. The secondary-school age-distortion measure also declines. These are exactly the kinds of outcomes one would expect to respond if local governments increased attention to regular schooling, school management, or complementary services that keep students attached to school.

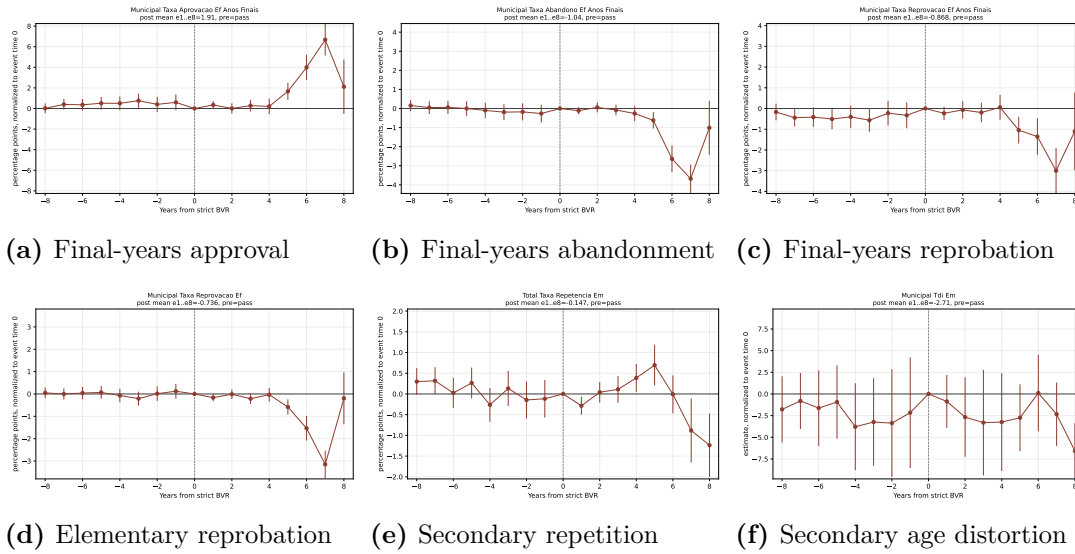
The achievement and teacher-profile outcomes in Figure 11 are also consistent with a schooling response, although they should be interpreted more cautiously. Municipal

Figure 9: BJS Event Studies: Child-Oriented Education Spending



Note: The figure plots BJS/imputation event-study estimates normalized to event time zero. Strict BVR timing is used, ever-hybrid municipalities are excluded, and standard errors are clustered by municipality. Only pass-pretrend outcomes are shown.

Figure 10: BJS Event Studies: School Flow and Progression



Note: Each panel reports a pass-pretrend BJS event study normalized to event time zero. Outcomes are annual municipal or total public-school indicators from the education screen.

and public final-years IDEB approval and flow indicators rise after BVR. Some grade-5 SAEB outcomes also move favorably. Teacher-profile indicators move as well, but their substantive interpretation is less direct because they combine assignment, training, and classification margins. I therefore treat them as supporting evidence that the regular-schooling environment changed, not as a clean stand-alone mechanism.

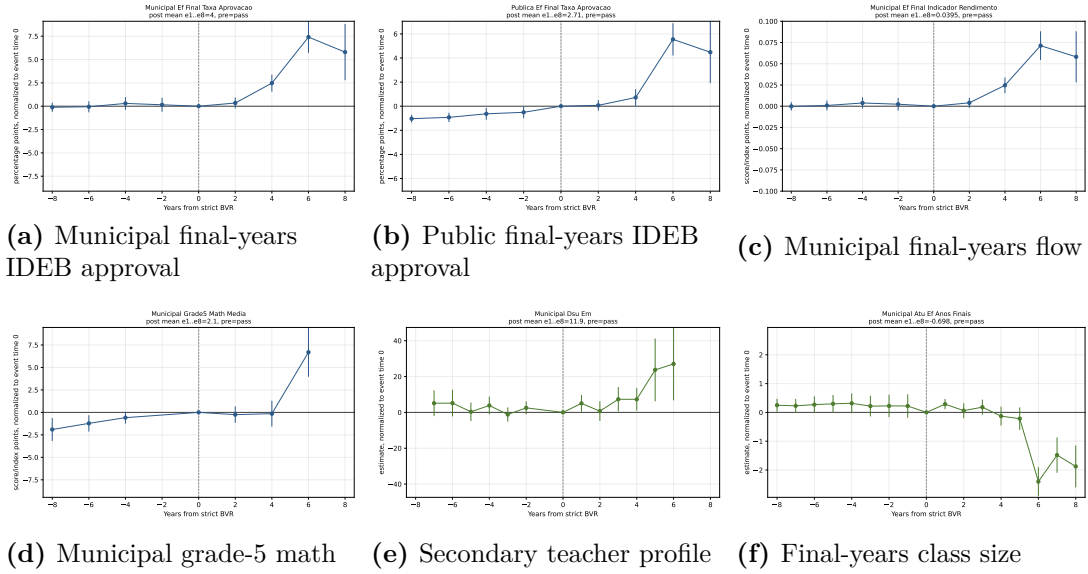
The education pattern fits the paper’s main representation logic. If BVR disproportionately removed lower-education citizens from the rolls while also updating stale schooling records among those who complied, the electorate that politicians observe in official returns becomes smaller and more educated. A standard political-economy view predicts that politicians respond to the preferences and salience of the electorate that can reward or punish them (Meltzer and Richard, 1981; Besley and Coate, 1997). In this setting, the surprising feature is not that policy shifts toward redistribution. Rather, the evidence suggests a tilt toward formal child-oriented services and school progression, which may be especially salient to a more administratively attached electorate.

This interpretation is closely related to, but directionally different from, the Brazilian electronic-voting result in Fujiwara (2015). Electronic voting lowered the cognitive cost of casting a valid ballot and de facto enfranchised less-educated voters, producing downstream health and spending responses favorable to poorer voters. BVR operates earlier in the participation chain. It raises the compliance burden required to remain registered, so the downstream policy response need not mirror the electronic-voting response. If the reform raises the relative weight of higher-education and more administratively attached registrants, the observed child-schooling pattern is consistent with a different kind of political response.

7.2. Municipal Expenditure and Visible Public Goods

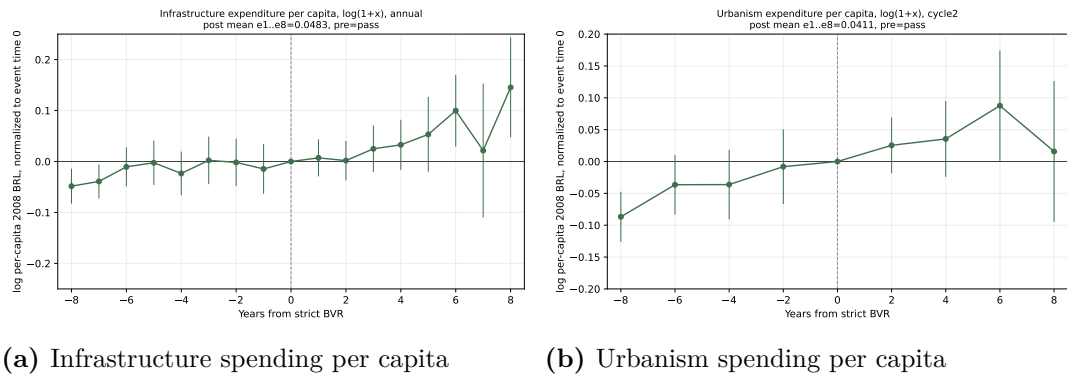
The second pass-pretrend pattern appears in municipal spending on visible local public goods. Figure 12 plots the two SICONFI log per-capita spending outcomes that pass pretrends and fit the mechanism: annual infrastructure spending and two-year-cycle urbanism spending. Infrastructure rises persistently after strict BVR in the annual panel. Urbanism rises in the two-year-cycle panel. A related two-year-cycle infrastructure result is near-pretrend but is not plotted here because the figure rule is restricted to pass-pretrend outcomes.

Figure 11: BJS Event Studies: Achievement, Flow, and Teacher Profile



Note: Biennial IDEB and SAEB outcomes are aligned so that the first post-BVR assessment corresponds to event time zero and adjacent waves are two-year steps. Only pass-pretrend outcomes are plotted.

Figure 12: BJS Event Studies: Infrastructure and Urbanism Spending



Note: The left panel uses the annual SICONFI spending panel. The right panel uses forward two-year cycles. Both panels show pass-pretrend BJS estimates normalized to event time zero.

These expenditure results fit two complementary explanations. The first is political. Infrastructure and urbanism are visible, locally attributable goods. If strict BVR changes which citizens remain electorally salient, local politicians may respond by emphasizing visible public goods that appeal to the active electorate. The second is administrative. BVR required local recadastramento, communication, and coordination with the electoral bureaucracy. That process may have improved the local state’s overview of the population and the places where citizens interact with public offices. Biometric reforms in other settings show that authentication technologies can improve delivery and reduce leakage when they also improve the information available to administrators (Muralidharan et al., 2016; Bossuroy et al., 2024).

The spending evidence also limits the mechanism. The pass-pretrend results are stronger for spending levels than for spending shares. In the broader screen, no SICONFI spending-share outcome passes the pretrend test, and the near-pretrend share estimates are small. This means the policy response should not be described as a clean budget reallocation toward infrastructure or education. It is more consistent with a scale or capacity response: selected municipal functions expand in per-capita terms, but the evidence is weaker that they take a larger share of the municipal budget.

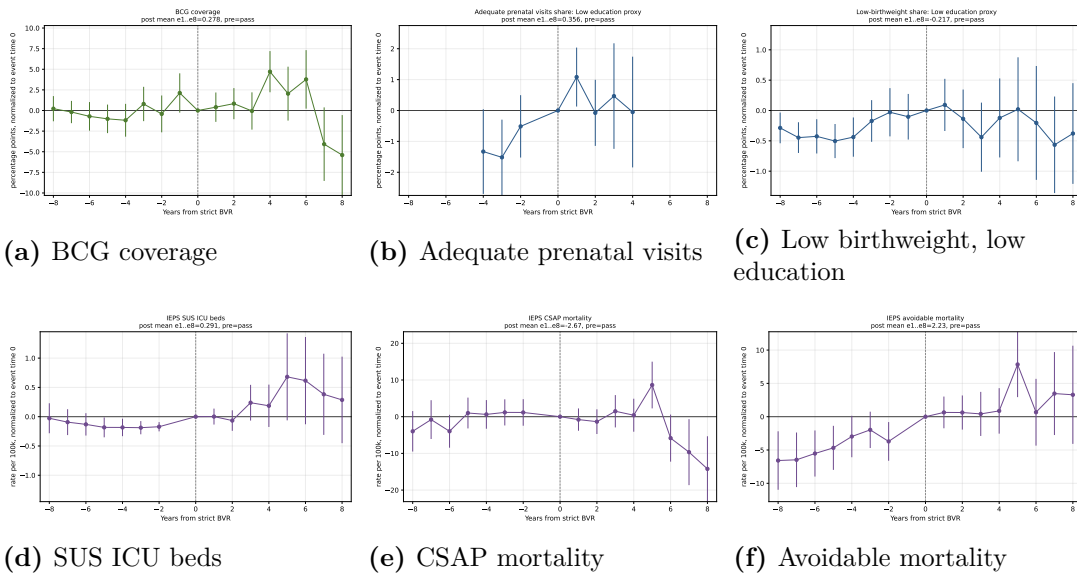
This distinction matters for the link to the voter-registry findings. A change in the administratively visible electorate could affect local policy even without a dramatic reprioritization of budget shares. If recadastramento improves records and changes which citizens are visible to politicians, local governments may expand services that are both visible and administratively easier to justify, while remaining constrained by earmarks, mandates, and existing budget rules. That is why the expenditure evidence is best read together with the education-flow evidence rather than as a separate fiscal story.

7.3. Health and Birth Outcomes

The health and birth outcomes provide a more mixed picture. Figure 13 plots pass-pretrend outcomes in maternal care, immunization, low birthweight, health infrastructure, and mortality. Some panels move in favorable directions, such as BCG coverage, adequate prenatal visits among low-education mothers, low-birthweight outcomes among low-education mothers, and ICU beds. Other pass-pretrend outcomes move in directions that are harder to interpret, especially avoidable mortality and SUS beds. The health

evidence therefore does not support a broad claim that BVR improved health-service delivery.

Figure 13: BJS Event Studies: Selected Birth and Health Outcomes



Note: The figure reports pass-pretrend BJS paths for selected birth and health outcomes. Percentage-point variables are kept in percentage points when that is how the source outcome is defined.

The mixed health pattern is informative because it contrasts with the electronic-voting evidence. Fujiwara (2015) finds that a voting technology that enfranchised less-educated voters increased state health spending and improved prenatal care and low-birthweight outcomes among less-educated mothers. The BVR estimates here do not reproduce a clean version of that chain. BVR is an integrity and registry technology that operates through pre-election compliance. Its health consequences are therefore likely to be weaker and more indirect than a reform that made ballots easier for low-education voters to cast and count.

The pass-pretrend health outcomes are still useful for interpretation. The favorable movements in BCG coverage, adequate prenatal care, and low-birthweight shares among lower-education mothers are consistent with the possibility that some local administrative upgrading spilled into basic health contact. The ICU-bed and mortality estimates, however, do not form a coherent service-delivery mechanism. Some of these outcomes may reflect broader health-system changes, data quality, or compositional changes in who appears in administrative records rather than direct BVR-induced health policy.

For that reason, I treat the health evidence as corroborating at most a weak administrative capacity story. It is not the main mechanism. The strongest downstream evidence remains in education and selected municipal spending. Health outcomes help bound the interpretation: BVR did not generate a general public-service transformation, and any policy response appears to have been narrower than a broad improvement in health administration.

7.4. Mechanisms

The results are most consistent with a representation channel operating through the administratively visible electorate. Strict BVR suppresses the registry, especially among lower-education voters, and the recadastramento process updates records among voters who comply. As a result, the electorate observed by politicians becomes smaller, more selected, and more educated in official data. In the model in Section 9, politicians respond to the registered electorate rather than to the full adult population. The pass-pretrend education and spending results are consistent with this mechanism: after the registry shifts, local policy appears to move toward regular schooling and selected visible public goods.

A second mechanism is administrative legibility. Biometric registration can improve the quality of administrative records by forcing citizens and the state to update information at the point of contact. This logic is consistent with evidence from biometric monitoring and smartcard interventions in India, where biometric systems improved delivery, reduced leakage, and strengthened administrative measurement (Muralidharan et al., 2016; Bossuroy et al., 2024). In the Brazilian BVR setting, the direct administrative output is a cleaner and more current voter registry. The downstream education and infrastructure patterns suggest that this improved overview may also have changed how local governments identify, justify, or target some services.

The third mechanism is political selection through administrative burden. Registration rules and procedural frictions can screen out citizens even when eligibility is formally universal. The disability-program evidence in Deshpande and Li (2019) shows that application costs can reduce take-up and worsen targeting when they screen out high-need applicants. The administrative-burden framework in Herd and Moynihan (2018) makes the same general point: procedural costs are policy instruments with distributional consequences. Applied to BVR, this means that the reform can simultaneously improve registry integrity and shift representation away from citizens with high compliance costs.

The null results sharpen the mechanism. The BJS screen does not support a clean school-transport channel, because the transport and access outcomes mostly fail pre-trends. It does not support a broad health-system channel, because the pass-pretrend health outcomes are mixed and most health measures fail. It also does not support a clean budget-share reallocation story, because no spending-share outcome passes pre-trends. The mechanism should therefore be stated narrowly: BVR may have shifted political attention and administrative capacity toward regular schooling and selected visible municipal goods, not toward all public services.

This interpretation also reconciles the downstream evidence with the paper’s welfare framework. A reform can increase integrity and improve some administrative outputs while still distorting representation if the compliance burden falls unevenly. The policy consequences documented here are therefore not a simple vindication or indictment of BVR. They illustrate the paper’s central tradeoff: biometric registration can clean and update the electorate that politicians see, but when that visible electorate becomes less representative of the underlying population, subsequent policy responses may reflect the new registry rather than the full citizenry.

8. Perceived Integrity and Democratic Legitimacy

The administrative results show that biometric registration changed both the size and the composition of the electorate. A natural next question is whether those changes were accompanied by shifts in how citizens viewed political institutions and democracy itself. Those downstream attitudinal effects are not obvious *ex ante*. A cleaner registry and more credible voter identification could strengthen confidence in elections and in the institutions that run them. But the same reform also imposed an additional compliance step before election day, and Sections 4 and 5 show that it disproportionately reduced the share of lower-education voters on the rolls. For that reason, the trust module is best treated as a secondary, more suggestive part of the paper: it asks whether exposure to electoral modernization coincided with different political attitudes, not whether the survey evidence alone can establish a clean causal effect. The model makes that caution explicit. Proposition 5 shows that higher observed trust among post-reform respondents can arise mechanically from selection into compliance rather than from a within-person attitudinal treatment effect.

8.1. Building Trust and Democracy Indices

Survey setting and outcomes. I use Brazil waves of the LAPOP *AmericasBarometer* linked to municipality-level biometric exposure through the respondent’s reported interview municipality (LAPOP Lab, 2023). The benchmark linked sample used in this section covers seven survey waves, 2006, 2008, 2010, 2012, 2014, 2016, and 2018, for a total of 11,008 respondents in 235 municipalities. The regression sample falls to 10,632 observations once I impose a common set of controls and require non-missing outcomes. I study two standardized outcomes. The first is an index of trust in institutions built from eight questions about support for and confidence in Brazil’s political institutions, municipal government, the president, political parties, and elections. The second is a trust-in-democracy index built from three questions on whether democracy is the best form of government, satisfaction with democracy, and whether those who govern care about what ordinary people think. Throughout this section, “trust” refers to trust in institutions, while trust in democracy is discussed separately.

Index construction. The trust-in-institutions and trust-in-democracy outcomes sum-

marize correlated survey items using principal component analysis after median imputation and standardization of the underlying responses. For respondent i ,

$$z_i^{\text{trust}} = \frac{PC_{1i}^{\text{trust}} - \overline{PC_1^{\text{trust}}}}{\text{sd}(PC_1^{\text{trust}})}, \quad z_i^{\text{dem}} = \frac{PC_{1i}^{\text{dem}} - \overline{PC_1^{\text{dem}}}}{\text{sd}(PC_1^{\text{dem}})}, \quad (7)$$

where PC_{1i}^{trust} is the first principal component of the eight-item trust-in-institutions block and PC_{1i}^{dem} is the first principal component of the three-item democracy block.

Table 3: PCA Decomposition of the Trust and Democracy Indices

Survey item	Trust in institutions		Trust in democracy	
	PC1	PC2	PC1	PC2
Institutions respected / Democracy is best	0.636	0.432	0.670	-0.370
Rights protected / Satisfied with democracy	0.735	0.290	0.673	-0.370
Proud of political system / Government cares about people like me	0.772	0.270	0.574	0.810
Support political system	0.763	0.235		
Trust parties	0.718	-0.169		
Trust municipal government	0.639	-0.382		
Trust president	0.654	-0.372		
Trust elections	0.672	-0.397		
Eigenvalue	3.928	0.869	1.231	0.910
Share explained	0.491	0.109	0.410	0.305

Note: Entries are PCA loadings after median imputation and standardization of the underlying LAPOP items. ‘Satisfied with democracy’ is reverse-coded before PCA so that larger values indicate more democratic satisfaction. The sample uses official public Brazil LAPOP waves with the full trust and democracy blocks: 2008, 2010, 2012, 2014, 2017, and 2019.

Table 3 reports the first two principal components for both survey blocks. Two features are worth emphasizing. First, the institutional-trust block is clearly more one-dimensional than the democracy block: the first trust component explains 49.1% of total variance, while the second explains only 10.9%. The loadings are positive and fairly uniform across items, which suggests that confidence in elections, local government, parties, the president, and the political system move together as part of a common latent dimension. Second, the democracy block is less tightly summarized by a single factor. Its first principal component explains 41.0% of total variance and the second still explains 30.5%, loading especially strongly on the political-efficacy item. That difference matters for interpretation. The trust-in-institutions outcome can be read more naturally as a common institutional-confidence index, whereas the trust-in-democracy outcome is better understood as a broader summary of democratic attachment and perceived political

voice. Appendix Section H plots the full explained-variance shares for both PCA objects.

These two indices therefore capture related but distinct dimensions of political attitudes. The first measures confidence in the machinery of governance and electoral administration. The second captures a broader orientation toward democracy as a regime and as a lived political system. Distinguishing between them is important for the interpretation of the results below. A reform like BVR could plausibly increase trust in elections and the institutions that administer them without generating equally strong changes in broader views about democracy.

8.2. BVR Exposure and Political Attitudes

The administrative results showed that BVR did not simply modernize election administration in the abstract. It changed who remained on the voter rolls, especially along education lines. That makes attitudinal heterogeneity particularly interesting. BVR could raise trust by signaling modernization, cleaner rolls, and more credible identity verification. But it could also dampen confidence if the reform was experienced as exclusionary, bureaucratically costly, or politically distant. I therefore examine whether municipal exposure to BVR is associated with differences in trust outcomes, and whether those associations vary along four salient respondent characteristics: gender, race, marital status, and education.

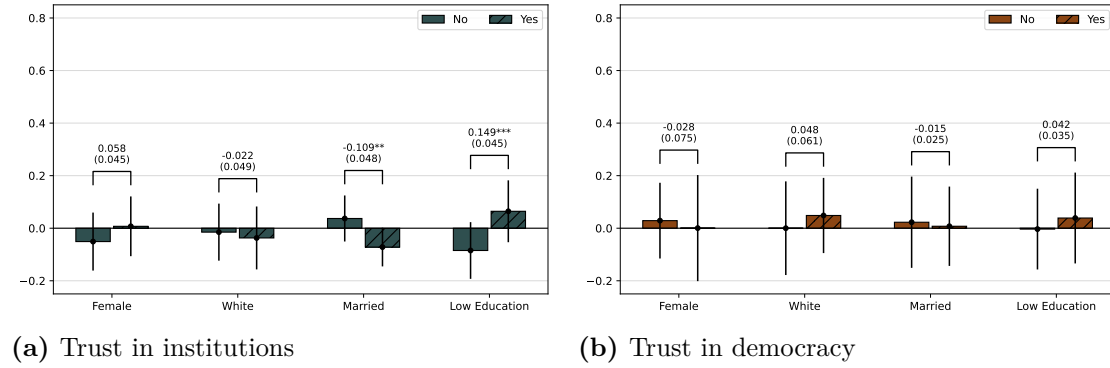
Empirical strategy. I estimate a separate interaction regression for each of four binary respondent characteristics: female, white, married, and low education. In each case,

$$y_{imt} = \beta \text{BVR}_{mt} + \lambda G_i + \theta (\text{BVR}_{mt} \times G_i) + X'_{imt} \Gamma + \alpha_m + \tau_t + \varepsilon_{imt}, \quad (8)$$

where y_{imt} is either z_i^{trust} or z_i^{dem} for respondent i in municipality m and survey year t , BVR_{mt} equals one if the municipality had entered the biometric regime by survey year t , and X_{imt} includes low education, female, white, married, working, age, log GDP per capita, and log municipality population. Municipality fixed effects α_m absorb time-invariant local characteristics and survey-year fixed effects τ_t absorb national shocks common to all respondents in a wave. Standard errors are clustered by municipality and year. The figures report two quantities for each subgroup: the baseline treatment effect for the omitted category and the combined treatment effect for the group with

$G_i = 1$. The printed number above each pair is the interaction coefficient θ . Because the outcomes are standardized, all reported effects are in standard-deviation units.

Figure 14: Biometric Registration and Political Trust by Respondent Characteristics



Note: Each panel reports coefficients from regressions of the relevant standardized trust outcome on biometric-registration exposure, a respondent characteristic, and their interaction, controlling for low education, female, white, married, working, age, log GDP per capita, and log municipality population, with municipality and survey-year fixed effects. Solid bars show the estimated BVR effect for the omitted category and hatched bars show the combined effect for the group indicated on the horizontal axis. Vertical lines report 95% confidence intervals. The number above each pair is the interaction coefficient with clustered standard error in parentheses. Standard errors are clustered by municipality and year.

Results. Figure 14 summarizes the linked-survey regressions. The figure is organized to mirror the underlying interaction specification. For each respondent characteristic, the solid bar reports the estimated BVR effect for the omitted category and the hatched bar reports the combined effect for the subgroup with the indicator equal to one. The number above each pair is the interaction term itself. This presentation is useful because it separates two questions: whether BVR is associated with a positive or negative shift in the outcome for the reference group, and whether that association differs systematically across respondent types.

Panel (a) shows the results for trust in institutions. The general pattern is that there is little evidence of heterogeneous effects by gender, race, or marital status. For women, the point estimate is higher than for men, but the difference is imprecise. For whites, the interaction is slightly negative, which suggests a somewhat smaller institutional-trust response than among non-whites, but again the estimate is far from precise enough to support a strong claim. Married respondents show a similarly small negative interaction. The clearest result concerns education. The low-education interaction is 0.123 standard deviations (s.e. 0.031), statistically significant at conventional levels. Substantively, this

means that the institutional-trust response to BVR is concentrated among respondents with less schooling. The estimated effect for the higher-education group is essentially zero, while the combined effect for lower-education respondents is positive, about 0.115 standard deviations.

That education pattern matters because it lines up with the administrative results in a revealing way. Sections 4 and 5 showed that BVR initially reduced the share of low-education voters in the registry. The survey evidence suggests that among lower-education respondents who remained politically connected to treated municipalities, exposure to the reform was associated with higher trust in institutions rather than lower trust. Proposition 5 offers a disciplined way to read that pattern: the positive trust estimate is consistent with the post-reform sample becoming selectively composed of lower-education citizens with unusually high latent attachment to the system. A more behavioral interpretation, in which direct interaction with a more modernized electoral bureaucracy improved perceptions of state capacity and procedural credibility, remains possible, but the model implies that the first interpretation has to be taken seriously.

Panel (b) reports the corresponding estimates for trust in democracy. Here the evidence is weaker. The point estimates for the subgroup-specific BVR effects are generally positive, but the interaction terms for female, white, and married respondents are all small and imprecisely estimated. The low-education interaction remains positive, about 0.054 standard deviations, but it is not statistically distinguishable from zero. In other words, once the outcome shifts from institutional trust to broader democratic attitudes, the heterogeneity becomes much noisier. This weakens the case for claiming that BVR substantially changed trust in democracy in the same way it appears to have shifted trust in institutions.

Taken together, the survey evidence points to a narrower conclusion than the title of the section might suggest. The most credible pattern is not a generalized increase in political trust across all groups. It is a more specific association between BVR exposure and higher trust in institutions among lower-education respondents. For democratic attitudes, the estimates move in the same general direction but are too imprecise to carry the same weight. That distinction is important for the overall message of the paper: the attitudinal results are best read as suggestive downstream evidence that complements the administrative findings, not as a standalone headline result.

8.3. Individual-level Exposure and Within-Municipality Variation

The municipality-level treatment assignment in equation (8) is deliberately coarse. It assigns the same BVR status to every respondent in a treated municipality-wave even though older and younger respondents need not have encountered the biometric regime at the same point in their voting lives. To sharpen that concern, I construct a respondent-level exposure proxy from interview age, interview year, and municipal rollout timing. The exercise uses benchmark LAPOP waves from 2006 through 2020, restricts attention to municipalities observed in every wave in that window, and drops respondents younger than 18 so that the exposure proxy matches the adult-eligibility rule. Let

$$\text{EligibleYear}_i = t_i - \text{Age}_i + 18, \quad \text{EventTime}_{im} = t_i - \max\{\text{TreatYear}_m, \text{EligibleYear}_i\}, \quad (9)$$

where t_i is the LAPOP interview year and TreatYear_m is the municipality’s first biometric election year. I then classify respondents into three groups. “Always-BVR” respondents satisfy $\text{EligibleYear}_i \geq \text{TreatYear}_m$ and therefore could only have entered the electorate under biometric registration. “Mixed” respondents were already adult voters when their municipality switched, so they experienced the transition as incumbents in the electorate. The residual category consists of respondents interviewed in not-yet-treated or never-treated municipalities. This design lets treatment vary within municipality-year cells because respondents of different ages imply different exposure histories, but the balanced-wave restriction is demanding: the final regression sample contains 2,425 respondents in 8 municipalities.

Two points are worth stating explicitly. First, the exposure proxy is approximate. LAPOP records age at interview, not actual registration year, and Brazil allows optional voting from age 16. I nevertheless follow the requested age-18 rule so that the new design is anchored to adult eligibility rather than adolescent optional registration. Second, the mixed cohort is substantively important and should not simply be dropped. Those respondents were exposed to BVR as adult voters, but they are not the same as people who first entered the electorate under the biometric regime. In the individual-level regressions below, I therefore retain the mixed cohort as a separate control group. Because exposure timing is itself imputed from age, the respondent-level specifications also add a cubic in age to flexibly partial out life-cycle differences in trust that a linear control might miss.

Table 4: Municipality-level and Respondent-level BVR Exposure in LAPOP

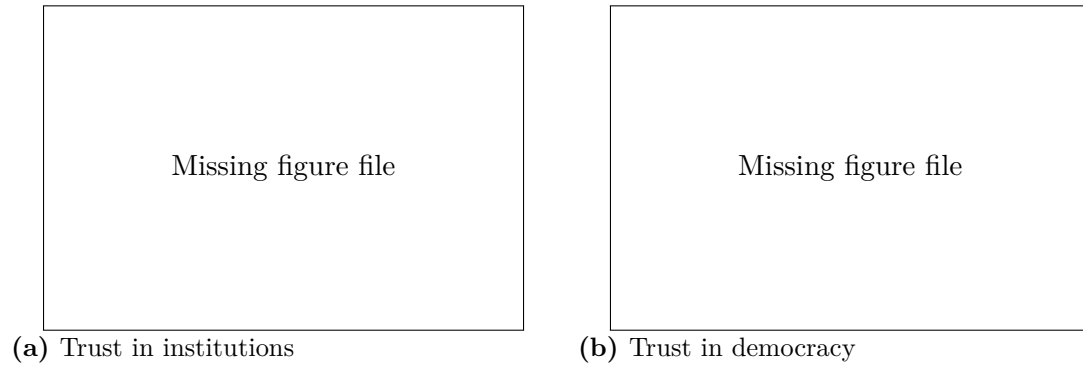
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Table 4 compares the original municipality-level treatment assignment to two respondent-level alternatives on this balanced sample. Column (1) re-estimates the municipal treatment design. Column (2) replaces municipality exposure with an indicator for having only ever been eligible under BVR, while controlling separately for the mixed cohort. Column (3) interacts that respondent-level exposure with low education. The main lesson is cautionary. For trust in institutions, the municipality-level treatment coefficient is positive, 0.251 standard deviations (s.e. 0.106), but its low-education interaction is small, 0.041 (s.e. 0.093). The respondent-level intent-to-treat estimate is also positive, 0.235 standard deviations, but imprecise (s.e. 0.184). The triple-difference interaction turns negative, -0.310 , and remains statistically indistinguishable from zero (s.e. 0.219). This is a severe support problem: after requiring complete wave coverage and age-18 eligibility, there are only 28 “always-BVR” adults, and only 2 of them have low education.

For trust in democracy, the respondent-level exercise is similarly uninformative about a clean cohort mechanism. The municipality-level coefficient is positive on the balanced sample, 0.428 standard deviations (s.e. 0.146), but the individual intent-to-treat estimate is close to zero, 0.097 (s.e. 0.243), and the low-education triple-difference estimate is noisy, 0.358 (s.e. 0.370). The table therefore does not provide evidence that the low-education trust pattern in the main municipality-level analysis is driven by respondents who first entered the electorate under BVR. Instead, it reinforces the narrower interpretation of the section: the survey evidence is suggestive and consistent with Proposition 5, but the respondent-level age proxy is too thinly supported to carry the mechanism by itself.

Figure 15 plots the respondent-level event-study coefficients. The dynamic path is informative precisely because it is not especially clean. For trust in institutions, several leads are already positive, including the -6 bin, and post-treatment coefficients are also

Figure 15: Respondent-level Event Studies for Trust and Democracy



Note: Never-treated municipalities are retained as controls, with all respondent-level event-time indicators set to zero. Event time is binned into two-year intervals from -8 to $+8$, with endpoints absorbing more distant exposure and event time -2 omitted as the reference category. All specifications include municipality and survey-year fixed effects, the same respondent controls as Table 4, a cubic in age, and standard errors clustered by municipality-year cell. The no-hybrid versions of these figures are produced separately; in this balanced sample they are identical because no hybrid municipalities remain.

positive in multiple bins. For trust in democracy, the -4 lead is negative while later post-treatment bins become positive. I read that pattern as a warning against over-interpreting respondent-level timing as literal registration timing. The event study is useful because it exposes the limits of the age-based proxy and the thin support behind the always-BVR cohort. It is not strong enough to overturn the broader selection-based reading of the LAPOP results.

A final placebo is reassuring. The corrected LAPOP file includes Brazil waves in both 2006 and 2008, but I follow the requested pre-rollout exercise and use the 2008 wave alone, coding placebo treatment as whether a respondent's municipality would adopt BVR by 2014. In that placebo regression, the state-fixed-effects specification is essentially zero for both trust in institutions (-0.039 , s.e. 0.103) and trust in democracy (-0.015 , s.e. 0.119). That does not prove parallel trends, but it does make it harder to argue that municipalities slated to adopt BVR within a few years were already mechanically more trusting before the main rollout wave.

Interpretation and limitations. These trust patterns should be interpreted carefully. The municipality-level treatment indicator is assigned at the municipality-year level rather than from a respondent's observed biometric registration status, so the baseline estimates combine direct exposure, compositional change, and broader local equilibrium

effects. The respondent-level exercise in Section 8.3 addresses part of that concern by exploiting variation in implied registration cohorts within municipality-year cells, and the null 2008 placebo further reduces the worry that future-adopter municipalities were already mechanically more trusting. But that stricter design does not fully solve the problem. The data are repeated cross-sections rather than a respondent panel, age is measured at interview rather than at registration, the age-18 eligibility rule is only an approximation in a setting with optional voting from age 16, and the mixed cohort sits between clean pre- and post-BVR exposure statuses. This is especially important given the administrative evidence in Section 4: if biometric registration changed who remained on the rolls, some of the attitudinal differences may still reflect selection into the post-reform electorate rather than a pure change in beliefs. Proposition 5 is therefore not a side note but the central interpretive frame for this section. The trust-in-democracy outcome is also less conceptually tight than the institutional-trust index because, as Table 3 shows, the first principal component captures a smaller share of the underlying variation. For those reasons, I treat the LAPOP evidence as suggestive downstream evidence that complements the administrative results, not as the paper’s primary causal contribution.

8.4. Backlash and Electoral Integrity

The trust results above suggest that BVR was not generally experienced as a threatening reform, but a distinct concern remains: electoral modernization can itself become a focal point for partisan suspicion. In Brazil this concern became especially salient during the Bolsonaro presidency, when attacks on the TSE and on electronic voting helped turn election administration into an object of political conflict. If those messages stuck, respondents aligned with Bolsonaro should be less willing to say that votes are counted correctly, and that skepticism could be sharper in municipalities where biometric registration made the digital electoral infrastructure more visible.

Data and outcomes. The reproducible public LAPOP file available for this exercise is the Brazil 2021 wave. That file includes an electoral-integrity battery but does not include individual presidential vote recall, so I use approval of the executive as the respondent-level proxy for Bolsonaro alignment. The first outcome is a binary indicator equal to one if the respondent says votes are counted correctly *always* and zero if the answer is *sometimes* or *never*. The second outcome is a binary indicator equal to one

if the respondent says politicians can identify how each person voted *never*, and zero otherwise; higher values therefore correspond to stronger belief in ballot secrecy. Because these integrity items were asked in a narrower 2021 survey module, the usable control set is more limited than in the main trust regressions. I therefore retain low education, female, white, age, urban residence, and log municipality population.

Empirical strategy. I estimate the following cross-sectional linear probability model:

$$y_{i,m} = \beta_1 \text{BVR}_m + \beta_2 \text{ApproveBolsonaro}_i + \beta_3 (\text{BVR}_m \times \text{ApproveBolsonaro}_i) + \mathbf{X}'_{i,m} \Gamma + \varepsilon_{i,m}, \quad (10)$$

where $y_{i,m}$ is one of the two integrity outcomes for respondent i in municipality m , BVR_m equals one if the municipality had entered the biometric regime by 2021, and $\text{ApproveBolsonaro}_i$ equals one when the respondent rates the executive's performance as good or very good. The coefficient on $\text{ApproveBolsonaro}_i$ therefore captures the baseline integrity gap between Bolsonaro-aligned and other respondents in municipalities without BVR exposure, while β_3 measures whether that gap differs in municipalities already under BVR. A negative interaction would be consistent with backlash: Bolsonaro-aligned respondents in BVR municipalities would be especially skeptical. A positive interaction would imply the opposite. All specifications use LAPOP survey weights, and standard errors are clustered at the municipality level.

Table 5: Backlash and electoral-integrity regressions, LAPOP Brazil 2021

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Note: Weighted linear-probability models using the public Brazil 2021 LAPOP wave. Columns (1)–(2) use an indicator for saying votes are counted correctly *always*; columns (3)–(4) use an indicator for saying politicians can identify how people voted *never*. ‘BVR’ equals one when the respondent’s municipality had entered the biometric regime by 2021. ‘Bolsonaro approval’ equals one when the respondent evaluates the executive as ‘good’ or ‘very good’. All specifications control for low education, female, white, age, urban residence, and log municipality population. Standard errors clustered by municipality appear in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Results. Table 5 shows two patterns. First, there is a clear Bolsonaro-alignment gap in perceptions of vote counting. In column (1), respondents who approve of Bolsonaro are about 22 percentage points less likely to say that votes are always counted correctly, even after conditioning on the available controls. In the interaction specification, the baseline approval gap is even larger in untreated municipalities: column (2) estimates a coefficient of -0.324 on ‘Bolsonaro approval’ ($p < 0.01$).

Second, the interaction does not support a backlash story in which BVR deepens that skepticism. For the vote-count outcome, the interaction term in column (2) is positive, about 0.198, and marginally significant at conventional levels ($p = 0.063$). Substantively, this means that the fair-count gap between Bolsonaro-aligned respondents and others is *smaller*, not larger, in municipalities already exposed to BVR. Put differently, the cross-sectional evidence does not suggest that biometric exposure amplified distrust of the count among respondents aligned with Bolsonaro; if anything, it attenuated that distrust. For ballot secrecy, the estimates are much weaker. Columns (3)–(4) show little evidence that either BVR or the interaction between BVR and Bolsonaro approval systematically shifts whether respondents believe politicians can trace individual votes.

Discussion. These estimates should be interpreted cautiously. The analysis relies on a single public cross-section from 2021, so it is not a direct post-2022 election test and cannot distinguish pre-existing local differences from attitudinal responses to BVR itself. The individual alignment measure is also imperfect: approval of Bolsonaro’s government is a reasonable proxy for partisan affinity, but it is not the same as verified vote choice. In addition, BVR is still measured at the municipality level rather than the respondent level, which raises the usual exposure-mismeasurement concerns. Finally, the integrity items were asked in a narrower module, which limits the available controls and keeps the estimation sample modest. Even with those caveats, the results are informative. They show a sizable Bolsonaro-alignment gap in trust that votes are counted correctly, but they do not show that BVR exposure intensified that gap. The most defensible reading is therefore a restrained one: the public-wave evidence points to skepticism among Bolsonaro-aligned respondents, but not to a clear BVR-specific backlash channel.

9. A Welfare Framework for Electoral Monitoring

This section develops the paper’s central theoretical framework. The framework treats electoral modernization reforms—voter registration laws, biometric verification, identification requirements—as pre-participation ordeals that screen the electorate. Three features distinguish the framework from existing ordeal theory and from existing models of costly voting. First, the verification intensity is chosen by a welfare-maximizing state, not taken as given. Second, verification operates before participation rather than at the ballot box, so it shapes the *set* of citizens whose preferences are aggregated rather than the *rate* at which a fixed set of citizens turn out. Third, the resulting composition shift feeds back into policy through probabilistic voting, so verification has both a direct cost (excluded citizens) and an indirect cost (policy distortion from an unrepresentative electorate).

The main theoretical result (Proposition 4) characterizes how the socially optimal verification intensity depends on the correlation between compliance costs and policy preferences. Even when this correlation is zero, verification distorts policy through a between-group composition channel; when the correlation is positive, an additional within-group drift channel reinforces the distortion and lowers the optimal intensity further. The cost-preference correlation ρ is the framework’s central novel parameter; it is estimable from the data and forms the basis of the paper’s calibration exercise.

The framework deliberately keeps the formal apparatus light. The contribution is conceptual rather than technical: a single new parameter, the cost-preference correlation, organizes a wide range of comparative statics that the existing literature has examined piecemeal. Formal proofs of all propositions, regularity conditions, and extensions appear in Appendix A.

9.1. Environment

There is a continuum of citizens indexed by $i \in [0, 1]$. Each citizen is characterized by a vector of primitives (e_i, c_i, v_i, g_i^*) .

The variable $e_i \in \{L, H\}$ denotes the citizen’s education category, with L standing for low

education and H for high education. Let $\pi \in (0, 1)$ denote the share of low-education citizens. The dichotomous treatment of education is a simplification; the framework extends naturally to richer schemas, which we discuss in Appendix A.

The variable $c_i \geq 0$ is the citizen's idiosyncratic cost of complying with verification. Within education group $e \in \{L, H\}$, c_i is drawn from distribution F_e with density f_e . The cost captures, in reduced form, the time, information, and bureaucratic friction a citizen must absorb to register under a given verification regime: the opportunity cost of travel to a registration office, the cognitive cost of understanding new procedures, and the psychic cost of interacting with the state.

The variable $v_i > 0$ is the citizen's value of being registered to vote. We treat v_i as an idiosyncratic draw from a distribution H with density h , independent of all other primitives. The variable captures the bundle of motives that lead a citizen to value electoral participation: civic attachment, expressive value, instrumental belief in influencing outcomes, and social signaling.

The variable $g_i^* \in \mathbb{R}$ is the citizen's preferred one-dimensional policy. Within education group e , g_i^* is drawn from distribution G_e with mean $\bar{g}_e$. Higher values of g_i^* correspond to more redistributive or more leftist policies; lower values correspond to less redistributive or more rightist policies.

Three assumptions structure the environment.

Assumption 1 (Differential compliance). *F_L is first-order stochastically dominated by F_H : $F_L(c) \leq F_H(c)$ for all $c \geq 0$, with strict inequality on a set of positive measure.*

Assumption 1 states that low-education citizens face higher compliance costs in the FOSD sense. This is the formal counterpart to the long-documented empirical regularity that registration burdens fall disproportionately on less-educated citizens (Rosenstone and Wolfinger, 1978; Nagler, 1991; Ansolabehere and Konisky, 2005; Hajnal et al., 2017). The dominance is reduced-form: it can arise from differential information, time costs, mobility constraints, or cognitive load, and the model takes no stand on the underlying mechanism.

Assumption 2 (Ordered preferences). *High-education citizens have lower mean preferred policy: $\bar{g}_H < \bar{g}_L$.*

Assumption 2 is the standard education-redistribution gradient from the political economy literature (Meltzer and Richard, 1981). It is empirically well-documented in Brazil (Hunter, 2010; Terron and Soares, 2010) and in cross-country evidence on the income-redistribution preference gradient (Alesina et al., 2018). The assumption can be relaxed to allow for heterogeneous education-preference gradients across regions or time periods; we discuss this extension in Section 9.9.

Assumption 3 (Within-group cost-preference correlation). *Within each education group, compliance costs and policy preferences have correlation $\rho_e \in [-1, 1]$. For tractability, we assume $\rho_L = \rho_H = \rho$.*

Assumption 3 introduces the framework’s central novel parameter. The correlation ρ captures whether, controlling for education, citizens with higher administrative friction costs also have more leftist preferences ($\rho > 0$), more rightist preferences ($\rho < 0$), or are preference-neutral ($\rho = 0$). The assumption that $\rho_L = \rho_H$ simplifies the algebra; in Appendix A we show the main results extend to heterogeneous group-specific correlations under regularity conditions.

The economically interesting case is $\rho > 0$. Within a given education group, the citizens with the highest compliance costs are typically those most marginally attached to the political process—those who live farthest from registration offices, have the least flexible work schedules, or lack documentation that makes interaction with the state easier. These same citizens have, in many empirical settings, more leftist political preferences. The case $\rho < 0$ is logically possible (high-cost citizens have rightist preferences) but uncommon in the empirical literature on developing democracies.

9.2. Registration Stage

The state chooses a verification intensity $\tau \in [0, 1]$, interpreted as the administrative demandingness of registration. Each citizen registers if and only if the value of being registered net of the scaled compliance cost is non-negative:

$$\text{Register}_i = 1 \iff v_i \geq \tau c_i. \quad (11)$$

For $\tau = 0$, every citizen registers because the compliance cost vanishes. For $\tau > 0$, regis-

tration is selective: only citizens whose value of registering exceeds the scaled compliance cost actually register. Denote by $R_e(\tau)$ the registration rate in group e :

$$R_e(\tau) = \int_0^\infty \int_0^\infty \mathbf{1}\{v \geq \tau c\} f_e(c) h(v) dc dv = \int_0^\infty f_e(c) \bar{H}(\tau c) dc, \quad (12)$$

where $\bar{H}(x) = 1 - H(x) = \Pr(v \geq x)$ is the survival function of v . The first proposition establishes the heterogeneous compliance result that drives the empirical predictions on registry composition.

Proposition 1 (Heterogeneous compliance). *Under Assumption 1, $R_L(\tau) \leq R_H(\tau)$ for all $\tau > 0$, with strict inequality whenever the FOSD is strict on a set reached by v/τ with positive H -probability. Under the additional regularity condition that $f_L(c)/f_H(c)$ is non-increasing in c (the monotone likelihood ratio property, MLRP), the relative low-education registration rate $R_L(\tau)/R_H(\tau)$ is weakly decreasing in τ .*

The proof, given in Appendix A, follows from integrating the conditional registration probability over the value distribution and exploiting the FOSD ordering of cost distributions; the ratio result follows from a likelihood-ratio argument under MLRP. Proposition 1 is the registration-stage prediction that the paper’s empirical evidence bears on directly. The decomposition in Section 5 identifies $R_L(\tau)$ and $R_H(\tau)$ for the Brazilian setting from event-time-zero registry effects.

9.3. Electoral Stage

After registration closes, the registered electorate selects policy through a two-candidate probabilistic voting game in the tradition of Lindbeck and Weibull (1987) and Dixit and Londregan (1996). Each candidate A, B commits to a policy $g \in \mathbb{R}$. A registered voter i receives utility $u_i(g) = -(g - g_i^*)^2$ from policy g . Voters also have additive candidate-specific valence shocks drawn from a smooth common distribution.

Under standard regularity conditions for probabilistic voting equilibrium—continuity of candidate utility in policy, compact policy space, sufficient density of swing voters in each group—both candidates converge to the policy that maximizes the swing-voter-weighted

sum of registered voters' utilities. The equilibrium policy is

$$g^*(\tau) = \frac{\pi R_L(\tau)\phi_L \bar{g}_L(\tau) + (1 - \pi)R_H(\tau)\phi_H \bar{g}_H(\tau)}{\pi R_L(\tau)\phi_L + (1 - \pi)R_H(\tau)\phi_H}, \quad (13)$$

where ϕ_e is the density of swing voters in group e and $\bar{g}_e(\tau)$ is the mean preferred policy among registered citizens in group e at verification intensity τ . Equation (13) is standard. The novel feature of the framework is that $\bar{g}_e(\tau)$ depends on τ through the registration-stage selection: when $\rho \neq 0$, verification selects on c_i , which is correlated with g_i^* .

Lemma 1 (Within-group policy drift). *Under Assumption 3 with $\rho > 0$, $\bar{g}_e(\tau) < \bar{g}_e(0)$ for $\tau > 0$. That is, verification shifts each group's registered mean preference rightward (toward lower policy values).*

Lemma 1 identifies a within-group drift channel that complements the between-group composition shift implied by Proposition 1. Verification not only reshapes who is registered by education—it also reshapes within-education preferences when costs and preferences correlate. Both channels operate simultaneously and reinforce each other when $\rho > 0$.

We now define the central quantity for the welfare analysis: the *representation distortion* induced by verification.

$$d(\tau) \equiv g^*(\tau) - g^*(0). \quad (14)$$

When $\tau = 0$, every citizen is registered and $g^*(0)$ is the unrestricted democratic equilibrium. When $\tau > 0$, verification excludes some citizens and the post-registration equilibrium policy differs from $g^*(0)$. The distortion has two components: a between-group component driven by the differential compliance result (Proposition 1) and a within-group component driven by the cost-preference correlation (Lemma 1). The between-group component is present whenever Assumption 1 holds, regardless of ρ ; the within-group component is present only when $\rho \neq 0$.

Proposition 2 (Drift direction and sensitivity to ρ). *Under Assumptions 1–3:*

- (i) *If $\rho \geq 0$, $d(\tau) \leq 0$ for all $\tau \geq 0$, with strict inequality whenever $\tau > 0$ and either Assumption 1 or Assumption 3 holds strictly.*

(ii) $|d(\tau)|$ is non-decreasing in ρ at any fixed $\tau > 0$.

(iii) $|d(\tau)|$ is non-decreasing in τ at any fixed $\rho \geq 0$.

The proof appears in Appendix A. Proposition 2 establishes that the representation distortion is monotone in both verification intensity and preference correlation. The empirical work in Section 5 estimates the magnitude of the composition shift; the calibration exercise then infers ρ from cross-municipality variation and bounds the implied $|d(\tau)|$.

9.4. Welfare

The social planner's objective combines three components: integrity, inclusion, and procedural representation. Because welfare is determined only up to a positive affine transformation, three independent positive weights would over-determine the model: doubling all weights leaves the optimum τ^* unchanged. We therefore normalize the weights to lie on the unit simplex. Let $\alpha, \beta \geq 0$ with $\alpha + \beta \leq 1$, and define $\gamma \equiv 1 - \alpha - \beta \geq 0$. The welfare function is

$$W(\tau; \alpha, \beta) = \alpha \cdot \Phi(\tau) - \beta \cdot X(\tau) - (1 - \alpha - \beta) \cdot d(\tau)^2. \quad (15)$$

The integrity term reflects the social value of cleaner electoral records, more secure voter authentication, and reduced impersonation risk. The function Φ satisfies $\Phi(0) = 0$, $\Phi'(\tau) > 0$, and $\Phi''(\tau) < 0$. The concavity reflects diminishing returns: the first marginal increase in verification eliminates the most egregious registry errors, while later increases pick up smaller residual gains.

The exclusion term reflects the direct welfare cost of citizens being kept off the rolls,

$$X(\tau) = \pi(1 - R_L(\tau)) + (1 - \pi)(1 - R_H(\tau)),$$

which is the standard cost in ordeal theory (Nichols and Zeckhauser, 1982; Alatas et al., 2016): every excluded citizen represents a loss of civic participation that the planner regrets at rate β .

The representation term is the framework’s novel addition. The squared distortion $d(\tau)^2$ from equation (14) reflects the standard convex-loss view of departures from the unrestricted democratic equilibrium; small distortions are tolerable, larger ones are increasingly bad. The squared form simplifies the algebra; results extend to general convex penalties under regularity conditions detailed in Appendix A. The representation term operationalizes a procedural interpretation of democratic welfare: the planner values policy outcomes that reflect what a fully registered electorate would have produced. This is in the spirit of Kolm (1977), Sen (1985), and the deliberative-democratic tradition emphasizing equal political voice as a constitutive feature of legitimacy. Verification is costly not only because it excludes individuals but because it systematically distorts the aggregation of preferences.

The simplex parametrization reduces the calibration problem to a two-dimensional region (α, β) in the unit triangle, with $\gamma = 1 - \alpha - \beta$ inferred residually. The vertices of the simplex correspond to extreme planner types: $(\alpha, \beta, \gamma) = (1, 0, 0)$ is an integrity-only planner who chooses $\tau^* = 1$; $(0, 1, 0)$ is an inclusion-only planner who chooses $\tau^* = 0$; $(0, 0, 1)$ is a representation-only planner who chooses τ^* to minimize $|d(\tau)|$, which under Assumption 1 also implies $\tau^* = 0$ (since $d(0) = 0$). Interior (α, β, γ) produces interior τ^* .

Assumption 4 (Well-behaved integrity function). $\Phi : [0, 1] \rightarrow \mathbb{R}_+$ is twice continuously differentiable with $\Phi(0) = 0$, $\Phi'(0) > 0$ finite, $\Phi'(\tau) > 0$, and $\Phi''(\tau) < 0$ for $\tau \in (0, 1]$.

Under Assumption 4 and standard regularity of the compliance and preference distributions, $W(\tau; \alpha, \beta)$ is continuous and bounded on $[0, 1]$, so it attains a maximum.

Proposition 3 (Existence and interior optimum). *Suppose Assumptions 1–4 hold and the boundary derivatives satisfy $W'(0) > 0$ and $W'(1) < 0$. Then there exists an interior verification intensity $\tau^* \in (0, 1)$ satisfying $W'(\tau^*) = 0$. Under sufficient curvature conditions on Φ , τ^* is unique.*

The boundary conditions are economically sensible: $W'(0) > 0$ says the first marginal unit of verification delivers integrity gains exceeding combined inclusion and representation costs; $W'(1) < 0$ says full verification is excessive. Sufficient conditions for strict concavity of W involve curvature comparisons between Φ and the cost-and-distortion terms; these are spelled out in Appendix A.

9.5. The Main Result

The paper’s central theoretical contribution is a comparative static on the optimal verification intensity with respect to the preference-cost correlation ρ .

Proposition 4 (Optimal verification and preference correlation). *Under Assumptions 1–4 and the regularity conditions of Proposition 3, the optimal verification intensity τ^* is non-increasing in ρ for any fixed weights (α, β) in the unit simplex with $\gamma = 1 - \alpha - \beta > 0$. Strict monotonicity holds whenever the within-group drift channel of Lemma 1 is operative.*

Proposition 4 is the framework’s central theoretical claim. A society in which compliance costs correlate with political preferences—plausibly, most societies, because education and income are correlated with both administrative friction and political orientation—should choose less intense verification than otherwise. Note that the result does not say that $\rho = 0$ reduces the welfare problem to a standard ordeal-theoretic benchmark with no representation term; the between-group composition channel persists at $\rho = 0$, since it depends only on the FOSD ordering of cost distributions. Rather, the result says that the within-group drift channel, governed by ρ , adds to the always-present between-group channel, and the optimal intensity is monotonically lower as ρ rises.

The result has bite for the comparative analysis of voter ID and registration reforms. In settings where compliance costs are nearly orthogonal to political preferences (perhaps small homogeneous democracies with high state capacity), only the between-group channel operates and verification can be welfare-improving for modest representation weights. In settings where compliance costs correlate strongly with preferences (developing democracies with sharp class-based political cleavages, US states with racially polarized voting and unequal access to identification), the within-group drift channel adds to the between-group channel and the optimal intensity is meaningfully lower. Brazil falls in the latter category, as we document empirically.

Corollary 1 (Welfare-improvement threshold). *For an observed verification intensity τ_0 and weights (α, β) in the unit simplex, τ_0 is welfare-improving relative to no verification ($\tau = 0$) if and only if*

$$\alpha \cdot \Phi(\tau_0) > \beta \cdot X(\tau_0) + (1 - \alpha - \beta) \cdot d(\tau_0)^2. \quad (16)$$

Corollary 1 is operational. For any parameter vector (α, β, ρ) and observed τ_0 , one can check whether inequality (16) holds. Calibrating the inequality for Brazil characterizes the parameter region in which BVR is welfare-improving. The two-dimensional simplex parametrization makes the welfare region naturally representable as a contour over the (α, β) unit triangle, with ρ serving as a parameter that shifts the contour.

9.6. Trust as Selection

The framework also generates a prediction relevant for interpreting the survey evidence in Section 8.

Proposition 5 (Trust is a selection artifact). *Let $T_i = h_T(v_i)$ with $h'_T(\cdot) > 0$ denote observed trust in democratic institutions. For any $\tau > 0$,*

$$\mathbb{E}[T_i \mid \text{Register}_i = 1] \geq \mathbb{E}[T_i \mid \text{Register}_i = 0],$$

with strict inequality when the compliance constraint binds on a set of positive measure.

Proposition 5 provides the interpretive frame for the paper’s survey evidence. Higher observed trust in post-reform treated municipalities is consistent with either a behavioral change in beliefs or a selection effect from verification. The paper’s empirical work cannot fully separate these, but Proposition 5 establishes that the selection interpretation is always present and is the conservative null against which behavioral interpretations should be judged.

9.7. Comparative Statics and Testable Implications

The framework generates four predictions that the empirical sections evaluate.

Prediction I (Heterogeneous compliance, Proposition 1). Low-education registration rates fall more sharply than high-education rates as verification intensity rises. The decomposition in Section 5 estimates $R_L(\tau)$, $R_H(\tau)$ from event-time-zero coefficients in dynamic event studies on count outcomes.

Prediction II (Policy drift direction, Proposition 2). Verification induces a rightward drift in the post-registration electorate’s preferences, with magnitude increasing in ρ . Municipality-level vote shares before and after BVR provide the natural empirical setting for this prediction.

Prediction III (Selection into trust, Proposition 5). Observed trust levels among registered citizens in treated municipalities may be higher than in untreated ones for reasons that do not require behavioral updating. Section 8 interprets the trust evidence through this lens.

Prediction IV (Welfare sensitivity to ρ , Proposition 4). The welfare evaluation of verification depends critically on the correlation between compliance costs and political preferences. The calibration exercise estimates ρ and evaluates the welfare implications using Corollary 1.

9.8. Intuition: Voters, Politicians, and Policymakers

The formal apparatus is austere but the underlying intuitions describe how real actors reason about electoral verification reforms.

The voter’s decision. A citizen contemplating registration under a new biometric regime weighs two things implicitly. The first is the value of being on the rolls: the policy stake she has in the next election, the expressive value of casting a ballot, the social cost of being unregistered in a community that votes. The second is the friction of completing the procedure: the time it takes to travel to the registration office during business hours, the cognitive burden of understanding new requirements, the small but real anxiety of bureaucratic encounter. The model’s compliance condition $v_i \geq \tau c_i$ is a stylized rendering of this calculation. What casual reasoning often misses is the ratio structure: when verification intensity τ doubles, the registration condition becomes twice as demanding for every citizen, but it bites disproportionately on those with high c_i . Two citizens with identical v_i can end up on opposite sides of the registration threshold simply because one lives closer to a registration office or has a more flexible work schedule. This is what political-science writing means by “administrative burden as a censitary mechanism”—a mechanism that effectively imposes a property qualification by other

means (Herd and Moynihan, 2018). Even a benevolent state choosing τ to maximize welfare will produce censitary effects whenever Assumption 1 holds, which is to say, almost always.

The politician’s incentive. Politicians do not observe v_i , c_i , or g_i^* at the individual level. What they observe is the composition of the registered electorate and survey indicators of constituent preferences. The probabilistic-voting framework makes the connection between registry composition and policy explicit: the equilibrium policy weights group preferences by registered population share times swing-voter density. When verification intensifies and R_L falls relative to R_H , the weight on low-education preferences in the policy formula declines, and politicians responding optimally to the registered electorate shift platforms accordingly. This shift is not a moral failure: within the electoral incentives politicians face, attending to the registered electorate’s preferences is what wins elections. The mechanism operates at the level of statistical aggregates, but the welfare consequences are real. Lemma 1 and Proposition 2 together capture this: even within an education group, the citizens who continue to register are systematically different in their preferences from those who do not.

The policymaker’s choice. The state’s decision to adopt and intensify verification reflects a balancing of values that is rarely articulated as such. Election administrators emphasize integrity: the credibility of the registry, the prevention of fraud, the modernization of state capacity. Civil society advocates emphasize inclusion: the cost of exclusion to the disenfranchised, the democratic ideal of universal suffrage. Constitutional courts increasingly emphasize representation: whether reforms produce systematic disparities that violate equal protection or non-discrimination norms. The welfare function in equation (15) formalizes this triad through the simplex weights (α, β, γ) .

The framework reveals a sharp prediction about how the optimal intensity depends on which value the planner prioritizes. A planner with high α relative to β and γ will choose intense verification regardless of ρ . A planner with low α and high β will choose modest verification, again largely regardless of ρ . The interesting case—where ρ matters most—is when the simplex weights are interior. In this case, the optimal τ^* is interior, and Proposition 4 delivers its bite: the higher is ρ , the lower is τ^* .

This has direct implications for how policymakers should think about reforms. A jurisdiction considering biometric registration or a new voter ID law should not only ask “how much fraud will this prevent?” and “how many citizens will be excluded?” but also “how strongly do compliance costs correlate with political preferences in our setting?” The third question is empirically tractable: it can be answered by examining whether the demographic groups facing the highest administrative friction are also the groups with the most distinctive policy preferences.

9.9. Discussion and Extensions

The framework simplifies in several respects. We discuss four points.

Independence of value and cost. The framework assumes v_i is independent of c_i . In reality, citizens facing high compliance costs may also have lower subjective value of registration: a self-protective psychological adjustment, or a learned response to repeated bureaucratic frustration. Appendix A discusses the formal extension. If v_i and c_i are positively correlated, the comparative statics in Proposition 1 sharpen and the trust selection in Proposition 5 also sharpens. The main result on optimal τ^* in Proposition 4 remains qualitatively similar.

Politicians’ strategic response. The framework treats candidate platforms as endogenous to the registered electorate’s preferences but holds candidate selection and mobilization spending fixed. In richer settings, parties most affected by exclusion have incentives to invest in mobilization, voter education about new requirements, and outreach to marginal compliers. These responses are documented in the US voter-ID literature (Grimmer and Yoder, 2021) and would attenuate the direct compositional effect. The framework’s prediction on τ^* should be interpreted as conditional on a fixed strategic environment.

Heterogeneous group correlations. We assume $\rho_L = \rho_H$ for tractability. In reality, the cost-preference correlation may differ across groups. The main result extends

to heterogeneous correlations under regularity conditions detailed in Appendix A; the operative parameter is a population-weighted average $\bar{\rho}$.

Dynamic considerations. The framework is static. In reality, citizens excluded at the moment of reform can re-register in subsequent cycles, and the registered electorate evolves over time. The empirical evidence in Section 5 shows partial registry recovery at horizons of 4–8 years. The dynamic extension of the framework is straightforward: replace $R_e(\tau)$ with $R_e(\tau, t)$ where t is time-since-reform, and integrate welfare over horizon. The qualitative comparative statics survive.

9.10. Position in the Literature

The framework synthesizes three literatures that have evolved largely separately.

The first is ordeal theory, beginning with Nichols and Zeckhauser (1982) and developed structurally by Alatas et al. (2016), Finkelstein et al. (2025), and Deshpande and Li (2019). This literature analyzes administrative friction as a screening device in transfer programs: the standard result is that ordeals can improve targeting if costs are higher for non-target types and can worsen targeting if costs correlate negatively with type. Our framework imports this structure into the electoral domain and adds the policy feedback channel that is absent in transfer settings.

The second is the costly-voting literature, including Börgers (2004) and Krasa and Polborn (2008). This literature treats participation costs as exogenous features of the voter’s environment and analyzes welfare implications of voluntary versus compulsory voting. Our framework treats verification intensity as a state-chosen instrument operating before the vote, conceptually distinct from the costs the voter faces at the ballot box on election day.

The third is the political economy of state capacity, including Besley and Persson (2011), Acemoglu and Robinson (2019), and applications to electoral administration in Fujiwara (2015) and Marx et al. (2021). This literature emphasizes how administrative capacity shapes policy outcomes through institutional channels but has not formalized the specific mechanism by which verification intensity reshapes the electorate’s preference

distribution.

The framework’s contribution is to combine these literatures around a single new parameter—the cost-preference correlation ρ —that organizes the welfare analysis of electoral verification reforms in a way that none of the constituent literatures has done. The empirical work in the remainder of the paper estimates ρ for Brazil and applies the framework to evaluate the welfare consequences of biometric registration.

10. Calibration: Optimal Electoral Monitoring

10.1. Calibration Objects

10.2. Integrity Benefits

10.3. Inclusion Costs

10.4. Representation Costs

10.5. Optimal Monitoring Intensity

11. Conclusion

This conclusion is provisional scaffold text. It should be rewritten after the empirical findings, contribution claims, and limitations are fully established.

This paper is designed to study biometric voter registration in Brazil as both an administrative reform and a political event. The central premise is that changes to election infrastructure can affect not only the mechanics of registration, but also the social composition of the electorate and the way citizens think about institutions and democratic procedures. The project therefore links administrative data and survey evidence within a common framework organized around staggered municipal adoption.

The intended contribution is to keep together questions that are often studied separately. One strand of analysis focuses on electorate size and composition, while another focuses on trust, democratic support, and beliefs about electoral integrity. Bringing these strands into the same paper may clarify when administrative modernization appears primarily technical, when it appears politically consequential, and where the strongest inferential limits remain.

At this stage, the manuscript is a structured starting point rather than a finished argument. The next steps are to verify treatment timing, harmonize municipality identifiers, document data provenance, choose estimators transparently, and replace scaffold prose with evidence-backed claims. If the project succeeds, the final paper should offer a careful account of how biometric registration altered electoral administration in Brazil and what those changes may have meant for democratic confidence and skepticism.

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A. Proofs and Regularity Conditions for Section 9

This appendix provides formal proofs of the propositions in Section 9, states the regularity conditions under which the results obtain, and discusses several extensions referenced in the main text. Throughout, we use the notation established in Section 9: citizens are indexed by $i \in [0, 1]$ with primitives (e_i, c_i, v_i, g_i^*) , where $e_i \in \{L, H\}$, π is the share of low-education citizens, F_e and f_e are the cost CDF and density in group e , H and h are the value CDF and density (with survival function $\bar{H} = 1 - H$), and G_e is the within-group preference distribution. The welfare function is

$$W(\tau; \alpha, \beta) = \alpha \cdot \Phi(\tau) - \beta \cdot X(\tau) - (1 - \alpha - \beta) \cdot d(\tau)^2,$$

with (α, β) in the unit simplex and $\gamma \equiv 1 - \alpha - \beta$.

A.1. Regularity Conditions

The following technical conditions are maintained throughout. They guarantee that integrals defining $R_e(\tau)$ and $\bar{g}_e(\tau)$ are well-defined and that derivatives with respect to τ can be passed under the integral sign.

- Condition 1** (Regularity of distributions). (a) H has support $[\underline{v}, \infty)$ for some $\underline{v} > 0$, with continuously differentiable density h that is bounded on compact subsets and satisfies $\int_0^\infty v h(v) dv < \infty$.
- (b) For each $e \in \{L, H\}$, F_e has support $[0, \bar{c}_e]$ for some $\bar{c}_e \leq \infty$, with continuously differentiable density f_e that is bounded on compact subsets and satisfies $\int_0^\infty c f_e(c) dc < \infty$.
- (c) The conditional distribution of g_i^* given (e_i, c_i) has finite second moments and continuously differentiable conditional density.
- (d) (MLRP) The likelihood ratio $f_L(c)/f_H(c)$ is non-increasing in c on the common support of F_L and F_H .

Condition 1(d) is the monotone likelihood ratio property (MLRP). It strengthens Assumption 1 (FOSD) but is satisfied by standard parametric families used in the empirical

literature on participation costs (log-normal, Weibull, exponential mixtures). MLRP is needed for the ratio result in Proposition 1 (that $R_L(\tau)/R_H(\tau)$ is monotone in τ). The bare FOSD assumption is sufficient for the level result $R_L(\tau) \leq R_H(\tau)$.

Condition 2 (Joint distribution of cost and preference). *Within each education group e , the joint distribution of (c, g^*) has a continuously differentiable joint density $j_e(c, g)$ that satisfies the following.*

- (a) *The conditional expectation $\mu_e(c) \equiv \mathbb{E}[g^* \mid c, e]$ is monotone in c , with sign of $\mu'_e(c)$ equal to the sign of ρ .*
- (b) *For $\rho > 0$, $\mu_e(c)$ is strictly increasing in c and continuously differentiable; symmetrically for $\rho < 0$.*

Condition 2 formalizes Assumption 3 in a way that supports calculus arguments. Standard joint distributions used in applied work (bivariate normal, Gaussian copula, Plackett copula) satisfy this condition.

A.2. Proof of Proposition 1

Proof. We prove the level result $R_L(\tau) \leq R_H(\tau)$ first, then the monotonicity of the ratio under MLRP.

Step 1: Level result. By Fubini's theorem (valid under Condition 1(a)–(b)),

$$R_e(\tau) = \int_0^\infty \int_0^\infty \mathbf{1}\{v \geq \tau c\} f_e(c) h(v) dc dv = \int_0^\infty f_e(c) \bar{H}(\tau c) dc.$$

Equivalently, conditioning on v instead of c , $R_e(\tau) = \int_0^\infty F_e(v/\tau) h(v) dv$. Under Assumption 1, $F_L(c) \leq F_H(c)$ for all c , with strict inequality on a set of positive Lebesgue measure. Hence $F_L(v/\tau) \leq F_H(v/\tau)$ for all $v > 0$, and

$$R_L(\tau) = \int_0^\infty F_L(v/\tau) h(v) dv \leq \int_0^\infty F_H(v/\tau) h(v) dv = R_H(\tau).$$

The inequality is strict whenever the FOSD-strict set is reached by v/τ with positive H -probability.

Step 2: Ratio monotonicity under MLRP. Differentiating $R_e(\tau) = \int_0^\infty f_e(c)\bar{H}(\tau c) dc$ under the integral (Condition 1),

$$R'_e(\tau) = - \int_0^\infty c f_e(c) h(\tau c) dc < 0. \quad (17)$$

Define the proportional growth rate $\psi_e(\tau) \equiv R'_e(\tau)/R_e(\tau) < 0$. The log-ratio $\xi(\tau) \equiv \log R_L(\tau) - \log R_H(\tau)$ satisfies $\xi'(\tau) = \psi_L(\tau) - \psi_H(\tau)$. We show $\xi'(\tau) \leq 0$.

For any $\tau > 0$ and group e , define the probability measure $\nu_{e,\tau}$ on $[0, \infty)$ with density proportional to $f_e(c)\bar{H}(\tau c)$ and normalizing constant $R_e(\tau)$. Under this measure, citizens are weighted by their cost-type density and by the survival probability. For any non-negative integrand φ ,

$$\mathbb{E}_{\nu_{e,\tau}}[\varphi(c)] = \frac{1}{R_e(\tau)} \int_0^\infty \varphi(c) f_e(c) \bar{H}(\tau c) dc.$$

The proportional growth rate becomes

$$\psi_e(\tau) = - \frac{\int_0^\infty c f_e(c) h(\tau c) dc}{\int_0^\infty f_e(c) \bar{H}(\tau c) dc} = - \mathbb{E}_{\nu_{e,\tau}} \left[c \cdot \frac{h(\tau c)}{\bar{H}(\tau c)} \right].$$

The bracketed expression $\varphi(c) \equiv c \cdot h(\tau c)/\bar{H}(\tau c)$ is the product of c (increasing) and the hazard rate $h/\bar{H}$ evaluated at τc (non-decreasing under standard hazard regularity, which holds for the parametric families typically used in applied work). Hence $\varphi(c)$ is increasing in c .

Under MLRP (Condition 1(d)), the measure $\nu_{L,\tau}$ first-order stochastically dominates $\nu_{H,\tau}$. To see this, note that the likelihood ratio between the two densities is

$$\frac{d\nu_{L,\tau}/dc}{d\nu_{H,\tau}/dc} = \frac{f_L(c)\bar{H}(\tau c)/R_L(\tau)}{f_H(c)\bar{H}(\tau c)/R_H(\tau)} = \frac{R_H(\tau)}{R_L(\tau)} \cdot \frac{f_L(c)}{f_H(c)},$$

which is non-increasing in c by MLRP (the constant prefactor does not affect monotonicity). A non-increasing likelihood ratio implies that $\nu_{L,\tau}$ FOSD-dominates $\nu_{H,\tau}$.

Since $\varphi(c)$ is increasing in c , FOSD dominance gives

$$\mathbb{E}_{\nu_{L,\tau}}[\varphi(c)] \geq \mathbb{E}_{\nu_{H,\tau}}[\varphi(c)],$$

so $-\psi_L(\tau) \geq -\psi_H(\tau)$, i.e., $\psi_L(\tau) \leq \psi_H(\tau)$, i.e., $\xi'(\tau) \leq 0$. Therefore $\xi(\tau)$ is non-increasing, equivalently $R_L(\tau)/R_H(\tau)$ is non-increasing in τ . $\square$

A.3. Proof of Lemma 1

Proof. Fix education group e and verification intensity $\tau \geq 0$. The mean preferred policy among registered citizens in group e is $\bar{g}_e(\tau) = \mathbb{E}[g^* \mid e, v \geq \tau c]$.

Since v is independent of (c, g^*) within each group, we can integrate v out conditionally on (c, g^*) . The probability of registering given (c, e) is $\bar{H}(\tau c)$, so

$$\bar{g}_e(\tau) = \frac{\int_0^\infty \mu_e(c) \bar{H}(\tau c) f_e(c) dc}{\int_0^\infty \bar{H}(\tau c) f_e(c) dc}, \quad (18)$$

where $\mu_e(c) = \mathbb{E}[g^* \mid c, e]$ from Condition 2.

For $\tau = 0$, $\bar{H}(0) = 1$ and equation (18) reduces to

$$\bar{g}_e(0) = \int_0^\infty \mu_e(c) f_e(c) dc = \mathbb{E}[g^* \mid e] = \bar{g}_e.$$

For $\tau > 0$, $\bar{H}(\tau c)$ is strictly decreasing in c . Equation (18) expresses $\bar{g}_e(\tau)$ as a weighted average of $\mu_e(c)$ with weights $\bar{H}(\tau c) f_e(c)$, while $\bar{g}_e(0)$ is the same average with weights $f_e(c)$. The weights at $\tau > 0$ place relatively more mass on low- c values than the unweighted density.

Under Condition 2(b), with $\rho > 0$, $\mu_e(c)$ is strictly increasing in c . By the standard rearrangement inequality (a strictly increasing function's mean under a measure that places more mass on low values is strictly less than its mean under the original measure), we obtain

$$\bar{g}_e(\tau) = \frac{\int \mu_e(c) \bar{H}(\tau c) f_e(c) dc}{\int \bar{H}(\tau c) f_e(c) dc} < \int \mu_e(c) f_e(c) dc = \bar{g}_e(0).$$

The strict inequality uses Condition 2(b)'s strict monotonicity together with the non-constancy of $\bar{H}(\tau c)$ in c for $\tau > 0$.

The argument for $\rho < 0$ is symmetric, with the inequality reversed. For $\rho = 0$, μ_e is

constant in c (equal to $\bar{g}_e$), so $\bar{g}_e(\tau) = \bar{g}_e(0)$ for all τ . □

A.4. Proof of Proposition 2

Proof. Recall the equilibrium policy from equation (13). Define the swing-weighted registered shares

$$\omega_L(\tau) = \frac{\pi R_L(\tau)\phi_L}{\pi R_L(\tau)\phi_L + (1-\pi)R_H(\tau)\phi_H}, \quad \omega_H(\tau) = 1 - \omega_L(\tau),$$

so that $g^*(\tau) = \omega_L(\tau)\bar{g}_L(\tau) + \omega_H(\tau)\bar{g}_H(\tau)$.

Decompose the difference $d(\tau) = g^*(\tau) - g^*(0)$ by adding and subtracting $\omega_L(\tau)\bar{g}_L(0) + \omega_H(\tau)\bar{g}_H(0)$:

$$\begin{aligned} d(\tau) &= [\omega_L(\tau)\bar{g}_L(\tau) + \omega_H(\tau)\bar{g}_H(\tau)] - [\omega_L(0)\bar{g}_L(0) + \omega_H(0)\bar{g}_H(0)] \\ &= \underbrace{[\omega_L(\tau) - \omega_L(0)] \cdot [\bar{g}_L(0) - \bar{g}_H(0)]}_{\text{between-group: composition shift}} \\ &\quad + \underbrace{\omega_L(\tau)[\bar{g}_L(\tau) - \bar{g}_L(0)] + \omega_H(\tau)[\bar{g}_H(\tau) - \bar{g}_H(0)]}_{\text{within-group: drift}}. \end{aligned} \tag{19}$$

Part (i): Sign of $d(\tau)$ when $\rho \geq 0$.

Consider each component of equation (19).

Between-group component. Proposition 1 establishes that $R_L(\tau)/R_H(\tau)$ is non-increasing in τ . Since $\omega_L(\tau)$ is monotone increasing in $R_L(\tau)/R_H(\tau)$, $\omega_L(\tau)$ is non-increasing in τ , hence $\omega_L(\tau) \leq \omega_L(0)$ for $\tau > 0$. Assumption 2 gives $\bar{g}_L(0) > \bar{g}_H(0)$. The product is therefore non-positive,

$$[\omega_L(\tau) - \omega_L(0)] \cdot [\bar{g}_L(0) - \bar{g}_H(0)] \leq 0,$$

with strict inequality whenever Assumption 1 is strict on a set of positive measure.

Within-group component. Lemma 1 establishes $\bar{g}_e(\tau) \leq \bar{g}_e(0)$ for $\rho \geq 0$, with strict

inequality when $\rho > 0$. Each weighted term is non-positive, and their sum is non-positive. Strict inequality holds when $\rho > 0$.

Summing, $d(\tau) \leq 0$ for $\rho \geq 0$. The strict inequality holds whenever either Assumption 1 is strict (between-group strictness) or Assumption 3 is strict (within-group strictness).

Part (ii): $|d(\tau)|$ non-decreasing in ρ .

The between-group component does not depend on ρ at all: it depends only on the cost distributions and Assumption 2, which fix $\omega_L(\tau)$ and the difference $\bar{g}_L(0) - \bar{g}_H(0)$.

The within-group component depends on ρ through $\bar{g}_e(\tau) - \bar{g}_e(0)$. We show $\partial[\bar{g}_e(0) - \bar{g}_e(\tau)]/\partial\rho \geq 0$ for $\tau > 0$.

Parameterize the conditional expectation around $\rho = 0$ as

$$\mu_e(c; \rho) = \bar{g}_e + \rho \cdot \beta_e(c) + o(\rho),$$

where $\beta_e(c)$ is the structural sensitivity of within-group preferences to c , satisfying $\int \beta_e(c) f_e(c) dc = 0$ (so that the unconditional mean is $\bar{g}_e$ at $\rho = 0$) and $\beta'_e(c) > 0$ (so that increasing c shifts μ_e in the direction of ρ).

Substituting into equation (18),

$$\bar{g}_e(\tau; \rho) - \bar{g}_e(0; \rho) = \rho \cdot \left[\frac{\int \beta_e(c) \bar{H}(\tau c) f_e(c) dc}{\int \bar{H}(\tau c) f_e(c) dc} - \int \beta_e(c) f_e(c) dc \right] + o(\rho).$$

The bracket is strictly negative when $\beta'_e > 0$ and $\bar{H}(\tau c)$ is strictly decreasing in c (which holds for $\tau > 0$): the weighted average of an increasing function under weights that favor low c is below the unweighted average. Therefore, to leading order in ρ , $\bar{g}_e(0; \rho) - \bar{g}_e(\tau; \rho)$ is increasing in ρ at rate equal to the absolute value of this bracketed expression.

Both within-group components in equation (19) therefore become more negative as ρ rises (their absolute values grow), so $|d(\tau)|$ is non-decreasing in ρ at any fixed $\tau > 0$.

Part (iii): $|d(\tau)|$ non-decreasing in τ .

Both components of equation (19) have absolute values non-decreasing in τ .

Between-group. $\omega_L(\tau)$ is non-increasing in τ , so $\omega_L(0) - \omega_L(\tau)$ is non-decreasing. The absolute value of the between-group product is therefore non-decreasing in τ .

Within-group. Differentiating equation (18) with respect to τ , $\bar{g}_e(\tau)$ is non-increasing in τ for $\rho \geq 0$ (the same monotonicity argument as Lemma 1 applied infinitesimally). Hence $\bar{g}_e(0) - \bar{g}_e(\tau)$ is non-decreasing, and the within-group component's magnitude is non-decreasing.

The sum of two non-decreasing-magnitude components has non-decreasing magnitude. $\square$

A.5. Proof of Proposition 3

Proof. Under Assumption 4 and Condition 1, Φ is twice continuously differentiable on $[0, 1]$, $X(\tau)$ is continuously differentiable in τ via the differentiability of $R_e(\tau)$ established in equation (17), and $d(\tau)^2$ is continuously differentiable in τ via the differentiability of $\bar{g}_e(\tau)$ and $R_e(\tau)$. Hence $W(\tau; \alpha, \beta)$ is continuously differentiable on $[0, 1]$.

By the boundary conditions $W'(0) > 0$ and $W'(1) < 0$, the intermediate value theorem implies the existence of $\tau^* \in (0, 1)$ with $W'(\tau^*) = 0$.

For uniqueness, compute

$$W''(\tau) = \alpha\Phi''(\tau) - \beta X''(\tau) - 2\gamma[d'(\tau)^2 + d(\tau)d''(\tau)], \quad (20)$$

where $\gamma = 1 - \alpha - \beta$. The first term is strictly negative under Assumption 4. A sufficient condition for $W''(\tau) < 0$ on $(0, 1)$ is

$$\alpha \cdot |\Phi''(\tau)| > \beta \cdot |X''(\tau)| + 2\gamma \cdot |d'(\tau)^2 + d(\tau)d''(\tau)|.$$

Under standard regularity (Condition 1, plus boundedness of X'' and d''), the right-hand side is bounded above on $[0, 1]$, so the curvature condition can be satisfied by choosing Φ with sufficiently negative second derivative. When this condition holds, W is strictly

concave on $(0, 1)$, and τ^* is the unique maximizer. $\square$

A.6. Proof of Proposition 4

Proof. We treat ρ as a parameter that affects only the within-group drift channel; the between-group composition shift in equation (19) does not depend on ρ .

Step 1: First-order condition. The optimal $\tau^*(\rho)$ at fixed simplex weights (α, β) with $\gamma = 1 - \alpha - \beta > 0$ satisfies

$$W'(\tau^*; \rho) = \alpha\Phi'(\tau^*) - \beta X'(\tau^*) - 2\gamma \cdot d(\tau^*; \rho) \cdot \frac{\partial d(\tau^*; \rho)}{\partial \tau} = 0. \quad (21)$$

$X'(\tau) = -\pi R'_L(\tau) - (1 - \pi)R'_H(\tau) > 0$ since $R'_e(\tau) < 0$ from equation (17). Equation (21) can be rearranged to read

$$\alpha\Phi'(\tau^*) = \beta X'(\tau^*) + 2\gamma \cdot d(\tau^*; \rho) \cdot \frac{\partial d(\tau^*; \rho)}{\partial \tau}.$$

The left-hand side is the marginal integrity benefit; the right-hand side is the marginal social cost.

Step 2: Sign of the cross-partial $\partial^2 W / \partial \tau \partial \rho$.

Write $D(\tau, \rho) \equiv d(\tau; \rho)$. From Proposition 2: for $\rho \geq 0$, $D(\tau, \rho) \leq 0$ for $\tau > 0$; $|D|$ is non-decreasing in ρ ; $|D|$ is non-decreasing in τ .

The product $D(\tau, \rho) \cdot \partial D / \partial \tau$ is non-negative for $\rho \geq 0$ (product of two non-positive quantities, since $\partial D / \partial \tau \leq 0$ for $\rho \geq 0$ as a consequence of part (iii) extended infinitesimally). It is non-decreasing in ρ because both factors grow in absolute value as ρ rises while preserving their non-positive sign.

Therefore $\partial / \partial \rho [D \cdot \partial D / \partial \tau] \geq 0$. Multiplying by $-2\gamma < 0$,

$$\frac{\partial^2 W}{\partial \tau \partial \rho} = -2\gamma \cdot \frac{\partial}{\partial \rho} \left[D \cdot \frac{\partial D}{\partial \tau} \right] \leq 0.$$

The cross-partial is non-positive; strictly negative when $\rho > 0$ and Condition 2(b) holds.

Step 3: Implicit function theorem.

Define $G(\tau, \rho) \equiv W'(\tau; \rho)$. The optimum $\tau^*(\rho)$ satisfies $G(\tau^*(\rho), \rho) = 0$. Under the strict concavity condition of Proposition 3, $\partial G/\partial \tau = W''(\tau) < 0$ at $\tau = \tau^*(\rho)$. By Step 2, $\partial G/\partial \rho = \partial^2 W/\partial \tau \partial \rho \leq 0$.

By the implicit function theorem,

$$\frac{d\tau^*}{d\rho} = -\frac{\partial G/\partial \rho}{\partial G/\partial \tau} = -\frac{(\leq 0)}{(< 0)} \leq 0.$$

The inequality is strict whenever the cross-partial is strictly negative, which holds under Condition 2(b) (strict monotonicity of μ_e in c).

Step 4: Behavior at $\rho = 0$.

When $\rho = 0$, Lemma 1 gives $\bar{g}_e(\tau) = \bar{g}_e(0)$ for all τ , so the within-group component of $d(\tau)$ in equation (19) vanishes. The between-group component remains, since it depends only on the FOSD ordering of cost distributions through Proposition 1. Therefore $d(\tau; \rho = 0) \neq 0$ in general: the representation distortion does not vanish at $\rho = 0$. The optimum $\tau^*(0)$ is the value that maximizes W when only the between-group channel operates, and $\tau^*(\rho) \leq \tau^*(0)$ for $\rho > 0$ with strict inequality under Condition 2(b).

This is the precise sense in which the framework's main result holds. The framework does not claim that $\rho = 0$ reduces the welfare problem to a standard ordeal-theoretic benchmark; the between-group composition channel persists at $\rho = 0$ via Proposition 1. Rather, the framework's claim is that the within-group drift channel, governed by ρ , adds to the always-present between-group composition channel, and the comparative static $d\tau^*/d\rho \leq 0$ captures this addition. $\square$

A.7. Proof of Corollary 1

Proof. τ_0 is welfare-improving relative to $\tau = 0$ if and only if $W(\tau_0; \alpha, \beta) > W(0; \alpha, \beta)$. By the boundary conditions $\Phi(0) = 0$, $X(0) = 0$, and $d(0) = 0$,

$$W(0; \alpha, \beta) = \alpha \cdot 0 - \beta \cdot 0 - (1 - \alpha - \beta) \cdot 0 = 0.$$

Therefore the welfare-improvement condition $W(\tau_0; \alpha, \beta) > 0$ becomes

$$\alpha\Phi(\tau_0) - \beta X(\tau_0) - (1 - \alpha - \beta)d(\tau_0)^2 > 0,$$

which rearranges to inequality (16). $\square$

A.8. Proof of Proposition 5

Proof. Let $T_i = h_T(v_i)$ with $h'_T(\cdot) > 0$, so T_i is a strictly increasing function of v_i .

We show $\mathbb{E}[v_i \mid \text{Register}_i = 1] \geq \mathbb{E}[v_i \mid \text{Register}_i = 0]$. Since h_T is monotone increasing, the same inequality follows for T_i .

Conditional on $c_i = c$ and $e_i = e$, registration occurs iff $v_i \geq \tau c$. The conditional distribution of v_i among compliers ($v_i \geq \tau c$) FOSD-dominates the conditional distribution among non-compliers ($v_i < \tau c$): for any threshold $\tau c > 0$, the upper truncation of any distribution has mean weakly above the lower truncation. Hence $\mathbb{E}[v_i \mid v_i \geq \tau c] \geq \mathbb{E}[v_i \mid v_i < \tau c]$ pointwise in (c, e) .

Integrating over the joint distribution of (c, e) in the population (using the law of total expectation),

$$\begin{aligned} \mathbb{E}[v_i \mid \text{Register}_i = 1] &= \int \mathbb{E}[v_i \mid v_i \geq \tau c, c, e] dF_{c,e \mid \text{Register}=1} \\ &\geq \int \mathbb{E}[v_i \mid v_i < \tau c, c, e] dF_{c,e \mid \text{Register}=0} \\ &= \mathbb{E}[v_i \mid \text{Register}_i = 0]. \end{aligned}$$

Strict inequality holds whenever the registration constraint binds on a set of positive measure, which obtains for any $\tau > 0$ such that $\Pr(v < \tau c) > 0$ in some part of the population.

Applying the monotone transformation h_T preserves the inequality. $\square$

A.9. Heterogeneous Group Correlations

The main text assumes $\rho_L = \rho_H = \rho$. The extension to heterogeneous group correlations (ρ_L, ρ_H) proceeds as follows. Define the population-weighted average correlation

$$\bar{\rho} = \pi\rho_L + (1 - \pi)\rho_H.$$

Proposition 6 (Generalized main result). *Under the heterogeneous-correlation extension and regularity conditions analogous to those of Proposition 4, the optimal verification intensity τ^* is non-increasing in $\bar{\rho}$ at any fixed value of $\rho_L - \rho_H$.*

Proof sketch. The within-group drift in group e scales with ρ_e rather than the common ρ . The aggregate drift in $g^*(\tau)$ is the registered-share-weighted sum of within-group drifts. Holding fixed the difference $\rho_L - \rho_H$, increases in $\bar{\rho}$ raise both group-specific drifts proportionally, so the aggregate drift magnitude $|d(\tau)|$ rises monotonically with $\bar{\rho}$. The remainder of the argument follows the proof of Proposition 4, with $\bar{\rho}$ in place of ρ . $\square$

A.10. Correlated Value and Cost

The main text assumes v_i is independent of c_i . The extension allowing for correlation σ between v_i and c_i proceeds as follows.

If $\sigma > 0$ (high-cost types also have high value), the differential compliance result is muted: high-cost low-education types have higher value, so they are more likely to comply at the margin. The composition shift $|R_L(0)/R_H(0) - R_L(\tau)/R_H(\tau)|$ is smaller than under independence.

If $\sigma < 0$ (high-cost types have low value), the differential compliance result is amplified. The composition shift is larger. This is the empirically more relevant case based on the political-science literature on participation.

Proposition 7 (Robustness to value-cost correlation). *Under the extension with correlated v_i and c_i with correlation σ , and maintaining the other assumptions:*

- (i) Proposition 1 continues to hold for all $\sigma > -1$.
- (ii) Proposition 2's sign result continues to hold for all $\sigma \in [-1, 1]$.
- (iii) The magnitude of $d(\tau)$ is non-increasing in σ .
- (iv) By Proposition 4's argument, τ^* is non-decreasing in σ .

The qualitative implications of the framework are robust to value-cost correlation, but the magnitudes of all comparative statics shift with σ .

A.11. Discussion of Functional Forms

The main text uses a quadratic representation penalty $\gamma \cdot d(\tau)^2$ where $\gamma = 1 - \alpha - \beta$. Several alternatives are worth noting.

Linear penalty $\gamma \cdot |d(\tau)|$. Under a linear penalty, the marginal representation cost is $\gamma \cdot \text{sign}(d(\tau))$, which is constant in magnitude. The first-order condition becomes simpler but the comparative-static analysis requires additional care because the marginal cost is not differentiable at $d(\tau) = 0$. The qualitative result on $d\tau^*/d\rho \leq 0$ survives.

General convex penalty $\gamma \cdot \psi(|d(\tau)|)$ with ψ convex increasing. The qualitative results extend with the relevant comparative statics replaced by their corresponding convex-function versions. Quadratic is a useful baseline because it is symmetric and tractable.

Asymmetric penalty. If there is reason to think left-drift costs differ from right-drift costs (e.g., societies place higher weight on protecting redistribution-supporting voters), an asymmetric form $\gamma_L(d(\tau)^-)^2 + \gamma_R(d(\tau)^+)^2$ can be used with $\gamma_L + \gamma_R = 1 - \alpha - \beta$. The framework extends mechanically; the operative parameter becomes the relevant directional weight.

A.12. Linearization for Empirical Implementation

The empirical implementation of the framework requires a closed-form expression for the representation distortion $d(\tau)$ that can be calibrated from estimable quantities.

A first-order expansion of equation (19) around $\tau = 0$ yields

$$d(\tau) \approx \underbrace{\Delta\omega_L(\tau) \cdot [\bar{g}_L - \bar{g}_H]}_{\text{between-group}} + \underbrace{\rho \cdot \tau \cdot \kappa}_{\text{within-group}} + O(\tau^2), \quad (22)$$

where $\Delta\omega_L(\tau) = \omega_L(\tau) - \omega_L(0)$ is the change in the swing-weighted low-education share, and κ is a structural constant depending on the joint distribution of (c, g^*) that can be estimated from cross-municipality variation.

This linearization makes the calibration tractable: $\Delta\omega_L(\tau)$ is observed (from the decomposition), $\bar{g}_L - \bar{g}_H$ is observed (from pre-BVR vote-share differences across education-stratified municipalities), and κ can be estimated using the main-text calibration procedure.

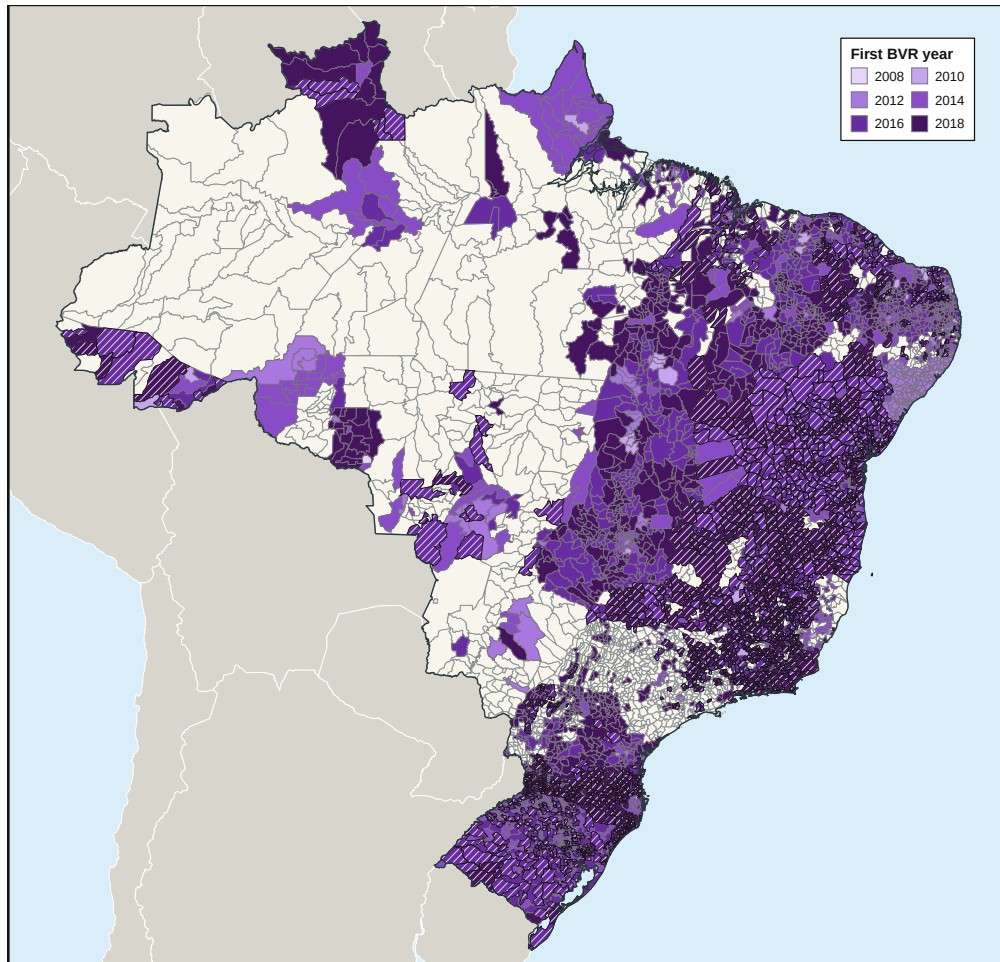
B. Maps

Figure 16 maps the municipal rollout of biometric registration by first observed BVR year, including municipalities observed with hybrid status. The figure complements the rollout timeline in the main text by showing the spatial concentration of later BVR waves and the municipalities flagged as hybrid in the official 2018 status source.

C. Appendix to Section 5: Technical Details and Additional Diagnostics

This appendix collects the technical pieces that support the main decomposition but are not needed to follow the paper’s central argument. The main text uses aggregate count accounting as the identification anchor and then uses the cohort framework to

Figure 16: Municipal Biometric Registration Rollout by First Treatment Year



Note: Municipalities are shaded by the first election year in which they are observed with any BVR status, including hybrid implementation. Unshaded municipalities have no observed BVR status through 2018 in the clean municipal panel. Diagonal hatching identifies municipalities flagged as hybrid in the official 2018 biometric-status source. The map uses the 2010 IBGE municipal geometry, so municipalities absent from that shapefile are not drawn.

allocate effects across places, ages, and regimes. The appendix documents the exact operator, implementation rules, and supplementary diagnostics that show where the cohort framework is most credible.

C.1. Exact cohort operator and education accounting

Let $N_{m,t,c,e}$ be the count of registered voters in municipality m , election year t , age cohort c , and education group $e \in \{L, H\}$. Denote total cohort size by $N_{m,t,c} = \sum_e N_{m,t,c,e}$. Let $A_{c'c}$ be the mechanical-aging matrix, where $A_{c'c}$ is the share of voters in source cohort c' at t who age into target cohort c at $t+2$. The matrix has at most two nonzero entries per row for all nonabsorbing bins, reflecting the fact that a two-year election cycle can leave part of a source cohort in place and move the remainder into the next bin. The exact entries follow the TSE age bins: single-year bins for ages 16–20, a four-year bin for ages 21–24, five-year bins thereafter, and an absorbing 100+ bin.

Given pre-treatment survival rates $\hat{\sigma}_{m,c'}$, the no-BVR counterfactual for total cohort size is

$$\hat{N}_{m,t+2,c}^0 = \sum_{c'} A_{c'c} \hat{\sigma}_{m,c'} N_{m,t,c'} + I_{m,t+2,c}, \quad (23)$$

where $I_{m,t+2,c}$ captures new registration into the youngest bins. This formulation makes explicit that a target cohort at $t+2$ can receive mass from more than one source cohort at t .

For the education split, let $P_{m,c',e \rightarrow e'}^0$ denote the pre-treatment transition probability from education group e in source cohort c' at t to education group e' at $t+2$ absent BVR. The corresponding no-BVR cell counts are

$$\hat{N}_{m,t+2,c,e'}^0 = \sum_{c',e} A_{c'c} \hat{\sigma}_{m,c'} N_{m,t,c',e} P_{m,c',e \rightarrow e'}^0 + I_{m,t+2,c,e'}, \quad (24)$$

where $I_{m,t+2,c,e'}$ allows inflows into the youngest bins by education category.

For ages 35 and older, the benchmark implementation sets the natural education-progression probability from low to high education to zero over a two-year cycle. For ages 25–34, that transition is allowed but expected to be small. For ages 16–24, it is estimated from pre-treatment transitions and the resulting record-updating estimates are treated

as auxiliary rather than central.

Cohort-level exit is

$$E_{m,t+2,c} = \max \left\{ 0, \hat{N}_{m,t+2,c}^0 - N_{m,t+2,c} \right\}. \quad (25)$$

Under Scenario B, all within-cohort exit is allocated to the low-education group. Under Scenario C, exit is allocated proportionally to the no-BVR education composition within the cohort. In both cases, record updating is the residual movement from low to high education required to reconcile observed and predicted cell counts after the exit allocation is fixed. Finally, record updating in any cell is capped by observed biometric uptake:

$$R_{m,t+2,c} \leq b_{m,t+2,c} N_{m,t+2,c}, \quad (26)$$

where $b_{m,t+2,c}$ is the share of the cohort with completed biometrics. This cap is usually slack in strict municipalities and informative in hybrid municipalities.

C.2. Implementation rules and interpretation

Survival-rate estimation. The preferred estimate uses municipality-specific pre-treatment transitions when the source cohort is large enough. When the source cell is too small, the algorithm falls back to the state-level estimate for the same cohort; if that is unavailable, it uses the national cohort mean from untreated observations. If the final code uses thresholds different from the ones below, replace them before integrating the section. In the current implementation, municipality-specific survival is used when the source cell contains at least 100 voters in the single-year young bins and at least 200 voters in the multi-year bins.

Interpretation of exit. The exit component is best read as BVR-attributable loss relative to the municipality's own pre-treatment cohort dynamics. Because abnormal post-adoption out-migration and administrative cancellation are not separately observed, the municipality-level exit estimate is an upper bound on true BVR-induced removal whenever post-adoption migration deviates from the pre-treatment baseline.

Interpretation of record updating. The cleanest record-updating estimates come from working-age cohorts that are old enough for natural schooling progression to be limited but not so old that survival-rate noise dominates. For that reason, the main text emphasizes the 25–44 and 35–64 patterns and treats the youngest and oldest bins as supporting rather than central evidence.

How to read the regime figure. The stacked regime figure is a summary of the share of the baseline electorate administratively affected by the reform through either removal or record updating. It is not a decomposition of net registry contraction, because record updating changes measured composition without shrinking the total count.

C.3. Supplementary tables

Table 6: Raw-count version of the cohort-based allocation by first-treatment regime

First regime	Municipalities	Exit count	Re-labeling (B)	Re-labeling (C)
Strict	2,475	6,210,453	3,607,795	4,958,520
Hybrid	1,846	1,561,570	550,381	719,475
All treated	4,321	7,772,023	4,158,176	5,677,995

Note: This is the raw-count counterpart to Table 2. Keeping counts in the appendix avoids mixing levels and rates in the main text while preserving the quantities needed for calibration or back-of-the-envelope comparisons.

Table 7: Working-age restrictions for the cohort-based allocation

Sample	Exit (upper bound)	Re-labeling (B)	Re-labeling (C)
All ages	8.35%	4.47%	6.10%
Ages 25–64	<i>[fill from code]</i>	<i>[fill from code]</i>	<i>[fill from code]</i>
Ages 35–64	<i>[fill from code]</i>	<i>[fill from code]</i>	<i>[fill from code]</i>

Note: Rates should be expressed as percentages of the restricted sample’s 2008 baseline electorate. This table is the most useful robustness check for the cohort framework because it shows whether the aggregate allocation changes materially once the noisiest age bins (16–24 and 65+) are removed. If the working-age estimates are close to the all-age estimates, say so directly in the text.

Table 8: Pre-treatment backtest of the cohort algorithm

Untreated cycle used as pseudo-treatment	Pseudo-exit	Pseudo-relabel B	Pseudo-relabel C
2008→2010	<i>[fill from code]</i>	<i>[fill from code]</i>	<i>[fill from code]</i>
2010→2012	<i>[fill from code]</i>	<i>[fill from code]</i>	<i>[fill from code]</i>
Pooled untreated bench- mark	<i>[fill from code]</i>	<i>[fill from code]</i>	<i>[fill from code]</i>

Note: Restrict each row to municipalities untreated at both dates and run the same cohort algorithm used in the main decomposition. The target is not exact zeros in every cell but small pseudo-effects relative to the post-adoption magnitudes. This is the most direct diagnostic of whether the framework is spuriously generating exit or record updating absent BVR.

Table 9: Implementation diagnostics for the cohort framework

Diagnostic	Value
Treated municipalities used in the allocation	4,321
Electorate-weighted share of cells using municipality-specific survival	<i>[fill from code]</i>
Electorate-weighted share of cells using state-level fallback	<i>[fill from code]</i>
Electorate-weighted share of cells using national fallback	<i>[fill from code]</i>
Electorate-weighted share of cells where the biometric-uptake cap binds	<i>[fill from code]</i>
Electorate-weighted share of cells where the non-negativity floor binds	<i>[fill from code]</i>
Bootstrap SE for aggregate exit	<i>[fill from code]</i>
Bootstrap SE for aggregate re-labeling (B)	<i>[fill from code]</i>
Bootstrap SE for aggregate re-labeling (C)	<i>[fill from code]</i>

Note: This table makes the implementation transparent. If the bootstrap is not yet available, drop those rows rather than reporting conventional standard errors that ignore first-stage uncertainty. The share of cells where the uptake cap or non-negativity floor binds is particularly informative about where the framework is doing mechanical cleanup rather than delivering clean signal.

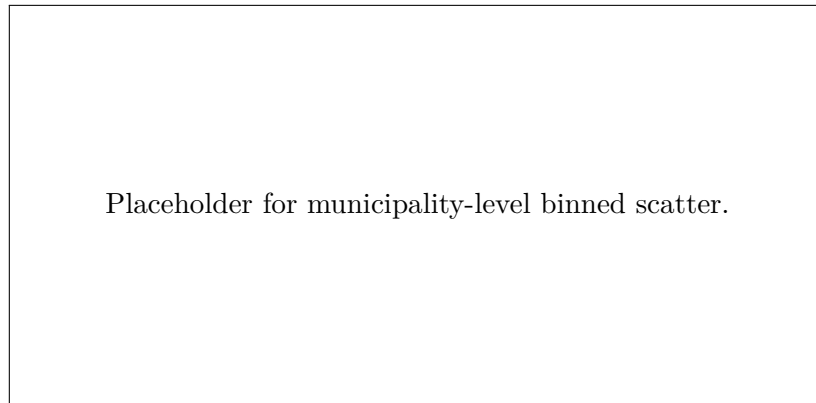


Figure 17: Municipality-level exit rate and baseline low-education share

Note: This figure is a descriptive complement to the main allocation results. Because the dependent variable is itself generated by the cohort framework, the figure should be read as supportive heterogeneity rather than independent identification.

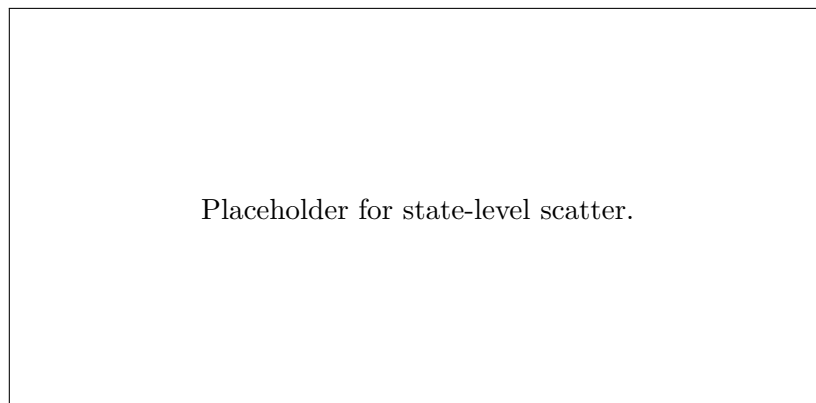


Figure 18: State-level exit rate and baseline low-education share

Note: State aggregation is useful for visual context but too coarse for identification. If retained, the text should emphasize the descriptive regional pattern rather than treat this figure as a test.

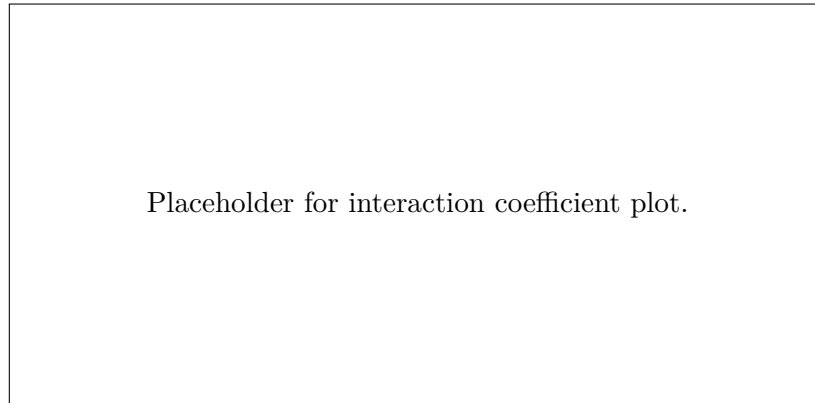


Figure 19: Supplementary interaction coefficients

Note: This figure should remain in the appendix unless the full pipeline is bootstrapped. Otherwise the reported uncertainty conditions on a generated dependent variable and does not capture first-stage uncertainty from the cohort allocation itself.

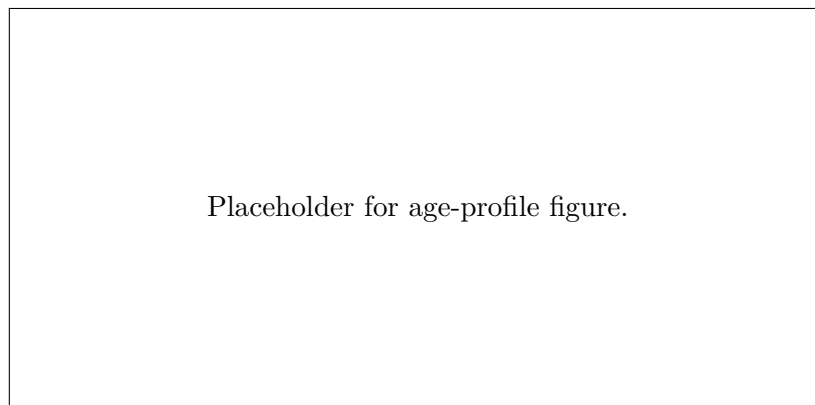
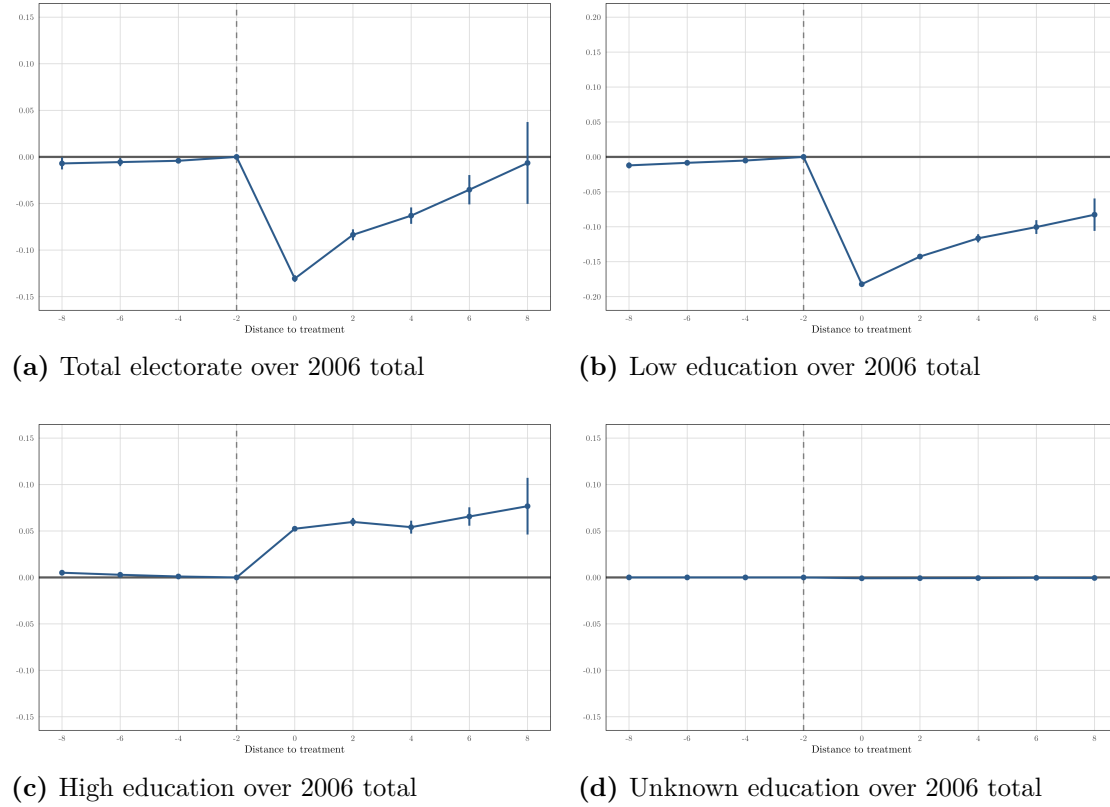


Figure 20: Full age profile of exit and record updating

Note: The main text should draw attention to the working-age concentration of re-labeling rather than to the noisy oldest bins. If a redesigned main-text age figure is later added, it should focus on ages 25–64 or 35–64 and relegate the full profile to this appendix.

Figure 21: Callaway–Sant’Anna decomposition event studies over the 2006 baseline



Note: Each panel plots Callaway–Sant’Anna dynamic event-study estimates using strict BVR municipalities as the treated sample and never-treated municipalities as the control group. Panel (a) uses total registered voters divided by the municipality’s 2006 registered voters. Panels (b)–(d) use the corresponding low-education, high-education, and unknown-education counts divided by the municipality’s 2006 total registered voters. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality. The component outcomes add to the total-electorate outcome by construction in the underlying data.

C.4. Additional figures

D. Including Hybrid Municipalities

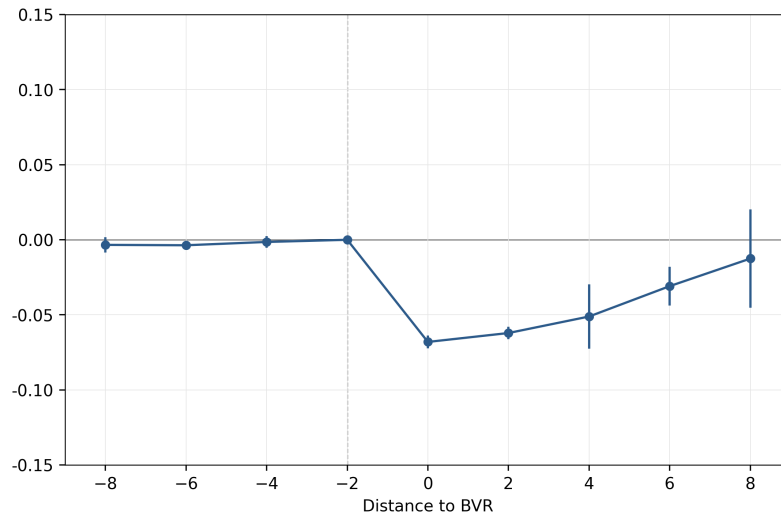
The main registry specifications use strict biometric registration as the treatment and exclude municipalities ever flagged as hybrid. This appendix repeats the Callaway–Sant’Anna event studies using a broader treatment definition that includes both strict and hybrid BVR municipalities. This is a lower-intensity specification: hybrid municipalities could have biometric updates before full enforcement, but voters were not yet subject to the same complete recadastramento requirement as in strict municipalities.

The hybrid-inclusive estimates preserve the same qualitative pattern as the strict-BVR estimates, but the impact effects are smaller. For the size of the electorate, the event-time 0 estimate moves from about -0.117 log points in the strict sample to about -0.068 log points when hybrid municipalities are included. For electorate composition, the low-education share falls by about 5.1 percentage points and the high-education share rises by about 5.1 percentage points at impact, compared with about 8.9 and 9.0 percentage points in the strict sample. The attenuation is consistent with hybrid BVR being a partial or lower-compliance-cost version of the reform rather than a different mechanism.

E. Alternative Estimator Comparisons

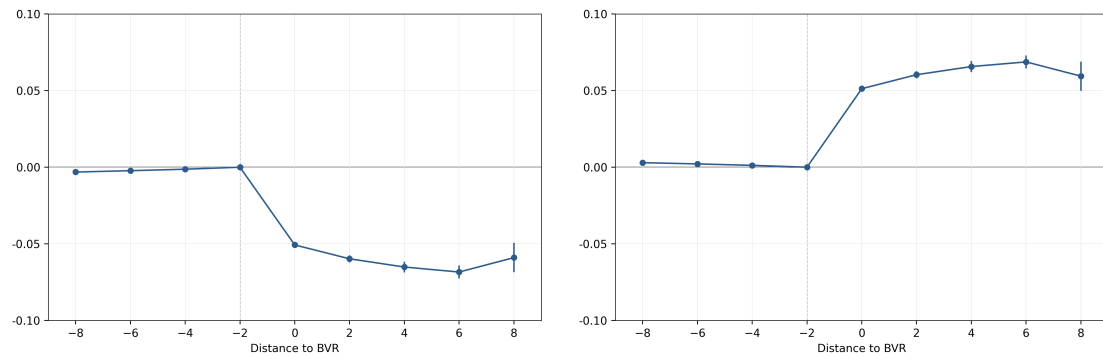
Because biometric registration rolls out at different times across municipalities, the main Callaway–Sant’Anna event studies are one way to summarize the dynamics under staggered adoption (Callaway and Sant’Anna, 2020). This appendix compares those main plots to dynamic TWFE, Borusyak–Jaravel–Spiess, and de Chaisemartin–D’Haultfoeuille alternatives (Borusyak et al., 2024; de Chaisemartin and D’Haultfoeuille, 2020). Across outcomes, the broad pattern is stable: BVR produces an immediate contraction in the registry and a persistent shift away from lower-education registrants and toward higher-education registrants. The main difference across estimators is precision at the longest horizons, where support becomes thin.

Figure 22: Including Hybrid Municipalities: Electorate Size



Note: The figure plots Callaway-Sant'Anna dynamic event-study estimates for the log number of registered voters after defining treatment as either strict or hybrid BVR. Never-treated municipalities are the control group. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality.

Figure 23: Including Hybrid Municipalities: Education Composition

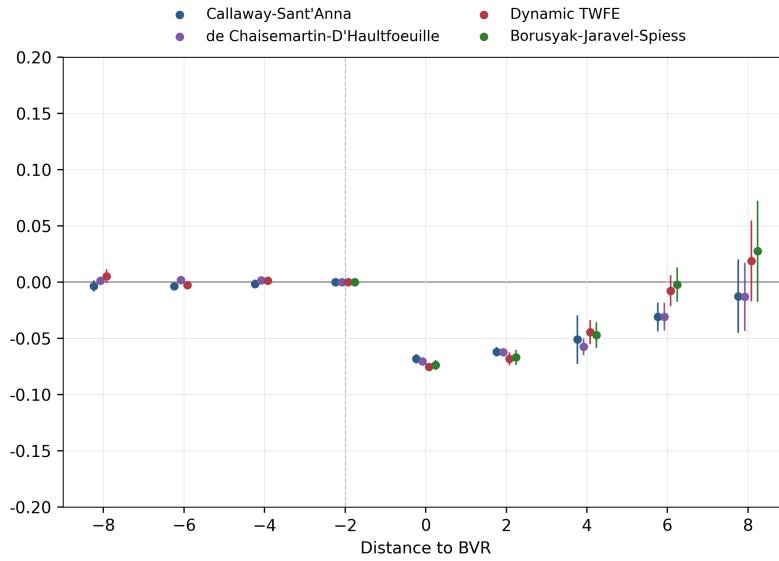


(a) Low-education share

(b) High-education share

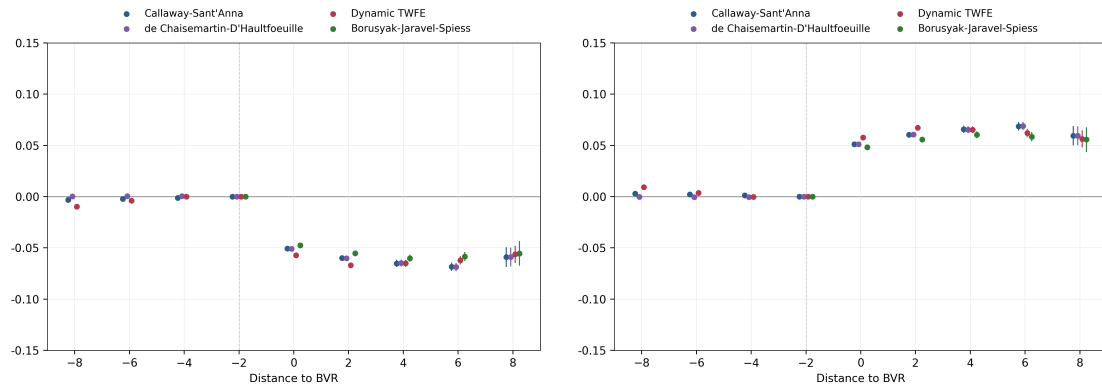
Note: Each panel plots Callaway-Sant'Anna dynamic event-study estimates after defining treatment as either strict or hybrid BVR. Panel (a) uses the share of registered voters with schooling below completed secondary education as the outcome. Panel (b) uses the share with incomplete secondary schooling or more. Never-treated municipalities are the control group. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality.

Figure 24: Including Hybrid Municipalities: Electorate Size Across Estimators



Note: Each point reports an event-time estimate and each whisker a 95% confidence interval for the hybrid-inclusive treatment definition. Colors identify the estimator. The Callaway–Sant’Anna series uses never-treated municipalities as the control group; the comparison series report dynamic TWFE, Borusyak–Jaravel–Spiess, and de Chaisemartin–D’Haultfoeulle alternatives.

Figure 25: Including Hybrid Municipalities: Education Composition Across Estimators

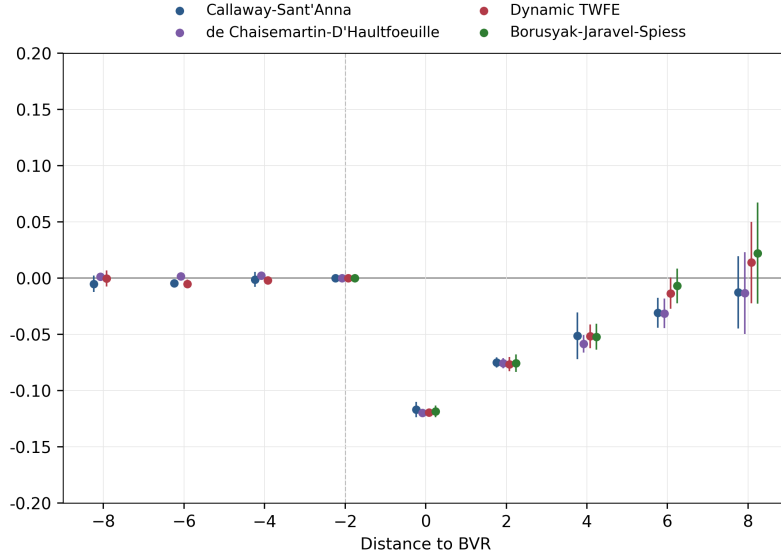


(a) Low-education share

(b) High-education share

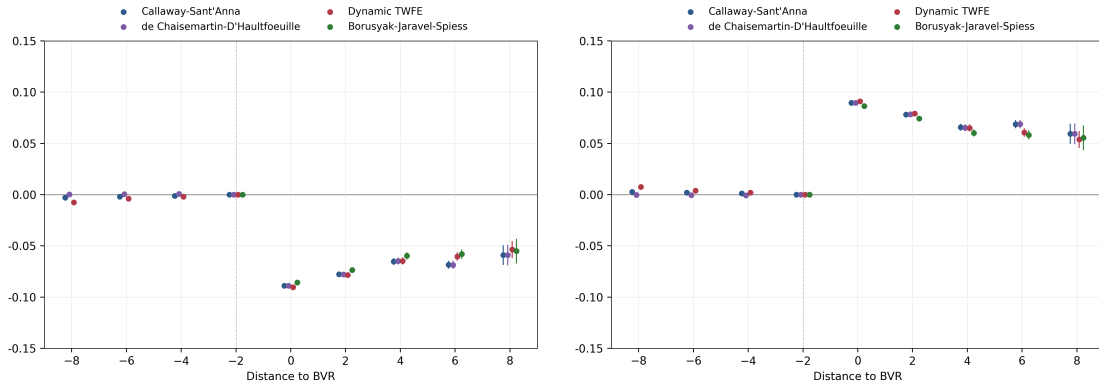
Note: Each point reports an event-time estimate and each whisker a 95% confidence interval for the hybrid-inclusive treatment definition. Colors identify the estimator. Panel (a) uses the share of registered voters with schooling below completed secondary education as the outcome. Panel (b) uses the share with incomplete secondary schooling or more.

Figure 26: Biometric Registration and Electorate Size: Alternative Estimators



Note: Each point reports an event-time estimate and each whisker a 95% confidence interval. Colors identify the estimator. The Callaway–Sant’Anna series corresponds to the main event study in the text. The comparison series report dynamic TWFE, Borusyak–Jaravel–Spiess, and de Chaisemartin–D’Haultfoeulle estimators, using the same strict-BVR sample where available.

Figure 27: Biometric Registration and Education Composition: Alternative Estimators



(a) Low-education share

(b) High-education share

Note: Each point reports an event-time estimate and each whisker a 95% confidence interval. Colors identify the estimator. Panel (a) uses the share of registered voters with schooling below completed secondary education as the outcome, and panel (b) uses the share with incomplete secondary schooling or more. Across Callaway–Sant’Anna, dynamic TWFE, Borusyak–Jaravel–Spiess, and de Chaisemartin–D’Haultfoeulle estimators, the low-education share falls around biometric adoption and the high-education share rises, with the largest differences appearing in the noisier long-run horizons.

F. Rambachan-Roth Sensitivity

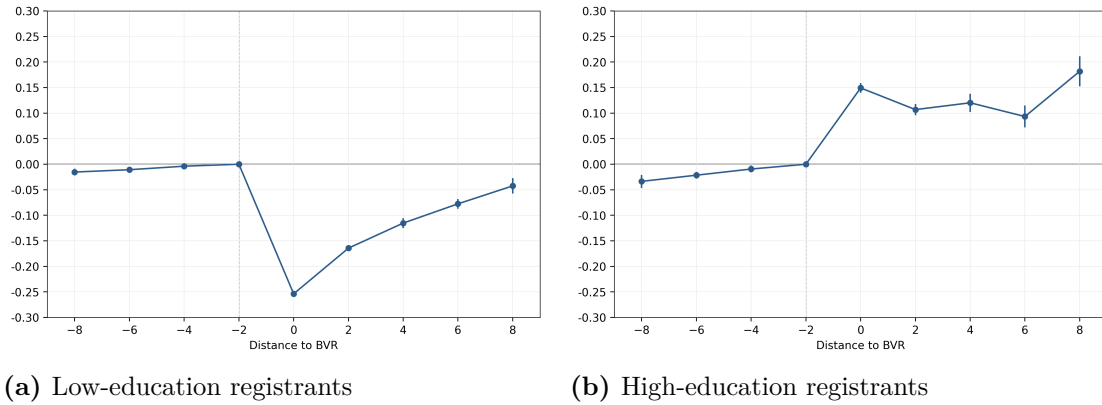
The most natural concern about the registry event studies is that the small pre-treatment coefficients in the main text may still leave room for latent violations of parallel trends. To quantify how much non-parallel drift would be required to overturn the impact estimates, I apply the Rambachan-Roth sensitivity framework (Rambachan and Roth, 2023) to the event-time 0 coefficient for the three main registry outcomes. The baseline confidence intervals are the same as in the main text. I then report identified sets under two alternative restrictions on the counterfactual trend. The first is a smoothness restriction that bounds second differences in the latent violation process. The second is a relative-magnitudes restriction that allows post-treatment violations to be up to $\bar{M}$ times as large as the largest pre-treatment deviation.

The plain-language takeaway is reassuring, but it depends on which class of violations one thinks is most plausible. Under the relative-magnitudes restriction, the main impact effects remain signed and statistically separated from zero even when post-treatment violations are allowed to be twice as large as the worst pre-trend visible in the data. The breakdown value therefore exceeds $\bar{M} = 2$ for the log size of the electorate and for both education-share outcomes. Put differently, the reviewer must allow confounding after adoption to be more than double the largest pre-treatment discrepancy before these baseline impact estimates lose statistical significance. The smoothness restriction is much harsher. Once one permits modest curvature in the latent untreated path, the identified sets widen quickly and can include zero by $M = 0.5$. I read that pattern as a reminder that the smoothness class is deliberately conservative, not as evidence against the baseline estimates: the relative-magnitudes exercise is the more natural metric when the visible pre-trends are already small, while the smoothness exercise asks whether sharply bending counterfactual paths could rationalize the observed break at adoption.

G. Additional Identification Checks

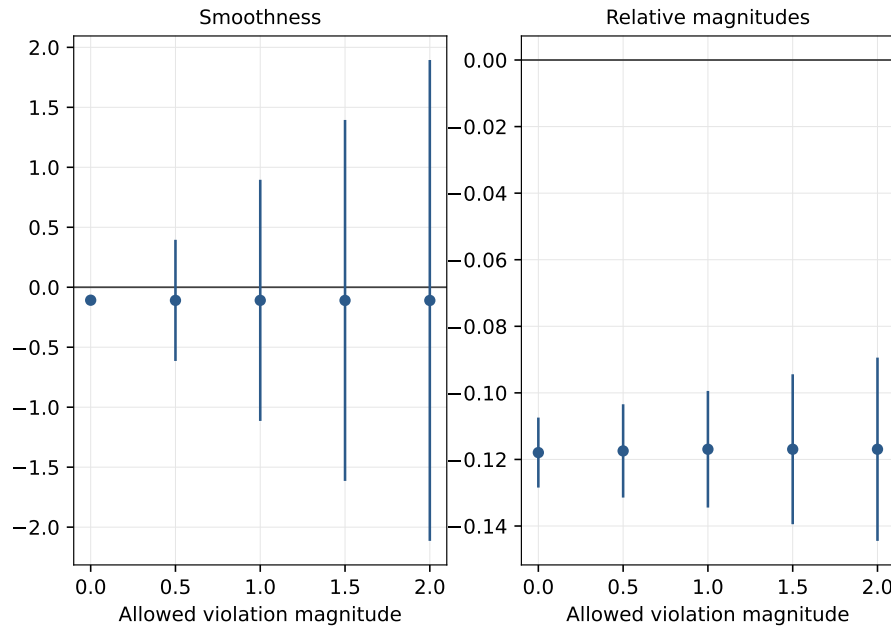
This section collects three further checks aimed at the same concern: that the main registry results might be driven by broader shocks or by estimator-specific features of the staggered-adoption design. First, I replace the common election-year fixed effects

Figure 28: Biometric Registration and Education-Specific Registry Counts



Note: Each panel plots Callaway-Sant'Anna dynamic event-study estimates for log education-specific registered-voter counts, using strict BVR municipalities as the treated sample and never-treated municipalities as the control group. Municipalities ever coded as hybrid are excluded. Event time -2 is the reference period and is shown at zero. Error bars report 95% confidence intervals with standard errors clustered by municipality. Both panels use the same y-axis range and tick marks.

Figure 29: HonestDiD Sensitivity: Electorate Size



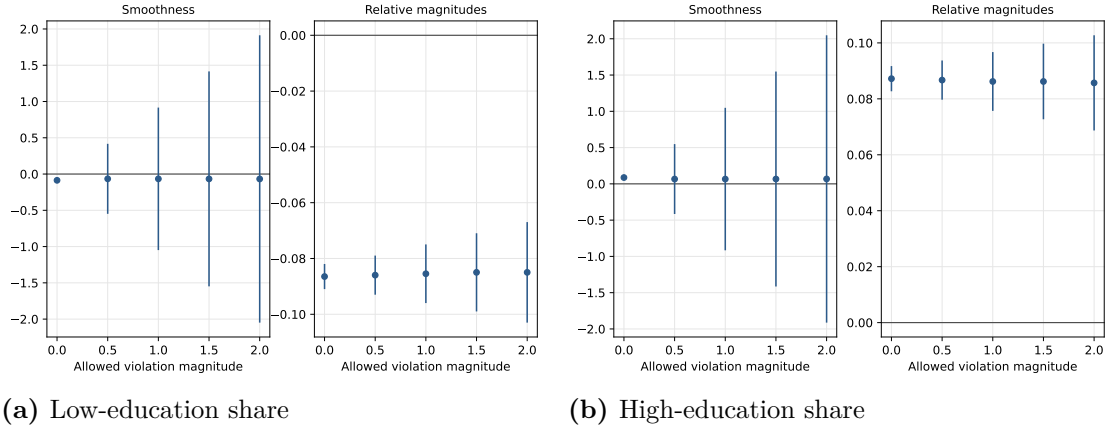
Note: The figure reports Rambachan-Roth identified sets for the event-time 0 coefficient from the baseline dynamic TWFE specification for the log number of registered voters. The left facet imposes smoothness bounds on second differences in the latent violation of parallel trends. The right facet imposes relative-magnitude bounds that scale the allowed post-treatment violation by the largest pre-treatment deviation. Under the relative-magnitudes restriction, the identified set remains strictly negative through $\bar{M} = 2$.

with state $\times$ election-year fixed effects in the dynamic TWFE robustness specification. I then apply an even stricter restriction that keeps only state-year cells with both treated and control municipalities, so every comparison is explicitly within-state and within-election. Figure 31 and Figure 32 show that the event-study paths are nearly unchanged. On impact, the coefficient on log registered voters moves from -0.118 in the dynamic TWFE companion to -0.118 with state-year fixed effects and -0.118 in the within-state-support sample. For the low-education share, the corresponding coefficients are -0.086 , -0.090 , and -0.086 . For the high-education share, they are 0.087 , 0.090 , and 0.087 . In proportional terms, every impact estimate remains within about 3.7 percent of the companion estimate. That is exactly the pattern one would want if the main estimates are not simply picking up state-level election shocks.

Second, I implement a randomization inference exercise that permutes first-treatment timing across municipalities 500 times while preserving the empirical cohort-size distribution. For all three outcomes, the actual event-time 0 coefficient lies deep in the tails of the placebo distribution: the permutation p-value is 0.002 for the log size of the electorate, 0.002 for the low-education share, and 0.002 for the high-education share. The implied placebo distribution is tightly centered near zero, so the observed impact effects are extreme even relative to many alternative staggered rollouts with the same cohort structure.

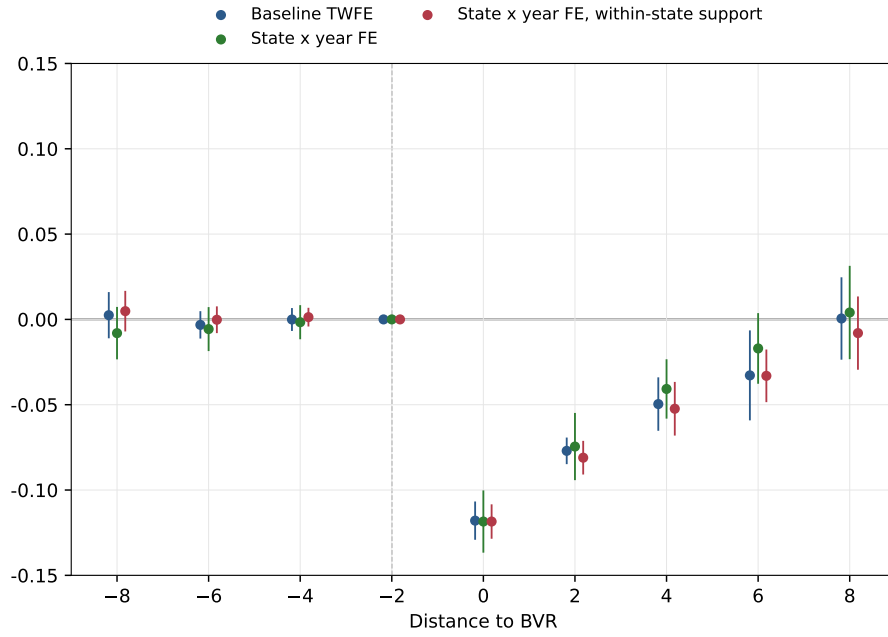
Third, the de Chaisemartin-D’Haultfoeuille weight diagnostic for the absorbing-treatment version of the dynamic TWFE design reports that only 3 of 5,460 ATT components receive negative weight, which is a share of 0.055 percent, and the absolute mass on those negative weights is effectively zero (de Chaisemartin and D’Haultfoeuille, 2020). Appendix E compares the main Callaway–Sant’Anna series to dynamic TWFE, Borusyak–Jaravel–Spiess, and de Chaisemartin–D’Haultfoeuille estimators directly. Finally, the cohort-balance exercise shows that treated cohorts differ meaningfully from never-treated municipalities in levels of population, income, and baseline education composition. Those level differences are precisely why the paper relies on within-municipality event-time comparisons rather than simple cross-sectional contrasts.

Figure 30: HonestDiD Sensitivity: Education Composition



Note: Each panel applies the same Rambachan-Roth procedure to the education-composition outcomes in the main text. At event time 0, the low-education share remains below zero and the high-education share remains above zero throughout the relative-magnitudes grid up to $\bar{M} = 2$, so the breakdown value exceeds that threshold for both outcomes. The smoothness bounds are again much wider, reflecting the fact that the smoothness class admits a broad family of sharply curved counterfactual paths.

Figure 31: Registry Size with State $\times$ Election-Year Fixed Effects



Note: The figure compares the dynamic TWFE companion event study to two robustness variants: one that replaces common election-year fixed effects with state $\times$ election-year fixed effects, and one that additionally restricts the sample to state-year cells containing both treated and control municipalities. Error bars report 95% confidence intervals with municipality and election-year clustering.

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H. Additional Graphs and Plots

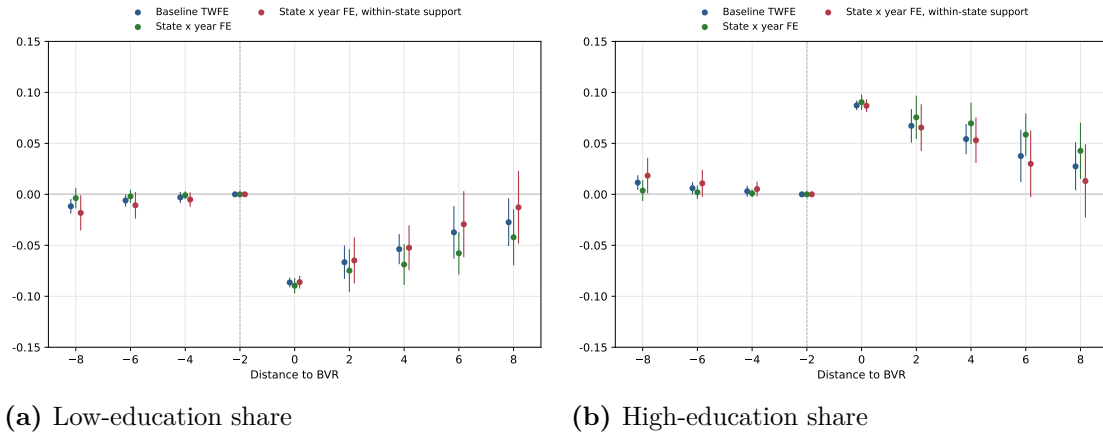
This appendix collects descriptive figures that complement the main-text survey discussion. Figures 34 and 35 report the share of explained variance by principal component for the institutional-trust and democracy survey blocks. The trust block is visibly more one-dimensional: its first component dominates the remaining components, while the democracy block leaves more variance for the second component. That pattern is one reason the democracy results in the main text are interpreted more cautiously.

I. Additional Tables

This section collects appendix tables that complement the main-text figures and discussion. When the generated table file is available, the first table reports the dynamic TWFE companion estimates, clustered standard errors, fixed-effects specification, and dependent-variable moments for the three registry outcomes studied in the main text.

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Figure 32: Education Composition with State $\times$ Election-Year Fixed Effects



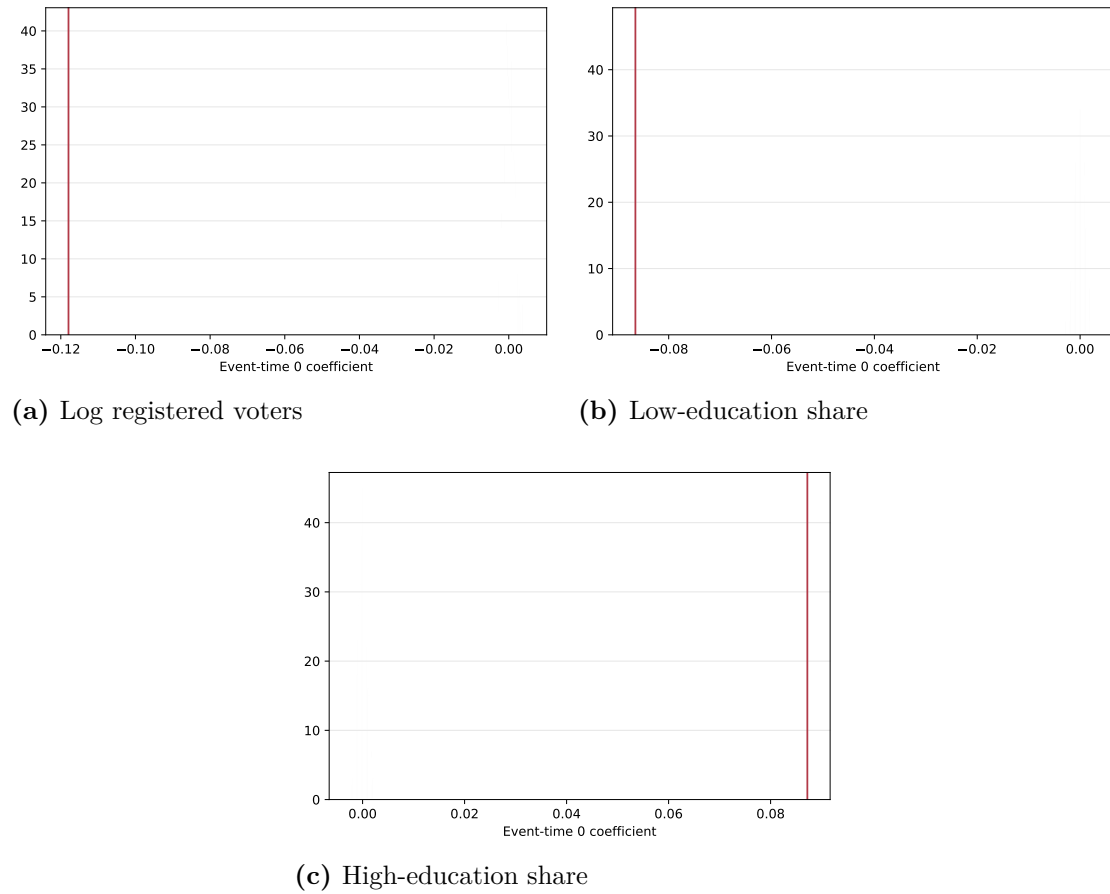
Note: Each panel overlays the dynamic TWFE companion event study with the same two robustness variants used in Figure 31. The education-composition effects remain visually and quantitatively close to the companion series, indicating that the composition results are not being driven by common state-level electoral shocks.

Table 10: Baseline differences by pre-BVR connectivity

	Low connectivity	High connectivity	Difference
Municipalities	1,321	1,321	0
Share treated by 2018	0.715	0.590	-0.125
Fixed broadband density, 2008	0.062	3.764	3.702
Log population, 2008	9.146	9.933	0.787
Log GDP per capita, 2008	8.439	9.463	1.024
North	0.166	0.041	-0.125
Northeast	0.624	0.058	-0.565
Center-West	0.020	0.204	0.183
Southeast	0.142	0.350	0.208
South	0.048	0.347	0.299
Low-education share, 2006	0.852	0.696	-0.156
High-education share, 2006	0.146	0.303	0.156
Log registered voters, 2006	8.776	9.588	0.813

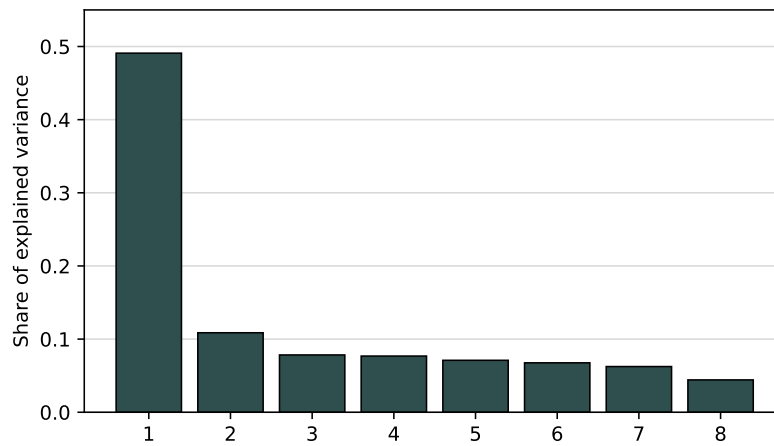
Note: The table compares municipalities in the bottom and top thirds of 2008 fixed-broadband density, after applying the main strict-BVR sample restrictions and excluding municipalities ever coded as hybrid. Region entries are shares. Population and GDP per capita are from the 2008 IBGE municipal panel. The difference column reports high minus low connectivity.

Figure 33: Permutation Distributions for the Event-Time 0 Coefficient



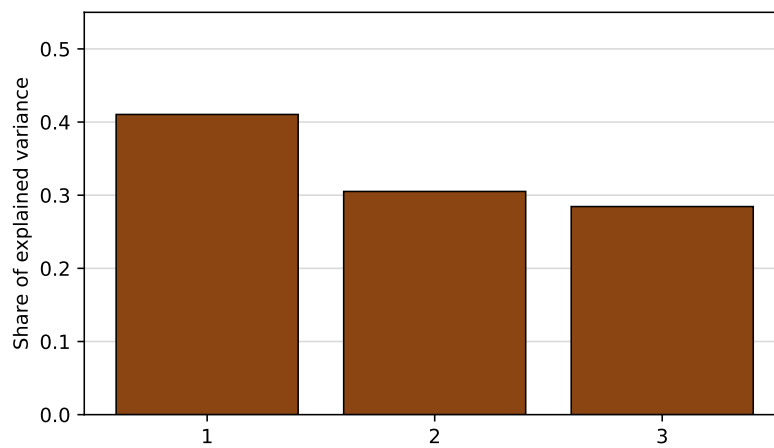
Note: Each histogram shows the distribution of the event-time 0 coefficient from 500 placebo assignments of treatment timing that preserve the observed cohort-size distribution. The vertical line marks the actual estimate from the dynamic TWFE companion event study. For all three outcomes, the observed estimate lies far in the tails of the placebo distribution, yielding permutation p-values of 0.002.

Figure 34: Trust Index: Share of Explained Variance by Principal Component



Note: The bars report the share of total variance explained by each principal component of the eight-item institutional-trust block. The first component explains 49.1% of the total variance, well above the remaining components.

Figure 35: Democracy Index: Share of Explained Variance by Principal Component



Note: The bars report the share of total variance explained by each principal component of the three-item democracy block. The first component explains 41.0% of the total variance and the second explains 30.5%, indicating a less sharply one-dimensional structure than in the institutional-trust block.

Table 11: Compositional regressions for decomposition interactions

Dependent Variables:	exit_rate_2008		cohort_exit_rate
Model:	Pooled	State FE	Within municipality
	(1)	(2)	(3)
<i>Variables</i>			
Constant	0.0456 (0.0373)		
Low-ed share (2008)	0.0210 (0.0252)	-0.0110 (0.0289)	
Strict	0.0213 (0.0268)	0.0298 (0.0276)	
Strict x low-ed share	0.1199*** (0.0392)	0.1344*** (0.0423)	
Log 2008 baseline	-0.0023 (0.0018)	-0.0045** (0.0021)	
Cohort low-ed share			-0.1582*** (0.0167)
Strict x cohort low-ed share			0.2926*** (0.0149)
<i>Fixed-effects</i>			
state		Yes	
ibge_municipality_id			Yes
cohort_band			Yes
<i>Fit statistics</i>			
Observations	4,321	4,321	93,484
R ²	0.59650	0.66372	0.56112
Within R ²		0.49456	0.10070
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			